

Question ID 7a327791

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 7a327791

PSAT2 RW 1.1

Novelist Leon Forrest admired William Faulkner’s writing style. Forrest’s novel *Divine Days* contains a long passage in tribute to Faulkner that is a perfect _____ of Faulkner’s style: anyone familiar with Faulkner’s writing would see the resemblance.

Which choice completes the text with the most logical and precise word or phrase?

- A. forgetting
- B. rejection
- C. imitation
- D. opinion

ID: 7a327791 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of novelist Leon Forrest’s admiration of William Faulkner’s writing style. In this context, "imitation" means something that resembles or reproduces something else. The text states that Forrest admired Faulkner’s style and that anyone familiar with Faulkner’s style would be able to see the stylistic similarities between a particular passage in Forrest’s novel *Divine Days* and Faulkner’s writing, a fact that supports the idea that Forrest’s novel pays tribute to Faulkner by reproducing, or imitating, Faulkner’s style.

Choice A is incorrect because in this context, a "forgetting" would mean an instance in which something is overlooked or not remembered. The text emphasizes Forrest’s admiration for Faulkner’s style and the fact that Forrest’s writing in some instances closely resembles Faulkner’s stylistically. It therefore wouldn’t make sense to say that in trying to pay tribute to Faulkner’s style, Forrest failed to remember it. Choice B is incorrect because in this context, a "rejection" would mean a dismissal of something as unworthy. The text emphasizes Forrest’s admiration for Faulkner’s writing style, stating that parts of Forrest’s writings indicate efforts to copy that style. It wouldn’t make sense therefore to suggest that Forrest had rejected or dismissed Faulkner’s style as unworthy. Choice D is incorrect because in this context, an "opinion" would mean a view or judgment. Although the text focuses on Forrest’s admiration of Faulkner, which suggests that Forrest had formed a positive judgment of Faulkner’s style, the word opinion wouldn’t make sense in this sentence: the sentence doesn’t say that the passage from Forrest’s novel expresses his view of Faulkner’s style; rather, it suggests that stylistically, the passage closely resembles, or imitates, Faulkner’s writing.

Question Difficulty: Easy

Question ID a7991da1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: a7991da1

PSAT2 RW 1.2

A unique dialect, or regional variety, of Spanish is spoken in Puerto Rico. It contains many words borrowed from the language of the Taínos, the Indigenous people of Puerto Rico. African languages also made important contributions to the Puerto Rican dialect. For example, the way certain vowel sounds are pronounced in it can be _____ to how they are pronounced in Yoruba, a West African language.

Which choice completes the text with the most logical and precise word or phrase?

- A. traced
- B. surrendered
- C. announced
- D. offered

ID: a7991da1 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of the origins of the unique dialect of Spanish spoken in Puerto Rico. In this context, "traced" means tracked back to. The text notes that the regional variety of Spanish spoken in Puerto Rico has its roots in the language of the Taínos and in the West African language of Yoruba. This context supports the idea that the way certain vowel sounds are pronounced in the Puerto Rican dialect can be traced back to Yoruba.

Choice B is incorrect because the text presents the statement about the relationship between vowel pronunciation in the Puerto Rican dialect of Spanish and in Yoruba as an example to support the claim that African languages have contributed to the Puerto Rican dialect. It therefore wouldn’t make sense to say that the pronunciation of certain vowel sounds in the Puerto Rican dialect can be "surrendered," or handed over on demand, to the pronunciation in Yoruba. Choice C is incorrect because the text gives no indication that the way certain vowel sounds are pronounced in the Puerto Rican dialect of Spanish has been "announced," or officially declared or proclaimed. According to the text, elements of the dialect have been inherited or borrowed from African languages, and the relationship between the pronunciation of certain vowels in the Puerto Rican dialect and in Yoruba is an example of this inheritance, suggesting that the pronunciation in the Puerto Rican dialect can be traced to Yoruba. Choice D is incorrect because the text presents the statement about the relationship between vowel pronunciation in the Puerto Rican dialect of Spanish and in Yoruba as an example to support the claim that African languages have contributed to the Puerto Rican dialect. It therefore wouldn’t make sense to say that the pronunciation of certain vowel sounds in the Puerto Rican dialect can be "offered," or presented as a gift, to the pronunciation in Yoruba.

Question Difficulty: Easy

Question ID fa014d2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: fa014d2d

PSAT2 RW 1.3

The invention in 1958 of the integrated circuit (or microchip) radically altered the semiconductor industry. In fact, some historians argue that it fundamentally _____ the industry by enabling it to take advantage of mass production methods for the first time.

Which choice completes the text with the most logical and precise word or phrase?

- A. overwhelmed
- B. bypassed
- C. obstructed
- D. transformed

ID: fa014d2d Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of the invention of the integrated circuit. As used in this context, "transformed" means substantially changed in some way. The text states that the invention radically altered the semiconductor industry, with some historians claiming that it had a fundamental effect on the industry by allowing for mass production for the first time. This context conveys that the invention of the integrated circuit transformed the semiconductor industry.

Choice A is incorrect. Although the text indicates that the invention of the integrated circuit significantly affected ("radically altered") the semiconductor industry, it doesn’t indicate that the invention "overwhelmed" the industry, which in this context would mean that it overcame the industry such that the industry struggled or was defeated. Instead, the text conveys that the invention’s effect was positive, since it allowed the semiconductor industry to begin making use of mass production methods. Choice B is incorrect because it wouldn’t make sense in context to say that the invention of the integrated circuit "bypassed" the semiconductor industry, which would mean it intentionally avoided the industry. The text indicates that the invention directly affected the semiconductor industry, since it made it possible for the industry to begin engaging in mass production methods. Choice C is incorrect because the text doesn’t indicate that the invention of the integrated circuit "obstructed" the semiconductor industry, which would mean that it blocked or hindered the industry in some way. Instead, the text indicates that the invention’s effect was positive, since it made it possible for the semiconductor industry to begin engaging in mass production methods.

Question Difficulty: Easy

Question ID 3654202f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 3654202f

PSAT2 RW 1.4

Radar and sonar can detect objects or people through walls, but they are too costly for consumer use. Researchers Fadel Adib and Dina Katabi have shown, however, that there may be a comparatively _____ alternative: small changes to the Wi-Fi technology used in smartphones can give users a cheap way to count people or even identify their gestures through walls.

Which choice completes the text with the most logical and precise word or phrase?

- A. infrequent
- B. inexpensive
- C. unmistakable
- D. impractical

ID: 3654202f Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of ways to detect people through walls. In this context, "inexpensive" means less costly or low in price. The passage describes two kinds of methods that can be used to "detect objects or people through walls": radar and sonar, and changes to the Wi-Fi technology used in smartphones. The word "comparatively" before the blank indicates that the best answer would be one that helps to describe a comparison between the two. Radar and sonar are described as "costly," while the method that uses Wi-Fi technology is described as "cheap"; therefore, it would be comparatively affordable, or inexpensive.

Choice A is incorrect because the text indicates that a small modification to the smartphones that many consumers already own would presumably result in consumers using that technology to detect objects or people through walls more regularly, or frequently; therefore, it would not be true that such technology would be comparatively infrequent, or uncommon. Choice C is incorrect because there is nothing in the text to indicate that either radar and sonar or changes to smartphone Wi-Fi technology are capable of being misunderstood or mistaken for something else; therefore, it would not make sense to describe changes to Wi-Fi smartphone technology as comparatively unmistakable, or indisputable. Choice D is incorrect because the text indicates that a small modification to the smartphones that many consumers already own and carry with them would be more functional or practical than radar and sonar, not more impractical, or unrealistic.

Question Difficulty: Easy

Question ID d8d33cc5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: d8d33cc5

PSAT2 RW 1.5

The early British postal system required the cost of mail delivery to be paid upon receipt, a system which encouraged inventive strategies by the intended recipient to avoid payment. To improve this system, _____ were proposed in 1837, including the use of a postage stamp, a small receipt pasted to the mail indicating that delivery costs had been paid by the sender.

Which choice completes the text with the most logical and precise word or phrase?

- A. investigations
- B. expansions
- C. reforms
- D. possessions

ID: d8d33cc5 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text’s discussion of the early British postal system. In this context, "reforms" most nearly means attempts at improving something. The text indicates that the pay-upon-delivery postage method led to fee avoidance on the part of recipients. It then says that action was taken to improve the postal system, giving examples such as the move to having the sender pay and affixing stamps to indicate payment had been received. These represent revisions to the system intended to improve its functioning; therefore, "reforms" most logically completes the text.

Choice A is incorrect. In this context, "investigations" most nearly means attempts to understand something. And although it is plausible that an attempt to improve the postal system would involve investigations, nothing in the text provides a logical basis for understanding the example of changes like affixing stamps to letters as an investigation. Choice B is incorrect. In this context, "expansions" most nearly means additions to something. Although the text indicates that some new policies were adopted, nothing in the text suggests that the overall number of policies increased, which expansions would seem to entail. Choice D is incorrect. In this context, "possessions" most nearly means items of personal property or things that otherwise belong to, or are part of, something. The text provides no logical context for understanding changes like affixing stamps to letters as possessions.

Question Difficulty: Easy

Question ID 7d8224f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	

ID: 7d8224f9

PSAT2 RW 1.6

In 1154, Muhammad al-Idrisi completed a collection of maps of the lands known to medieval Arabic and European scholars. This collection was titled *Al-Kitāb al-Rujārī (The Book of Roger)*, after the Norman king Roger II who hired him to create it. To create the collection, al-Idrisi consulted Arabic and Greek maps and interviewed travelers about the lands they visited. He included these travelers’ stories alongside the map illustrations.

Which choice best states the main purpose of the text?

- A. To discuss the benefits of studying mapmaking
- B. To explain how travelers created maps
- C. To describe a collection of medieval maps and how it was created
- D. To compare medieval Arabic and Greek mapmaking techniques

ID: 7d8224f9 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most accurately states the main purpose of the text, which is to describe a collection of medieval maps and how it was created. The text begins by mentioning Muhammad al-Idrisi’s collection of maps of lands known to medieval Arabic and European scholars. It then states that the Norman king Roger II hired al-Idrisi to create the collection and details al-Idrisi’s methods of creation: consulting Arabic and Greek maps and interviewing travelers. In short, the text presents a collection of medieval maps and then goes on to describe how that collection came to be.

Choice A is incorrect because the text describes a collection of maps and the process of creating that collection but does not discuss the benefits of studying mapmaking in general. Choice B is incorrect because though the text mentions that al-Idrisi interviewed travelers, the text does not describe how those travelers created maps. Choice D is incorrect because though the text mentions that al-Idrisi consulted Arabic and Greek maps, the text does not offer a comparison of Arabic and Greek mapmaking techniques.

Question Difficulty: Medium

Question ID 2af2016f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	

ID: 2af2016f

PSAT2 RW 1.7

A study by Dr. Paul Hanel and colleagues concluded that people are more likely to behave politely when listening to ideas they disagree with if they think about values before they engage in a discussion. Study participants were assigned to one of two groups. The experimental group spent a few minutes writing about one of their personal values before they had a group discussion on a controversial topic. And the control group spent a few minutes writing about a drink (tea, milk, etc.) before their group discussion on that topic. Hanel and colleagues found that the experimental group’s discussion was more civil than the control group’s discussion was.

Which choice best describes the main purpose of the text?

- A. To describe a widely held belief and how a study’s results support that belief
- B. To argue that researchers were surprised by the results of a certain study
- C. To suggest ways to improve a certain study’s experimental design
- D. To explain a study’s conclusion and how a research team arrived at that conclusion

ID: 2af2016f Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most accurately describes the main purpose of the text, which is to explain a study’s conclusion and how researchers involved in the study arrived at that conclusion. The text begins by summarizing the main conclusion of the study conducted by Paul Hanel and colleagues: when confronted with ideas they disagree with in discussions about controversial topics, people are more likely to respond politely if they think about their personal values before engaging in such discussions. The text then goes on to describe the design of Hanel and colleagues’ experiment. By comparing interactions between members of an experimental group (who had been prompted to write about their personal values beforehand) to those between members of a control group (who had been prompted to write about a beverage), the team found that people in the experimental group behaved more civilly, or politely, than people in the control group did during discussions about a controversial topic. This finding led to the conclusion described at the beginning of the text.

Choice A is incorrect. Although the text discusses the results of a study, it doesn’t provide any indication that the conclusion the study supported—that when facing disagreement, people behave more politely when they have thought about their values—is a belief that is widely held. Choice B is incorrect because the text doesn’t indicate that the researchers found the results of their study to be surprising, or contrary to what they expected. In fact, there’s no indication provided in the text about how the researchers felt about the study’s results or that the results should be considered surprising. Choice C is incorrect. Although the text discusses the experimental design of a study, it doesn’t suggest any improvements to that design; instead, it focuses on how the design enabled the researchers to draw a particular conclusion.

Question Difficulty: Medium

Question ID 4b4ab04e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-Text Connections	■ ■ ■

ID: 4b4ab04e

PSAT2 RW 1.8

Text 1

Mycoprotein is a fungal biomass that can be eaten as an alternative to meat. Studies of the environmental impact of its manufacture generally agree it is lower than that of beef and closer to that of chicken or pork. But the expense of producing mycoprotein restricts its availability to a few countries with postindustrial economies. Knowing that cost reductions would expand access to mycoprotein, biochemists are exploring solutions, such as a cheaper substrate to feed the mycoprotein as it grows.

Text 2

Cattle farming is a principal cause of global deforestation, and a study by Florian Humpenöder and his colleagues found that replacing 20% of beef consumption worldwide with consumption of mycoprotein would cut deforestation by half if accomplished over the next thirty years. However, this would likely involve only a small change in agricultural water consumption, since water once dedicated to raising cattle would be diverted to raising crops instead.

Based on the texts, how would the author of Text 1 most likely respond to the study findings mentioned in Text 2?

- By emphasizing that since agricultural water consumption would remain static in the event of replacing beef consumption with mycoprotein consumption, an effort must be made to substitute mycoprotein for chicken and pork in

A. diets as well
- By asserting that the development of a more inexpensive substrate for mycoprotein production would contribute to the

B. goal of decreasing worldwide deforestation over time
- By noting that most people would be more likely to use mycoprotein as a substitute for chicken or pork in their diets than

C. as a substitute for beef
- By pointing out that some countries are responsible for greater deforestation than others and thus, to have any

D. significant effect on the environment, will have to replace more than 20% of their beef consumption with mycoprotein

ID: 4b4ab04e Answer

Correct Answer: B

Rationale

Choice B is the best answer because it reflects how the author of Text 1 would most likely respond to the study findings described in Text 2. The author of Text 2 discusses a study by Florian Humpenöder and his colleagues that found that deforestation would be reduced by half over the next thirty years if 20% of the beef consumed worldwide were replaced with mycoprotein. The author of Text 1 points out that mycoprotein is not widely available because of its high production cost, but goes on to note that this problem could be addressed by the creation of a cheaper substrate to feed mycoprotein. This suggests that the author of Text 1 would assert that the development of a less expensive mycoprotein substrate would contribute to the reduction in deforestation described in the study findings discussed in Text 2: if reducing the cost of mycoprotein increases people's access to it, then mycoprotein may be able to replace beef in more people's diets, thereby reducing the deforestation associated with beef production.

Choice A is incorrect because the author of Text 1 indicates that the environmental impact of mycoprotein production is close to that of chicken or pork production, so there is no reason to think that the author would assert that replacing chicken or pork with mycoprotein would be environmentally beneficial: such a replacement would not lessen the total environmental impact of food manufacture. Additionally, the specific issue of agricultural water consumption is never mentioned in Text 1, so there is no evidence indicating what the author of Text 1 would say about that issue. Choice C is incorrect. Although Text 1 does compare the environmental effects of producing mycoprotein to those of producing chicken or pork, nothing in Text 1 suggests that the author believes that people are more likely to replace chicken or pork with mycoprotein than they are to replace beef with mycoprotein. Choice D is incorrect because Text 1 makes no mention of countries' varying contributions to deforestation, so there is no evidence that the author of Text 1 would respond to the finding described in Text 2 by saying that some countries will have to replace more than 20% of their beef consumption with mycoprotein.

Question Difficulty: Hard

Question ID 8fb67890

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: 8fb67890

PSAT2 RW 1.9

Archaeologists have discovered a runestone in Norway that may contain the earliest example of written words in Scandinavia. Carbon dating at the discovery site revealed that the stone was likely carved between 1 and 250 CE. Runologist Kristel Zilmer believes the stone will be helpful in learning more about the use of runic alphabets in early Iron Age Scandinavia.

Which choice best states the main topic of the text?

- A. Battles of the Iron Age
- B. A runestone found in Norway
- C. A new method for dating rock samples
- D. The research interests of Kristel Zilmer

ID: 8fb67890 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately states the main topic of the text. The text begins by stating that archaeologists in Norway have discovered what may be the oldest known runestone (a stone with an inscription in the runic alphabets used in the region in ancient times). According to the text, the object was created between 1 and 250 CE. The text then mentions a researcher who comments on the runestone’s significance to the study of runic alphabets. Thus, the main focus throughout the text is a runestone found in Norway.

Choice A is incorrect because the text mentions the Iron Age only to indicate the era when the runestone was created and when runic alphabets were in use; there’s no discussion of battles during this era. Choice C is incorrect because the text mentions the carbon dating method just once and does so only to indicate the age of the runestone; moreover, the text doesn’t suggest that this method is new. Choice D is incorrect. Although the text indicates that Zilmer is a runologist, or a scholar of runestones, it does so in a single sentence, so Zilmer’s scholarly interests are therefore not a primary focus of the text.

Question Difficulty: Medium

Question ID e47d2524

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: e47d2524

PSAT2 RW 1.10

Microplastics are tiny pieces of plastic waste. Areas of the ocean with higher concentrations of microplastic particles also have smaller and fewer waves. A study by Yukun Sun and colleagues found that the concentration of microplastic particles cannot be the only reason for this reduced wave activity because the concentration of particles that would have the observed effect is much higher than that found in these areas of the ocean. However, they found that surfactants, chemicals often used to manufacture plastics, are released into the water from microplastics and have a much stronger wave-reducing effect.

According to the text, what did Sun and colleagues discover about surfactants?

- A. They have a much stronger effect on wave activity than microplastics alone do.
- B. They are mainly composed of water.
- C. They are helpful for removing microplastics from the ocean.
- D. They can be used to contain microplastics within certain areas of the ocean.

ID: e47d2524 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents a statement about surfactants that is supported by the text. The text indicates that higher concentrations of microplastics have been associated with waves that are relatively small and few. According to the text, however, a study by Sun and colleagues found that concentrations of microplastic particles can't be the only reason for the reduced wave activity because a much higher concentration of particles is needed to produce that observed effect. Instead, the text indicates that Sun and colleagues found that surfactants are released from microplastics into the ocean and have a much stronger wave-reducing effect than microplastics alone do.

Choice B is incorrect because the text doesn't discuss what surfactants are made of. Instead, it indicates that surfactants are frequently used to manufacture microplastics and that Sun and colleagues discovered that surfactants are released from microplastics into the ocean. Choice C is incorrect because the text doesn't discuss how microplastics can be removed from the ocean. Instead, the text indicates that Sun and colleagues discovered that surfactants are released into the ocean from microplastics and have a much stronger wave-reducing effect than microplastics alone do. Choice D is incorrect because the text doesn't indicate that surfactants can be used to contain microplastics within certain areas of the ocean. Instead, the text indicates that Sun and colleagues discovered that surfactants are released from microplastics into the ocean and have a much stronger wave-reducing effect than microplastics alone do.

Question Difficulty: Medium

Question ID 0ec15b5a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 0ec15b5a

PSAT2 RW 1.11

Incorporated and Unincorporated Self-Employment Rates in Four Occupational Fields, 2015

Occupational field	Incorporated self-employment rate	Unincorporated self-employment rate
Construction and resource extraction	4.4%	14.8%
Installation, maintenance, and repair	2.7%	6.2%
Management, business, and financial services	8.9%	9.7%
Sales and related	5.8%	7.8%

Self-employed workers—individuals who are not employees of other individuals or businesses but instead earn their income as business owners, independent contractors, or freelance workers—make up an important part of the US labor force. In order to receive legal and tax benefits enjoyed by corporations, self-employed individuals may choose to incorporate their business. A 2015 survey of incorporated and unincorporated self-employment rates in four occupational fields showed that the highest incorporated self-employment rate occurred among people working in _____

Which choice most effectively uses data from the table to complete the text?

- A. sales and related occupations.
- B. installation, maintenance, and repair occupations.
- C. construction and resource extraction occupations.
- D. management, business, and financial services occupations.

ID: 0ec15b5a Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses data from the table to complete the text. The text discusses self-employed workers in the US, and the table shows the US incorporated and unincorporated self-employment rates in four occupational fields in 2015. According to the table, the incorporated self-employment rate was 8.9% for the management, business, and financial services occupational field, which is higher than the incorporated self-employment rate shown for any other occupational field.

Choice A is incorrect because according to the table, the incorporated self-employment rate was 5.8% for the sales and related occupational field, which is lower than the 8.9% for the management, business, and financial services occupational field. Choice B is incorrect because according to the table, the incorporated self-employment rate was 2.7% for the installation, maintenance, and repair occupational field, which is lower than the 8.9% for the management, business, and financial services occupational field. Choice C is incorrect because according to the table, the incorporated self-employment rate was 4.4% for the construction and resource extraction occupational field, which is lower than the 8.9% for the management, business, and financial services occupational field.

Question Difficulty: Medium

Question ID ed586475

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: ed586475

PSAT2 RW 1.12

There Is Confusion is a 1924 novel by Jessie Redmon Fauset. In the novel, the narrator portrays the character Joanna as someone who admires ambition in other people to the exclusion of all other qualities: _____

Which quotation from *There Is Confusion* most effectively illustrates the claim?

- A. "Joanna was mightily interested in people who had a 'purpose' in life. Otherwise not at all."
- B. "Indeed from the very beginning Joanna showed her preference for her father."
- C. "Joanna was like her father not only so far as ambition was concerned but also in her willingness to work."
- D. "She had a good sense of logic, a strong power of concentration, and a remarkably retentive and visualizing memory."

ID: ed586475 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively presents a quotation from *There Is Confusion* that illustrates the claim that the narrator portrays Joanna as admiring the quality of ambition, or the determination to achieve something, and no other qualities in other people. By describing Joanna as not being interested in people unless they have "a 'purpose' in life"—that is, a goal they are determined to achieve—in which case she is very interested in them, the quotation reveals that Joanna cares only about others' ambition.

Choice B is incorrect because this quotation indicates only that Joanna has a preference for her father, without giving a reason for that preference; therefore, it doesn't illustrate the claim that Joanna cares only about others' ambition. Choice C is incorrect because this quotation doesn't illustrate that Joanna greatly admires the quality of ambition in others; it refers to Joanna's similarity to her father "so far as ambition was concerned" but doesn't reveal how she (or her father) views that quality. Choice D is incorrect because this quotation describes qualities Joanna possesses—praising her logic, concentration, and memory—instead of Joanna's interest in other people's qualities.

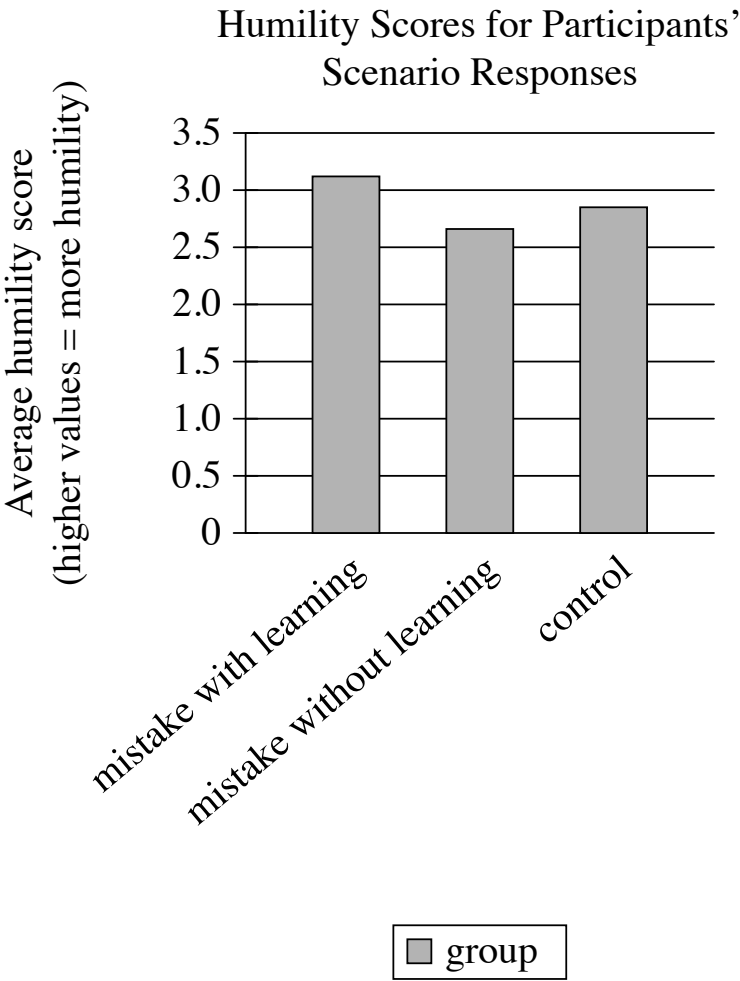
Question Difficulty: Medium

Question ID e99a38ec

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: e99a38ec

PSAT2 RW 1.13



Jia Hu and colleagues hypothesized that workplace leaders who reflect on lessons learned from past mistakes are likely to exhibit more humility than leaders who don't engage in such reflection. To test this, the team placed 301 managers in one of three groups. Participants in two experimental groups were asked to reflect on a mistake, one group focusing on a mistake that resulted in learning and the other group focusing on a mistake that didn't result in learning. Participants in a control group were asked to reflect on their daily routine. All the participants then described how they would respond to a workplace scenario. After evaluating the responses for evidence of humility, the researchers concluded that their hypothesis was correct.

Which choice best describes data in the graph that support the researchers' conclusion?

- A. None of the three groups' average humility scores exceeded 3.5.
- B. The managers in the control group exhibited only slightly less humility on average than the managers in the two experimental groups did.
- C. The managers who reflected on a past mistake that resulted in learning exhibited more humility on average than the managers in the other two groups did.

D. All three groups exhibited less humility on average than the researchers expected.

ID: e99a38ec Answer

Correct Answer: C

Rationale

Choice C is the best answer because it describes data in the graph that support the researchers' conclusion that their hypothesis about workplace leaders and humility was correct. According to the text, Jia Hu and colleagues "hypothesized that workplace leaders who reflect on lessons learned from past mistakes are likely to exhibit more humility than leaders who don't engage in such reflection." The bar graph shows the humility scores for managers who participated in Jia Hu and colleagues' study. It shows average humility scores for managers who reflected on a past mistake that they learned from, managers who reflected on a past mistake that they didn't learn from, and managers who were in the control group that, according to the text, simply reflected on their daily routines. The graph shows that managers who reflected on a past mistake they learned from received an average humility score between 3.0 and 3.5. Both the control group and managers who reflected on a mistake they didn't learn from received a lower average humility score of between 2.5 and 3.0. Thus, the managers who reflected on a past mistake that resulted in learning exhibited more humility on average than the managers in the other two groups did. This supports the researchers' conclusion that their initial hypothesis that workplace leaders who reflect on past mistakes they learned from exercise greater humility was correct.

Choice A is incorrect. Although according to the graph it's true that none of the group's humility scores exceeded 3.5, this idea does not support the researchers' conclusion that their initial hypothesis was correct. The statement that none of the three groups' average humility scores exceeded 3.5 does not distinguish between those leaders who reflected on lessons learned from past mistakes and those who didn't, so it would not support Hu and colleagues' hypothesis that workplace leaders who reflect on past mistakes they learned from exercise greater humility. Choice B is incorrect. While the graph shows that managers in the control group did exhibit less humility than the experimental group of managers who reflected on a mistake they learned from, the graph also shows that the control group exhibited more (not less) humility than the experimental group that reflected on a mistake they didn't learn from. Furthermore, the idea that the managers in the control group exhibited only slightly less humility on average than the managers in the two experimental groups did would not support the researchers' conclusion that their initial hypothesis that workplace leaders who reflect on past mistakes they learned from exercise greater humility was correct. Choice D is incorrect because there is no indication in the graph that all three groups of participants exhibited less humility on average than the researchers expected. The graph and text don't cite any specific expectations that the researchers had with regard to the humility scores for each group.

Question Difficulty: Medium

Question ID eca09a92

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: eca09a92

PSAT2 RW 1.14

If an animal has been extinct for a long time, how can scientists learn what color it was? One group of scientists came up with a possible answer. When the scientists examined the fossilized feather of an extinct bird, they found melanosomes in it. Melanosomes produce pigment, or grains of color, inside cells. Because melanosomes are shaped differently depending on which colors they produce, the scientists hypothesized that they could _____

Which choice most logically completes the text?

- A. show how melanosomes can be found in fossils belonging to animals from other extinct species.
- B. determine the colors of the bird based on the appearance of the melanosomes in the feather.
- C. explain why the melanosomes in the feather were so well preserved.
- D. identify the colors of extinct animals whose fossils lack melanosomes.

ID: eca09a92 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of a hypothesis by one group of scientists about how to determine the colors of a long-extinct animal. The text explains that the scientists found melanosomes in the fossilized feather of an extinct bird and that melanosomes are responsible for producing color inside cells. The text also explains that melanosomes have different shapes depending on the colors they produce. Given this information, it follows that the scientists hypothesized that they could determine the colors of the extinct bird by examining the shapes of the melanosomes in the feather.

Choice A is incorrect because the text never suggests that the scientists were seeking to show how melanosomes can be found in fossils belonging to animals from other extinct species. Rather, the text indicates that the scientists were seeking to identify an extinct bird’s colors, and the text strongly suggests that the scientists hypothesized that they could achieve their goal by examining the shapes of the melanosomes in the bird’s fossilized feather. Choice C is incorrect because the text never suggests that the scientists were seeking to explain why the melanosomes in the feather were so well preserved. Rather, the text indicates that the scientists were seeking to identify an extinct bird’s colors, and the text strongly suggests that the scientists hypothesized that they could achieve their goal by examining the shapes of the melanosomes in the bird’s fossilized feather. Choice D is incorrect because the text suggests only one method of identifying the colors of extinct animals: by examining the shapes of melanosomes found in fossils. The text doesn’t discuss other methods for learning the colors of extinct animals and therefore provides no support for the idea that the scientists could identify the colors of extinct animals whose fossils lack melanosomes.

Question Difficulty: Easy

Question ID 26f5c4ba

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 26f5c4ba

PSAT2 RW 1.15

In 2022, Crystal Reeck and colleagues studied whether the decision-making modes that guide consumers influence their choice between nonenvironmentally friendly standard electricity plans and environmentally friendly green plans that cap electricity usage. Study participants who self-reported using either an Affect Mode or Role Mode—which prioritize choices that have a stronger positive emotional or social impact, respectively—were more likely to select a green plan. Conversely, participants using a Calculation Mode—which aims to minimize both financial cost and personal inconvenience—were more likely to select a standard plan, even when the green option was cheaper. This finding suggests that participants using a Calculation Mode _____

Which choice most logically completes the text?

- _____ were equally unlikely to factor the financial savings of the green plan into their decision-making as were participants
- A. using either the Affect or Role Modes.
- B. may have been less strongly motivated to appear socially responsible with their choice of plan than they realized.
- _____ may have determined that the green plan imposed additional burdens on them that were not sufficiently offset by the
- C. potential financial savings.
- _____ were less likely to believe that the green plan was truly cost-effective than were participants using either the Affect or
- D. Role Modes.

ID: 26f5c4ba Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most logically completes the text's discussion of the influence of decision-making modes on consumers' choices of different electricity plans. The Affect Mode describes choices based on the decision-maker's preferred emotional outcome, the Role Mode describes choices based on the decision-maker's social motivations, and the Calculation Mode describes choices based on minimizing financial costs and maximizing the decision-maker's convenience. Thus, if those using a Calculation Mode disfavor the green plan, it would be due to some financial or convenience burden the green plan imposes that the other doesn't. The text indicates that the green plan had a lower financial cost but was nonetheless rejected by participants using the Calculation Mode. It therefore follows that the green plan likely imposed a convenience burden that outweighed potential financial savings for these participants.

Choice A is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, unlike the Affect or Role Modes, the Calculation Mode would almost certainly involve taking price differences into account. Choice B is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, there is no reason to think that participants using the Calculation Mode at any point took social responsibility into account, let alone think that they discovered a disparity between their expectations and reality. Choice D is incorrect. According to the text, the Affect Mode involves the decision-maker's preferred emotional outcome, the Role Mode involves social motivations, and the Calculation Mode is oriented toward minimizing financial costs and maximizing the decision-maker's convenience. Thus, the text doesn't indicate that cost-effectiveness is a factor for participants using the Affect or Role Modes, and there is no basis to compare their beliefs in the truthfulness of the green-plan description with those of the Calculation Mode users.

Question Difficulty: Hard

Question ID d4f173ec

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: d4f173ec

PSAT2 RW 1.16

To illustrate Albert Einstein’s special theory of relativity, picture two jugglers: one juggling on a steadily moving parade float, the other juggling while standing still on a sidewalk. The laws of physics are identical for both _____ motion relative to each other. But what, Einstein wondered, about the speed of light?

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. jugglers’, regardless of they’re
- B. jugglers, regardless of there
- C. juggler’s, regardless of their
- D. jugglers, regardless of their

ID: d4f173ec Answer

Correct Answer: D

Rationale

Choice D is the best answer. The convention being tested is the use of plural nouns and possessive determiners. The plural noun "jugglers" and the plural possessive determiner "their" correctly indicate that there are multiple jugglers whose motion is being discussed.

Choice A is incorrect because the context requires the plural noun "jugglers" and the plural possessive determiner "their," not the plural possessive noun "jugglers'" and the contraction "they're." Choice B is incorrect because the context requires the plural possessive determiner "their," not the word "there," which means "in that place." Choice C is incorrect because the context requires the plural noun "jugglers," not the singular possessive noun "juggler's."

Question Difficulty: Easy

Question ID 4f2ff5f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: 4f2ff5f2

PSAT2 RW 1.17

Tortoises can be found in many works of literature. For example, in Tom Stoppard’s 1993 play *Arcadia*, there is a tortoise that _____ by two names (Plautus and Lightning) and appears in both of the play’s parallel timelines. As a character, the tortoise symbolizes the connection between the past and present.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. goes
- B. will have gone
- C. went
- D. had gone

ID: 4f2ff5f2 Answer

Correct Answer: A

Rationale

Choice A is the best answer. The convention being tested is the use of verbs to express tense in a sentence. In this choice, the present tense verb "goes" is consistent with the other present tense verbs ("appears" and "symbolizes") used to describe the tortoise in Stoppard’s play. Furthermore, it’s conventional to use the present tense when discussing a literary work.

Choice B is incorrect because the future perfect tense verb "will have gone" isn’t consistent with the other present tense verbs used to describe the tortoise in Stoppard’s play. Choice C is incorrect because the past tense verb "went" isn’t consistent with the other present tense verbs used to describe the tortoise in Stoppard’s play. Choice D is incorrect because the past perfect tense verb "had gone" isn’t consistent with the other present tense verbs used to describe the tortoise in Stoppard’s play.

Question Difficulty: Medium

Question ID 20ea68b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: 20ea68b7

PSAT2 RW 1.18

It can take time for proposed amendments to the US Constitution to become law. For example, the Twenty-Second Amendment, which limits the number of _____ can serve, was first proposed in 1947 but wasn't approved by the required three-fourths majority of state legislatures until 1951.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. terms presidents
- B. term's presidents
- C. term's president's
- D. terms president's

ID: 20ea68b7 Answer

Correct Answer: A

Rationale

Choice A is the best answer. The convention being tested is the use of plural nouns. The plural nouns "terms" and "presidents" correctly indicate that the amendment involves multiple terms and applies to presidents in general.

Choice B is incorrect because the context requires the plural noun "terms," not the singular possessive noun "term's." Choice C is incorrect because the context requires the plural nouns "terms" and "presidents," not the singular possessive nouns "term's" and "president's." Choice D is incorrect because the context requires the plural noun "presidents," not the singular possessive noun "president's."

Question Difficulty: Medium

Question ID ace95f84

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	

ID: ace95f84

PSAT2 RW 1.19

By analyzing the level of radioactive decay within a fossil specimen, scientists can establish the age of that fossil with a high degree of precision. When radioactive elements aren't present, scientists turn to _____ analysis of Earth's sediment layers (strata)—to estimate how old a fossil is based on the age of the strata in which the fossil is found.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. stratigraphy—the
- B. stratigraphy (the
- C. stratigraphy: the
- D. stratigraphy, the

ID: ace95f84 Answer

Correct Answer: A

Rationale

Choice A is the best answer. The convention being tested is the use of punctuation within a sentence. The dash after "stratigraphy" pairs with the dash after "(strata)" to separate the supplementary element "the analysis of Earth's sediment layers (strata)" from the rest of the sentence. This supplementary element functions to define the term "stratigraphy," and the pair of dashes indicates that this element could be removed without affecting the grammatical coherence of the sentence.

Choice B is incorrect because a parenthesis can't be paired with a dash in this way to separate the supplementary element from the rest of the sentence. Choice C is incorrect because a colon can't be paired with a dash in this way to separate the supplementary element from the rest of the sentence. Choice D is incorrect because a comma can't be paired with a dash in this way to separate the supplementary element from the rest of the sentence.

Question Difficulty: Medium

Question ID a5079e0d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	

ID: a5079e0d

PSAT2 RW 1.20

Horsepower is a unit of measurement used to determine how much power a vehicle produces. The measurement is based on how much and how quickly weight can be _____. One unit of mechanical horsepower is equivalent to the amount of power it takes to lift 550 pounds one foot off the ground in one second.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. moved, for example,
- B. moved,
- C. moved; for example,
- D. moved

ID: a5079e0d Answer

Correct Answer: C

Rationale

Choice C is the best answer. The convention being tested is the use of punctuation within a sentence. This choice uses a semicolon in a conventional way to join the first main clause ("The measurement...moved") and the second main clause ("for example...second").

Choice A is incorrect because it results in a comma splice. Without a coordinating conjunction, a comma can't be used in this way to join the two main clauses ("The measurement...moved" and "for example...second"). Choice B is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join the two main clauses ("The measurement...moved" and "one unit...second"). Choice D is incorrect because it results in a run-on sentence. The two main clauses ("The measurement...moved" and "one unit...second") are fused without punctuation and/or a conjunction.

Question Difficulty: Medium

Question ID 9eb43963

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: 9eb43963

PSAT2 RW 1.21

Water in the North Atlantic Ocean is pushed eastward by powerful winds, but the rotation of Earth and interference from nearby landmasses together cause _____ to swirl into a massive, churning whirlpool—also called the North Atlantic Gyre—that spins clockwise.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. these
- B. those
- C. them
- D. it

ID: 9eb43963 Answer

Correct Answer: D

Rationale

Choice D is the best answer. The convention being tested is pronoun-antecedent agreement. The singular pronoun "it" agrees in number with the singular antecedent "water" and clearly indicates that the water swirls into a whirlpool.

Choice A is incorrect. The plural pronoun "these" neither agrees in number with the singular antecedent "water" nor clearly indicates that the water—not another plural noun in the sentence, such as "landmasses" or "winds"—swirls into a whirlpool. Choice B is incorrect. The plural pronoun "those" neither agrees in number with the singular antecedent "water" nor clearly indicates that the water—not another plural noun in the sentence, such as "landmasses" or "winds"—swirls into a whirlpool. Choice C is incorrect. The plural pronoun "them" neither agrees in number with the singular antecedent "water" nor clearly indicates that the water—not another plural noun in the sentence, such as "landmasses" or "winds"—swirls into a whirlpool.

Question Difficulty: Medium

Question ID 37ec26e7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: 37ec26e7

PSAT2 RW 1.22

In order to create the Global Positioning System (GPS), scientists had to develop an accurate mathematical model of Earth’s shape that accounted for various forces, such as tides. _____ it was mathematician Gladys West who wrote the computer program that could perform these necessary calculations.

Which choice completes the text with the most logical transition?

- A. Ultimately,
- B. In other words,
- C. Secondly,
- D. In addition,

ID: 37ec26e7 Answer

Correct Answer: A

Rationale

Choice A is the best answer. "Ultimately" logically signals that West’s completion of the computer program was the conclusion of the process described in the previous sentence, in which scientists working on GPS sought to develop a mathematical model of Earth.

Choice B is incorrect because "in other words" illogically signals that the information about West’s program is a restatement of the information about the scientists’ efforts to develop a mathematical model of Earth. Instead, West’s program was the conclusion of those efforts. Choice C is incorrect because "secondly" illogically signals that West’s completion of the computer program was merely the next step in the scientists’ efforts to develop a mathematical model of Earth. Instead, West’s program was the conclusion of those efforts. Choice D is incorrect because "in addition" illogically signals that West’s completion of the computer program was merely additional information related to the scientists’ work on GPS. Instead, West’s program was the conclusion of the scientists’ efforts to develop a mathematical model of Earth.

Question Difficulty: Easy

Question ID 9fe7315b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: 9fe7315b

PSAT2 RW 1.23

Few New Yorkers have heard Thelma Pollard’s name. _____ many have seen her work in the famous Broadway musical *The Phantom of the Opera*. As the musical’s longtime makeup supervisor, Pollard put the iconic makeup on the face of the Phantom himself.

Which choice completes the text with the most logical transition?

- A. Similarly,
- B. Therefore,
- C. For example,
- D. However,

ID: 9fe7315b Answer

Correct Answer: D

Rationale

Choice D is the best answer. "However" logically signals that the claim in this sentence—that many New Yorkers have seen Pollard’s work in a famous Broadway musical—is true despite the previous information that few New Yorkers have heard Pollard’s name.

Choice A is incorrect because "similarly" illogically signals that the claim in this sentence is similar to the previous information that few New Yorkers have heard Pollard’s name. Instead, the claim that many New Yorkers have seen Pollard’s work is true despite the previous information. Choice B is incorrect because "therefore" illogically signals that the claim in this sentence is a result of the previous information that few New Yorkers have heard Pollard’s name. Instead, the claim that many New Yorkers have seen Pollard’s work is true despite the previous information. Choice C is incorrect because "for example" illogically signals that the claim in this sentence exemplifies the previous information that few New Yorkers have heard Pollard’s name. Instead, the claim that many New Yorkers have seen Pollard’s work is true despite the previous information.

Question Difficulty: Easy

Question ID 0839a4b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: 0839a4b9

PSAT2 RW 1.24

The Coastal Futures Conservatory in Virginia is known for creating aural representations of ecological data. One such effort combines underwater audio recorded in seagrass beds with data that track rising carbon levels in the seagrass. As carbon levels increase, the audio is correspondingly distorted; _____ listeners can “hear” the changes in the carbon levels.

Which choice completes the text with the most logical transition?

- A. furthermore,
- B. by comparison,
- C. for instance,
- D. thus,

ID: 0839a4b9 Answer

Correct Answer: D

Rationale

Choice D is the best answer. "Thus" logically signals that the information in this clause—that listeners can "hear" carbon levels increasing—is a result of the previous information about the audio distorting as carbon levels increase.

Choice A is incorrect because "furthermore" illogically signals that the information in this clause merely adds to the previous information about the audio distorting as carbon levels rise. Instead, the listeners’ ability to "hear" carbon levels increasing is a result of that distortion. Choice B is incorrect because "by comparison" illogically signals that the information in this clause is being compared to the information about the audio distorting as carbon levels rise. Instead, the listeners’ ability to "hear" carbon levels increasing is a result of that distortion. Choice C is incorrect because "for instance" illogically signals that the information in this clause is an example of how the audio distorts as carbon levels rise. Instead, the audio was distorted for the express purpose of representing ecological data; the listeners’ ability to "hear" carbon levels increasing is a direct, intended result of the distortion, not merely an example of it.

Question Difficulty: Medium

Question ID 878f835a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	

ID: 878f835a

PSAT2 RW 1.25

While researching a topic, a student has taken the following notes:

- Some atoms contain an excess of neutrons. Often, these neutrons form a “skin” on the atom’s surface.
- An atom of lead-208 has a neutron skin.
- The thickness of its neutron skin is approximately 0.28 trillionths of a millimeter.

The student wants to emphasize the thickness of lead-208’s neutron skin. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The neutron skin surrounding an atom of lead-208 measures about 0.28 trillionths of a millimeter.
- B. Atoms with excess neutrons will often acquire a neutron skin.
- C. An atom of lead-208, like some other atoms, is surrounded by a neutron skin.
- D. Neutrons surround the surface of an atom of lead-208.

ID: 878f835a Answer

Correct Answer: A

Rationale

Choice A is the best answer. The sentence emphasizes the thickness of lead-208’s neutron skin, noting that it is about 0.28 trillionths of a millimeter thick.

Choice B is incorrect. The sentence makes a generalization about atoms, stating that atoms with excess neutrons will often acquire a neutron skin; it doesn’t emphasize the thickness of lead-208’s neutron skin. Choice C is incorrect. The sentence states that lead-208 has a neutron skin; it doesn’t emphasize the thickness of that skin. Choice D is incorrect. The sentence indicates that lead-208 is surrounded by neutrons; it doesn’t emphasize the thickness of lead-208’s neutron skin.

Question Difficulty: Easy

Question ID 5a2a4b36

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	

ID: 5a2a4b36

PSAT2 RW 1.26

While researching a topic, a student has taken the following notes:

Pauline Hopkins (1859-1930) was an African American writer. In her career, she created many genre-defining stories. Her serialized novel *Hagar’s Daughter* was published from 1901 to 1902 in *The Colored American Magazine*. It is considered the first African American mystery novel.

The student wants to introduce Pauline Hopkins to an audience unfamiliar with her career. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Pauline Hopkins is the author of *Hagar’s Daughter*, a serialized novel published from 1901 to 1902.
- African American writer Pauline Hopkins created many genre-defining stories during her career, including the first African
- B. American mystery novel, *Hagar’s Daughter*.
- Published in *The Colored American Magazine* from 1901 to 1902, *Hagar’s Daughter* is a serialized novel by Pauline
- C. Hopkins.
- D. *Hagar’s Daughter* by Pauline Hopkins is considered the first African American mystery novel.

ID: 5a2a4b36 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The sentence introduces Pauline Hopkins to an audience unfamiliar with her career, identifying her as a genre-defining African American writer and noting a career accomplishment.

Choice A is incorrect. The sentence emphasizes that Pauline Hopkins wrote a serialized novel but doesn’t introduce Hopkins to an audience unfamiliar with her career. Choice C is incorrect. The sentence introduces Hopkins’s novel; it doesn’t introduce Hopkins. Choice D is incorrect. The sentence explains that a novel written by Hopkins is considered the first African American mystery novel; it doesn’t introduce Hopkins to an audience unfamiliar with her career.

Question Difficulty: Easy

Question ID c92ea686

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	

ID: c92ea686

PSAT2 RW 1.27

While researching a topic, a student has taken the following notes:

- The Ramayana* is a Sanskrit epic poem from ancient India.
- In *The Ramayana*, the character Kaikeyi is often portrayed as a villain. *Kaikeyi* is a 2022 novel by Vaishnavi Patel.
- The novel is a retelling of the epic poem from Kaikeyi’s point of view. It often portrays Kaikeyi as heroic.

The student wants to emphasize whose point of view the novel is told from. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. From the point of view of *The Ramayana*, the character Kaikeyi is often a villain.
- B. Vaishnavi Patel often portrays the character as heroic.
- C. *Kaikeyi* is a retelling of *The Ramayana* from the character Kaikeyi’s point of view.
- D. *The Ramayana* is an epic poem that features the character Kaikeyi.

ID: c92ea686 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The sentence emphasizes whose point of view the novel *Kaikeyi* is told from: the character Kaikeyi’s.

Choice A is incorrect. While the sentence does discuss point of view, it focuses on that of the epic poem rather than the novel. Choice B is incorrect. While the sentence seems to be referring to Patel’s novel, it doesn’t establish whose point of view the novel is told from. Choice D is incorrect. The sentence discusses the character Kaikeyi in the context of the epic poem; it doesn’t discuss the novel’s point of view.

Question Difficulty: Medium

Question ID 5d48f889

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 5d48f889

PSAT2 RW 2.1

The following text is adapted from Mercedes de Acosta’s 1921 poem “Spring and You.”

Now, Love and April, and the gold of your hair, Are all mingled together Like the blending of an exotic dream plant
With the fragrant perfume of a strange, frail flower.

As used in the text, what does the word “mingled” most nearly mean?

- A. Identified
- B. Combined
- C. Remembered
- D. Celebrated

ID: 5d48f889 Answer

Correct Answer: B

Rationale

Choice B is the best answer because as used in the text, "mingled" most nearly means combined. In the text, the speaker refers to several concepts mingling together and then compares this to "the blending of" two other things (an exotic plant and the fragrance of a strange flower). This comparison conveys that the speaker views the concepts of love, April, and the gold of someone’s hair as being blended together, or combined.

Choice A is incorrect because "mingled" means combined, not identified. The speaker names several concepts (love, April, and the gold color of someone’s hair) and then moves to a comparison of their mingling together to "the blending of" a plant and the scent of a flower, conveying the idea that the speaker views the concepts as blended together, or combined. Choice C is incorrect. The text as a whole may refer to a memory, but the comparison of the mingling of several concepts (love, April, and the gold color of someone’s hair) to "the blending of" a plant and the scent of a flower conveys that the speaker views those concepts as actually blended together, or combined, and not simply remembered at the same time. Choice D is incorrect. The speaker refers to several concepts (love, April, and the gold color of someone’s hair) and then compares the mingling together of those concepts to "the blending of" a plant and the scent of a flower; rather than suggesting that the concepts are celebrated, this comparison conveys that the speaker views the concepts as blended together, or combined.

Question Difficulty: Easy

Question ID 6286dd96

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 6286dd96

PSAT2 RW 2.2

The following text is adapted from Frances Hodgson Burnett’s 1905 novel *A Little Princess*. Sara is a young student at a school in London.

Sara not only could tell stories, but she adored telling them. When she sat or stood in the midst of a circle and began to invent wonderful things, her green eyes grew big and shining, her cheeks flushed, and, without knowing that she was doing it, she began to act and made what she told lovely or alarming by the raising or dropping of her voice.

As used in the text, what does the word “invent” most nearly mean?

- A. Mislead
- B. Disguise
- C. Create
- D. Discover

ID: 6286dd96 Answer

Correct Answer: C

Rationale

Choice C is the best answer because as used in the text, "invent" most nearly means create or make up. The text begins by stating that Sara likes to tell stories and thoroughly enjoys telling them. The rest of the text describes how she looks and acts when she invents, or creates, "wonderful things" in her stories ("what she told").

Choice A is incorrect. Although the text suggests that Sara makes up "wonderful things," nothing suggests that she does so to mislead, or intentionally deceive, others; there is no reason to assume that Sara’s listeners don’t know that she’s creating the stories she shares. Choice B is incorrect because nothing in the text suggests that Sara is disguising, or trying to cover up or conceal, "wonderful things" when she tells stories; rather, she is creating them and presenting them in ways that make them "lovely or alarming" for her listeners. Choice D is incorrect because nothing in the text suggests that Sara is discovering, or unexpectedly finding, "wonderful things" in the stories she loves to tell; rather, that she is making things up as she tells the stories.

Question Difficulty: Easy

Question ID 53542eb5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 53542eb5

PSAT2 RW 2.3

Shoppers can help keep money cycling within a community by making purchases at small local businesses instead of large retailers. Some cities are _____ programs to encourage this behavior, establishing reward points and other incentives for shopping at small businesses.

Which choice completes the text with the most logical and precise word or phrase?

- A. instituting
- B. occupying
- C. underestimating
- D. encountering

ID: 53542eb5 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of programs that encourage money to cycle within a community. In this context, "instituting" means initiating. The text indicates that making purchases at small local businesses can help keep money cycling within a community. The text goes on to state that some cities are establishing incentives to promote small business shopping. This context supports the idea that some cities are instituting programs to encourage this consumer behavior.

Choice B is incorrect because saying that some cities are "occupying," or forcibly holding possession or control of, programs that encourage shoppers to make purchases at small local businesses wouldn’t make sense in context. The text doesn’t discuss who controls these programs but instead indicates that some cities are establishing incentives that encourage shoppers to make purchases at small local businesses. Furthermore, it wouldn’t make sense to say that a city is occupying a program. Choice C is incorrect because nothing in the text suggests that cities are "underestimating," or failing to understand the value of, programs that encourage consumers to shop at small local businesses. The text indicates that some cities are establishing incentives to encourage small business shopping, which suggests that these cities understand the value of such programs. Choice D is incorrect because saying that some cities are "encountering," or coming into contact with, programs that encourage consumers to shop at small local businesses wouldn’t make sense in context. The text indicates that some cities are already establishing incentives to encourage shoppers to make purchases at small businesses. This context suggests that the cities are instituting the programs, not that they are encountering them.

Question Difficulty: Easy

Question ID 01de46bb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 01de46bb

PSAT2 RW 2.4

The Menominee and Anishinaabe peoples have been growing wild rice—known as *manoomin* in the Ojibwe language—in the Great Lakes region of North America for centuries, but climatic changes are causing lakes to get deeper, thereby threatening wild rice. These plants are extremely _____ to water depth during the “floating leaf” stage of development, and if the water is too deep, the buoyancy of the young wild-rice plants can literally uproot them from the lake bottom, destroying them.

Which choice completes the text with the most logical and precise word or phrase?

- A. immune
- B. sensitive
- C. limited
- D. receptive

ID: 01de46bb Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of wild rice. As used in this context, "sensitive" means vulnerable or easily affected. The text indicates that climate change is causing the lakes to get deeper, which consequently threatens the wild rice that grows in the area. The text goes on to state that deep water can cause young wild-rice plants to uproot from the bottom of the lake. This context supports the idea that the plants are sensitive to water depth during early stages of development.

Choice A is incorrect because "immune" means protected or shielded, which wouldn’t make sense in this context. The text explains that deep water can cause young wild-rice plants to become uprooted and destroyed, indicating that these plants are harmed by deep water, not protected from it. Choice C is incorrect because "limited" means confined or restrained, which wouldn’t make sense in this context. The text explains that water depth can negatively affect young wild-rice plants but doesn’t suggest that water depth limits or restrains the plants. Choice D is incorrect because "receptive" means open or tolerant, which wouldn’t make sense in this context. The text doesn’t indicate that the wild-rice plants are open to or tolerant of an increase in water depth but rather the opposite: an increase in water depth can harm young wild-rice plants.

Question Difficulty: Easy

Question ID eaea6f8f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	

ID: eaea6f8f

PSAT2 RW 2.5

Ordinary soap bubbles usually exist for a minute or less before popping due to either a rupture forced by gravity-induced drainage or the evaporation of the liquid from which the bubble is composed. But physicist Aymeric Roux and colleagues discovered ways to mitigate these factors, resulting in bubbles that can last for a year or more. For example, glycerol tends to adhere to water molecules, so a bubble with a shell that contains both water and glycerol is able to draw additional water molecules from the surrounding air and thereby compensate for evaporation.

Which choice best states the purpose of the underlined portion in the text as a whole?

- A. It describes the effects of a process devised by researchers that increases the longevity of an object discussed in the text.
- B. It details the circumstances that prompted the research discussed in the text.
- C. It presents a reason why the phenomenon discussed in the text that the researchers wanted to avoid will inevitably occur.
- D. It mentions a method discussed in the text that researchers intend to test in future experiments.

ID: eaea6f8f Answer

Correct Answer: A

Rationale

Choice A is the best answer because it describes how the underlined portion of the sentence functions in the text as a whole. The text begins by discussing ordinary soap bubbles, objects that usually exist for less than a minute before popping due to either evaporation or gravity-induced drainage. The text then goes on to mention work by researchers Aymeric Roux and colleagues, who have discovered how to increase the longevity of bubbles by altering their chemical composition. Finally, the underlined portion of the text describes how this process works: since the bubble contains both water and glycerol, it can draw water molecules from the surrounding air to replace those lost to evaporation. Therefore, the underlined portion of the sentence describes the effects of a process devised by researchers that increases the longevity of an object described in the text.

Choice B is incorrect because the circumstance that prompted the research is discussed in the first sentence of the text, not the underlined portion. Choice C is incorrect because the underlined portion of the text does not address why soap bubbles must eventually rupture. Rather, it explains how the longevity of soap bubbles can be extended. Choice D is incorrect because the underlined portion of the text does not mention any future experiments.

Question Difficulty: Easy

Question ID 422c5068

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	

ID: 422c5068

PSAT2 RW 2.6

Generally it takes Tule geese about four days to migrate south for the winter. From their summer breeding grounds in Cook Inlet, Alaska, the birds begin by flying over the Gulf of Alaska, keeping about 100 miles from the Canadian shore. They pause to rest on the Pacific Ocean, then fly toward Summer Lake, Oregon, before finally arriving at their winter destination of Sacramento Valley, California. In 2020, however, it took the geese over twice as long to make their way from Cook Inlet to Sacramento Valley. According to researchers, the reason was airborne pollutants.

Which choice best states the function of the underlined portion in the text as a whole?

- A. It illustrates a change in Tule geese’s usual flight behavior.
- B. It suggests an explanation for why Tule geese breed in Alaska.
- C. It describes part of the Tule geese’s typical winter migration journey.
- D. It compares Tule geese to other birds that migrate south for the winter.

ID: 422c5068 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it best states how the underlined portion functions in the text as a whole. The first sentence states that Tule geese typically take four days to migrate south. Then in the next sentence, which contains the underlined portion, the text describes the first part of that journey, which begins with the geese flying over the Gulf of Alaska while keeping about 100 miles from the Canadian shore. The rest of the text details the remainder of the geese’s typical journey and then mentions circumstances in 2020 that resulted in the geese taking an unusually long time to complete it. Thus, the underlined portion describes part of the Tule geese’s typical winter migration journey.

Choice A is incorrect. Although the text concludes by stating that in 2020, Tule geese took twice as long to complete their typical migration, which suggests a change in their usual flight behavior, the underlined portion doesn’t discuss this. Instead, the underlined portion describes the first part of the Tule geese’s typical winter migration journey over the Gulf of Alaska. Choice B is incorrect. Although the sentence containing the underlined portion mentions that Tule geese breed in Alaska, the underlined portion doesn’t explain why the geese breed in that location. Rather, the text mentions Alaska to explain that it’s the starting point of the Tule geese’s typical winter migration. Choice D is incorrect because the underlined portion doesn’t discuss any other birds that migrate south for the winter. In fact, the text is only concerned with the migration of Tule geese.

Question Difficulty: Medium

Question ID cf956802

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: cf956802

PSAT2 RW 2.7

When fashion designer Lloyd Henri Kiva New opened his store in Scottsdale, Arizona, in 1945, he quickly became known for creating delicately crafted leather goods, like belts and hats. He was perhaps most renowned for his colorful handbags, which he made by hand using a long and painstaking process. As he gained more customers, New began using sewing machines and other tools to help him produce bags more efficiently, though he continued to handcraft the crucial details that made each bag unique.

Based on the text, what would have been the most likely consequence if New had not begun using sewing machines?

- A. He would have been unable to ensure that each bag included unique, handcrafted details.
- B. He would have struggled to meet the increasing demand for his bags.
- C. He would have had to individually design each bag he produced.
- D. He would not have been able to generate as much interest in his bags.

ID: cf956802 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the most likely consequence if New had not begun using sewing machines. The text states that New gained new customers and that sewing machines allowed him to make bags more efficiently, or in less time than he could when sewing by hand. It’s reasonable to conclude that if New hadn’t reduced the time it took to make each bag by starting to use sewing machines, it would have been hard for him to keep up with the increased demand.

Choice A is incorrect because the text indicates that New added unique, handcrafted details to his bags before he started using sewing machines and continued to do so after he started using them. Choice C is incorrect because the text doesn’t suggest that individually designing each bag would have been a consequence of not using sewing machines, since New was already designing unique details for each bag before he started using sewing machines and continued to do so after he started using them. Choice D is incorrect because the text indicates that people were already interested in New’s bags before he started using sewing machines. Rather than allowing New to generate more interest in his bags, sewing machines helped New keep up with the interest that had already grown.

Question Difficulty: Easy

Question ID 82d2436a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: 82d2436a

PSAT2 RW 2.8

The following text is from Laila Lalami’s 2019 novel *The Other Americans*. The narrator is a member of her middle school’s jazz band.

One day the jazz band was invited to perform at the Summer Festival in Palm Springs. Walking across the stage to the piano, I did what my teacher had advised. Pretend you’re only playing for one person. That way you won’t be so nervous. I glanced at my father, who sat in the front row, leaning his head just so, waiting. Then I closed my eyes, and began to play.

©2019 by Laila Lalami

According to the text, what does the narrator do as she walks across the stage?

- A. She thinks about the differences between playing the piano alone and playing in a band.
- B. She feels excitement about playing in a music festival for the first time.
- C. She remembers how hard she has practiced for the performance.
- D. She follows her teacher’s advice about managing her nervousness.

ID: 82d2436a Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents a statement about what the narrator is doing that is directly supported by the text. According to the text, as the narrator walks across the stage to her piano, she recalls her teacher’s advice to pretend that she is playing for only one person. The text indicates that this advice was intended to help manage any nerves related to performing. Moreover, the text implies that the narrator follows this advice by deciding to focus on her father sitting in the audience. Thus, the text supports the idea that as she walks across the stage, the narrator follows her teacher’s advice about managing her nervousness.

Choice A is incorrect because the text doesn’t indicate that the narrator is thinking about how playing the piano alone is different from playing the piano in a band. Rather, the narrator thinks about her teacher’s advice to pretend that she is playing the piano for only one person to help ease her nervousness about performing. Choice B is incorrect because the text provides no indication that this is the first music festival that the narrator has performed in. Furthermore, the text indicates that the narrator is nervous, not excited, to play the piano at the festival. Choice C is incorrect because the text doesn’t suggest that the narrator is remembering how hard she practiced for the performance. Rather, she is thinking about what her teacher told her to do to help ease her nerves about playing the piano in front of an audience.

Question Difficulty: Easy

Question ID 2287b040

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: 2287b040

PSAT2 RW 2.9

In the 1960s, Chavela Vargas became an unlikely star in ranchera, a style of traditional Mexican music. Most ranchera singers had clear, polished voices and performed with a full band. But Vargas accompanied her raspy voice with just her guitar. Dressed in men’s trousers and a poncho, she would perform classic songs that had been written from a male point of view and were usually sung by men. She also altered those songs by performing them much more slowly than other ranchera singers did. The slower tempo allowed her to express the emotional quality of the lyrics more fully.

According to the text, what is one way that Vargas differed from other ranchera singers?

- A. She possessed a voice that was clear and polished.
- B. She avoided singing songs written from a male point of view.
- C. She disliked performing classic songs.
- D. She altered classic songs by slowing them down.

ID: 2287b040 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it accurately describes an aspect of Chavela Vargas that the text indicates is unusual among ranchera artists. According to the text, Vargas differed from other ranchera singers by slowing the tempo of classic songs, which allowed her to express the emotional quality of the lyrics more fully.

Choice A is incorrect because the text states that Vargas possessed an unusually raspy voice for a ranchera singer, not a clear and polished voice. Choice B is incorrect because the text indicates that Vargas was known to perform songs written from a male point of view, not that she avoided such songs. Choice C is incorrect because nothing in the text suggests that Vargas disliked performing classic songs. Rather, it suggests that she found innovative ways of performing them to emphasize their emotional content, which seems to suggest an affinity for such songs, not dislike.

Question Difficulty: Easy

Question ID 72c7dafd

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 72c7dafd

PSAT2 RW 2.10

Happy House is a 1920 novel by Jane Abbott. The narrator presents a young woman as being unimpressed with the house she is visiting: _____

Which quotation from *Happy House* most effectively illustrates the claim?

- A. "Her first feeling was of disappointment; in the square lines of the house there was little claim to beauty."
- B. "Someone had opened one of the blinds so here there was more light."
- C. "The door, built squarely in the middle of the house, opened almost directly upon a stone-flagged path that led in a straight line to the road."
- D. "She tip-toed through the hall and opened the door on the right."

ID: 72c7dafd Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses a quotation from *Happy House* to illustrate the claim that the young woman is unimpressed with the house she is visiting. In the quotation, it says the young woman's "first feeling was of disappointment" and that "there was little claim to beauty" when referring to the house. This disappointment and the overall lack of beauty in the house suggest that the young woman isn't impressed by the house she is visiting.

Choice B is incorrect because the quotation doesn't describe what the young woman thinks about the house. Instead, it indicates that there is more light because window blinds have been opened. Choice C is incorrect because the quotation describes a door and a path outside of the house but doesn't include anything to indicate the young woman's feelings about the house. Choice D is incorrect because the quotation states what the young woman did in the house ("tip-toed through the hall" and "opened the door"), but it doesn't show what she thinks about it.

Question Difficulty: Easy

Question ID 81498c6a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 81498c6a

PSAT2 RW 2.11

Average Prices Received by US Growers for Citrus Fruits, 2020–2021 (dollars per box)

Fruit	June 2020	June 2021	July 2020	July 2021
Grapefruits	\$13.80	\$23.72	\$16.13	\$22.98
Oranges	\$16.15	\$11.09	\$15.53	\$13.79
Lemons	\$19.50	\$28.94	\$21.01	\$31.78

An employee of a citrus grower in the United States is analyzing the prices of several varieties of fruit over the course of a growing season. The employee wishes to know how much a grower could have expected to receive on average for a box of lemons in July 2021. Consulting the table, the employee finds that this amount is _____

Which choice most effectively uses data from the table to complete the statement?

- A. \$31.78.
- B. \$23.72.
- C. \$22.98.
- D. \$28.94.

ID: 81498c6a Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most effectively uses data from the table to complete the text’s discussion of how much a grower could have expected to receive on average for a box of lemons in July 2021. The table lists the average prices received by US citrus growers in dollars per box for grapefruits, oranges, and lemons in June 2020, June 2021, July 2020, and July 2021. According to the table, the average price of a box of lemons in July 2021 was \$31.78.

Choice B is incorrect because the text indicates that the employee of a citrus grower wishes to know the average price for a box of lemons in July 2021. According to the table, \$23.72 was the average price for a box of grapefruits in June 2021, not for a box of lemons in July 2021. Choice C is incorrect because the text indicates that the employee of a citrus grower wishes to know the average price for a box of lemons in July 2021. According to the table, \$22.98 was the average price for a box of grapefruits in July 2021, not for a box of lemons in July 2021. Choice D is incorrect because the text indicates that the employee of a citrus grower wishes to know the average price for a box of lemons in July 2021. According to the table, \$28.94 was the average price for a box of lemons in June 2021, not in July 2021.

Question Difficulty: Easy

Question ID 3091f805

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	

ID: 3091f805

PSAT2 RW 2.12

Ochre sea stars live in tidal pools along the shoreline of the Pacific Ocean. At night, they move to higher shore levels in search of prey. But scientists Corey Garza and Carlos Robles noticed that ochre sea stars stayed at lower levels at night after heavy rains. Garza and Robles hypothesized that a layer of fresh water formed by rainfall was a barrier to the sea stars. To test their hypothesis, the scientists did an experiment. They placed some sea stars in a climbable tank of seawater and other sea stars in a similar tank of seawater with a layer of fresh water on top. Then, the scientists watched the sea stars’ behavior at night.

Which finding from the experiment, if true, would most directly support Garza and Robles’s hypothesis?

- None of the sea stars climbed to the tops of the tanks, but sea stars in the tank with only seawater moved around the

A. bottom of the tank more than sea stars in the other tank did.
- Sea stars in the tank with only seawater climbed to the top of the tank, but sea stars in the other tank stopped climbing

B. just below the layer of fresh water.
- Both groups of sea stars climbed to the tops of the tanks, but sea stars in the tank with only seawater climbed more

C. slowly than sea stars in the other tank did.
- Sea stars in the tank with only seawater mostly stayed near the bottom of the tank, but sea stars in the other tank

D. climbed into the layer of fresh water.

ID: 3091f805 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would support Garza and Robles’s hypothesis that a layer of fresh water forms a barrier to ochre sea stars. The text explains that the sea stars tend to move to higher shore levels at night in search of prey, but after a heavy rain, the sea stars stay at lower shore levels. Garza and Robles hypothesize that rainfall results in a layer of fresh water that the sea stars don’t cross. To determine whether fresh water forms a barrier to sea stars, Garza and Robles observed how sea stars behaved in a tank of only seawater and in a tank of seawater with a layer of fresh water on top. If the sea stars climbed to the top of the tank with only seawater but stopped climbing just below the layer of fresh water in the other tank, that would suggest that fresh water does indeed serve as a barrier to the sea stars, thereby supporting Garza and Robles’s hypothesis.

Choice A is incorrect because finding that sea stars in the tank with only seawater moved around the bottom of the tank more than sea stars in the other tank did but that none of the stars in either tank climbed to the top would be irrelevant to Garza and Robles’s hypothesis. Such a finding would reveal nothing about whether fresh water serves as a barrier to sea stars. Choice C is incorrect because finding that sea stars climbed to the top of both tanks would weaken, not support, Garza and Robles’s hypothesis, since it would indicate that the layer of fresh water wasn’t a barrier to the sea stars. Choice D is incorrect because finding that sea stars in the tank with only seawater mostly stayed near the bottom of the tank but sea stars in the other tank climbed into the layer of fresh water wouldn’t support Garza and Robles’s hypothesis. Instead, such a finding would suggest that the layer of fresh water wasn’t a barrier to the sea stars, thereby weakening Garza and Robles’s hypothesis.

Question Difficulty: Easy

Question ID 93cad661

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: 93cad661

PSAT2 RW 2.13

A main goal of the Association for the Advancement of Creative Musicians (AACM), an arts organization founded in 1965, is to advance new works by Black musicians. The AACM achieves this goal in part by focusing on young artists. By having established musicians and composers serve as mentors, the AACM gives young artists the benefits of expert technical training and creative guidance. Numerous organizations offer similar kinds of support to new generations of painters, writers, and other artists, suggesting that _____

Which choice most logically completes the text?

- A. artists of all ages benefit more from technical training than from creative guidance.
- many arts organizations recognize the importance of providing opportunities for young artists to learn from experienced

B. mentors.
- C. most established artists could become even better artists by serving as mentors.
- D. finding a mentor is more important for musicians than it is for painters, writers, and other types of artists.

ID: 93cad661 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the relationship between the AACM’s use of mentors to support young Black musicians and similar support that other organizations offer their young artists. According to the text, in service of AACM’s goal to support new works by Black musicians, AACM mentors provide beneficial training and guidance to young artists. The text goes on to say that many other art organizations support new artists in similar ways, suggesting that these organizations recognize the importance of providing opportunities for young artists to learn from mentors who are established in their field.

Choice A is incorrect because the text gives no indication that technical training is more beneficial than creative guidance. The text states that AACM mentors offer both technical training and creative guidance to young artists and that other arts organizations offer similar kinds of support; however, nothing in the text suggests that young artists, or artists of any age, benefit more from technical training. Choice C is incorrect because the text is mainly concerned with the fact that established artists are mentoring young artists through AACM and other arts organizations. The text states that young musicians benefit from AACM mentors but does not address what mentors gain from the relationship, so there is no reason to conclude that most established artists could improve as artists by serving as mentors. Choice D is incorrect. Although the text indicates that mentors are beneficial to young musicians and that young people pursuing other kinds of art (including painting and writing) can also find mentors through arts organizations, the text gives no indication that mentors are more important for musicians than for other types of artists.

Question Difficulty: Easy

Question ID f495b554

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: f495b554

PSAT2 RW 2.14

Silicon-based photovoltaic cells account for 95% of the cells used in solar panels worldwide despite converting an average of only 18–22% of the sunlight that reaches them. In a study addressing this relative inefficiency, a team led by Laura Miranda-Pérez demonstrated that the addition of a thin layer of the mineral perovskite—which captures the blue range of light in the solar spectrum, whereas silicon captures the red range—allows the cells to convert 29.5% or more of the Sun’s energy into usable electricity. Cells made with only perovskite, however, are no more efficient than silicon-based ones. It’s reasonable to conclude, then, that _____

Which choice most logically completes the text?

- photovoltaic cells with both silicon and perovskite are more efficient because they make use of more of the solar
- A. spectrum.
- B. photovoltaic cells with only perovskite and no silicon would likely convert more than 29.5% of the Sun’s energy.
- C. solar power will remain elusive until photovoltaic cells are replaced with a more practical technology.
- D. researchers need to evaluate whether other minerals like perovskite are as effective as perovskite seems to be.

ID: f495b554 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it presents the conclusion that most logically completes the text's discussion of the efficiency of various photovoltaic cells. The text indicates that Miranda-Pérez and her team found that adding a layer of perovskite to silicon-based cells increased the percentage of sunlight converted to usable electricity relative to the percentage converted to electricity by silicon-based cells. Using cells with only perovskite and no silicon, however, resulted in no improvement in efficiency over silicon-based cells. The text also states that perovskite captures light in the blue range of the solar spectrum and silicon captures light in the red range of the spectrum. If perovskite and silicon capture different parts of the solar spectrum and the combination of perovskite and silicon results in greater efficiency than the use of either perovskite or silicon alone does, then it is reasonable to infer that the improved efficiency of perovskite-silicon cells could be attributable to their making use of more of the solar spectrum than perovskite-based or silicon-based cells do.

Choice B is incorrect because it is directly contradicted by the text, which says that silicon-based cells convert 18%–22% of the sunlight that reaches them to electricity and that cells with only perovskite are no more efficient than silicon-based cells are. Perovskite-only cells thus must convert 22% or less—not more than 29.5%—of the sunlight that reaches them to electricity. Choice C is incorrect because the text focuses on a possible improvement to photovoltaic cells and makes no mention of any alternatives to those cells, so there is no reason to conclude that photovoltaic cells must be replaced with some other technology. Additionally, the text presents solar power as something that currently exists, so it wouldn't make sense to conclude that solar power will remain elusive. Choice D is incorrect because nothing in the text suggests that there is a need to evaluate the effectiveness of other minerals than perovskite. In fact, the text gives no indication that it is even possible to use other minerals than silicon and perovskite in photovoltaic cells. Instead, the text is focused on the effect of combining perovskite and silicon and the different parts of the solar spectrum that perovskite and silicon capture.

Question Difficulty: Hard

Question ID 64a40675

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	

ID: 64a40675

PSAT2 RW 2.15

In 1953, a fellow performer tripped on legendary jazz musician Dizzy Gillespie’s trumpet, bending its bell upward. When Gillespie tested the damaged instrument, he realized that he _____ sound of a bent bell over that of a straight one.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. preferred; the
- B. preferred the
- C. preferred, the
- D. preferred. The

ID: 64a40675 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The convention being tested is punctuation use between a verb and its object. No punctuation is needed between the verb "preferred" and its object "the sound of a bent bell over that of a straight one." The object helps complete the idea of the verb—in this case, it explains what type of sound Gillespie preferred—and any punctuation between the two results in an ungrammatical sentence.

Choice A is incorrect because no punctuation is needed between the verb and its object. Choice C is incorrect because no punctuation is needed between the verb and its object. Choice D is incorrect because no punctuation is needed between the verb and its object.

Question Difficulty: Easy

Question ID cf08e2fd

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: cf08e2fd

PSAT2 RW 2.16

From 1912 to 1951, Charlotta Bass owned and operated the newspaper *The California Eagle*. While it was under Bass’s leadership, *The Eagle* _____ one of the US’s most influential Black-owned newspapers.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. will become
- B. became
- C. is becoming
- D. to become

ID: cf08e2fd Answer

Correct Answer: B

Rationale

Choice B is the best answer. The conventions being tested are the use of verbs to express tense and the use of verb forms within a sentence. In this choice, the past tense verb "became" is consistent with the other past tense verbs ("owned," "operated," and "was") used to indicate that, at a period of time in the past, Bass’s leadership resulted in her newspaper becoming one of the most influential Black-owned newspapers in the US. In addition, "became" supplies the main clause with a finite (tensed) verb, which is required to perform the action of the subject (in this case, "*The Eagle*").

Choice A is incorrect because the future tense verb "will become" isn’t consistent with the other past tense verbs used to discuss Bass and her newspaper. Choice C is incorrect because the present progressive verb "is becoming" isn’t consistent with the other past tense verbs used to discuss Bass and her newspaper. Choice D is incorrect because it results in an ungrammatical sentence. The nonfinite to-infinitive "to become" doesn’t supply the main clause with a finite verb.

Question Difficulty: Easy

Question ID e62241b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: e62241b7

PSAT2 RW 2.17

What is the correct pronunciation of Kiribati? In the Gilbertese language spoken by residents of the island nation, the letter combination -ti makes the -s sound; as a result, the country’s name _____ pronounced “Kiribas.”

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. are
- B. have been
- C. are being
- D. is

ID: e62241b7 Answer

Correct Answer: D

Rationale

Choice D is the best answer. The convention being tested is subject-verb agreement. The singular verb "is" agrees in number with the singular subject "the country’s name."

Choice A is incorrect because the plural verb "are" doesn’t agree in number with the singular subject "the country’s name." Choice B is incorrect because the plural verb "have been" doesn’t agree in number with the singular subject "the country’s name." Choice C is incorrect because the plural verb "are being" doesn’t agree in number with the singular subject "the country’s name."

Question Difficulty: Easy

Question ID bce9910f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: bce9910f

PSAT2 RW 2.18

Light of Truth is a bronze and marble sculpture by artist Richard Hunt. It honors civil rights icon Ida B. Wells. The sculpture _____ in a tree-lined plaza in Chicago, just a few blocks from where Wells lived.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. are standing
- B. have been standing
- C. stands
- D. were standing

ID: bce9910f Answer

Correct Answer: C

Rationale

Choice C is the best answer. The convention being tested is subject-verb agreement. The singular verb "stands" agrees in number with the singular subject "sculpture."

Choice A is incorrect because the plural verb "are standing" doesn't agree in number with the singular subject "sculpture." Choice B is incorrect because the plural verb "have been standing" doesn't agree in number with the singular subject "sculpture." Choice D is incorrect because the plural verb "were standing" doesn't agree in number with the singular subject "sculpture."

Question Difficulty: Easy

Question ID 486f03da

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	

ID: 486f03da

PSAT2 RW 2.19

The short story “Rogue Enchantments” by Isabel Ibañez appears in *Reclaim the _____* anthology of fantasy and science fiction written by authors of Latin American descent.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. *Stars*. An
- B. *Stars*, while an
- C. *Stars* an
- D. *Stars*, an

ID: 486f03da Answer

Correct Answer: D

Rationale

Choice D is the best answer. The convention being tested is the use of punctuation within a sentence. This choice correctly uses a comma to mark the boundary between the main clause ("The short...*Stars*") and the supplementary element ("an anthology...descent") that provides additional information about what *Reclaim the Stars* is.

Choice A is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "an anthology." Choice B is incorrect. Joining the main clause and the following supplementary element with the conjunction "while" results in a confusing and ungrammatical sentence. Choice C is incorrect because it fails to mark the boundary between the main clause and the supplementary element with appropriate punctuation.

Question Difficulty: Easy

Question ID 6bc907c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: 6bc907c9

PSAT2 RW 2.20

Novelist Jane Austen greatly admired the work of Fanny Burney, a popular English author. In fact, scholars believe that a passage from the last chapter of *Cecilia*, a 1782 novel by Burney, likely inspired the title of one of _____ *Pride and Prejudice*.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. Austen’s most famous novels,
- B. Austens’ most famous novels’,
- C. Austens most famous novels,
- D. Austen’s most famous novel’s,

ID: 6bc907c9 Answer

Correct Answer: A

Rationale

Choice A is the best answer. The convention being tested is the use of plural and possessive nouns. The singular possessive proper noun "Austen’s" and the plural noun "novels" correctly indicates that Austen wrote multiple novels that were famous.

Choice B is incorrect because the context requires the singular possessive proper noun "Austen’s" and the plural noun "novels," not the plural possessive nouns "Austens'" and "novels’." Choice C is incorrect because the context requires the singular possessive proper noun "Austen’s," not the plural proper noun "Austens." Choice D is incorrect because the context requires the plural noun "novels," not the singular possessive noun "novel’s."

Question Difficulty: Easy

Question ID 13fcf575

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	■ ■ ■

ID: 13fcf575

PSAT2 RW 2.21

Roughly 300 nights a year, when the cold air descending from the Andes Mountains meets the warm air rising from Venezuela’s coastal Lake Maracaibo, the result is a spectacular lightning storm, its strikes so bright, so localized, and so _____ that it has become known as “Maracaibo’s Lighthouse.”

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. dependable:
- B. dependable;
- C. dependable
- D. dependable,

ID: 13fcf575 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The convention being tested is the punctuation of an integrated relative clause. No punctuation is needed before the relative clause beginning with "that" because the content of the relative clause ("that...Lighthouse") is integral to the meaning of the coordinated adjectival phrase ("so bright, so localized, and so dependable") that it modifies.

Choice A is incorrect because no punctuation is needed between the coordinated adjectival phrase ("so bright...dependable") and the integrated relative clause that modifies it. Choice B is incorrect because no punctuation is needed between the coordinated adjectival phrase ("so bright...dependable") and the integrated relative clause that modifies it. Choice D is incorrect because no punctuation is needed between the coordinated adjectival phrase ("so bright...dependable") and the integrated relative clause that modifies it.

Question Difficulty: Hard

Question ID 6a5939c2

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: 6a5939c2

PSAT2 RW 2.22

Paleontologists once thought that early apes lived in tropical forests, but recent research suggests that they may have actually lived in savannas. Tropical forests are humid and have many trees spaced close together. _____ savannas are drier, and their trees are spaced further apart.

Which choice completes the text with the most logical transition?

- A. For instance,
- B. In comparison,
- C. Firstly,
- D. In conclusion,

ID: 6a5939c2 Answer

Correct Answer: B

Rationale

Choice B is the best answer. "In comparison" logically signals that the description of savannas in this sentence—that they are drier and their trees are spaced further apart—forms a comparison with the description of tropical forests in the previous sentence.

Choice A is incorrect because "for instance" illogically signals that the description of savannas in this sentence exemplifies the description of tropical forests in the previous sentence. Instead, this description forms a comparison with the description of tropical forests. Choice C is incorrect because "firstly" illogically signals that the description of savannas in this sentence indicates the first in a series of things. Instead, this description forms a comparison with the description of tropical forests in the previous sentence. Choice D is incorrect because "in conclusion" illogically signals that the description of savannas in this sentence concludes or summarizes information in the previous sentences. Instead, this description forms a comparison with the description of tropical forests in the previous sentence.

Question Difficulty: Easy

Question ID 9d4f331c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: 9d4f331c

PSAT2 RW 2.23

When sculptor Frédéric Auguste Bartholdi was designing the Statue of Liberty, he sought the advice of engineer Gustave Eiffel. Eiffel suggested that he make the statue’s arm thick and position it straight above the figure’s head. ____ Bartholdi decided to slim the arm and tilt it out at an angle.

Which choice completes the text with the most logical transition?

- A. Additionally,
- B. Instead,
- C. Thus,
- D. For example,

ID: 9d4f331c Answer

Correct Answer: B

Rationale

Choice B is the best answer. "Instead" logically signals that the information in this sentence—that Bartholdi decided to slim and tilt out the arm of the Statue of Liberty—contrasts with Eiffel’s design suggestions for the statue.

Choice A is incorrect because "additionally" illogically suggests that the information in this sentence is merely an additional fact related to Eiffel’s design suggestions in the previous sentence. Instead, it contrasts with those suggestions. Choice C is incorrect because "thus" illogically signals that the information that follows is a result of Eiffel’s design suggestions in the previous sentence. Instead, it contrasts with those suggestions. Choice D is incorrect because "for example" illogically signals that the information about Bartholdi’s design decisions in this sentence exemplifies Eiffel’s design suggestions in the previous sentence. Instead, it contrasts with those suggestions.

Question Difficulty: Easy

Question ID e0bd4f8a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: e0bd4f8a

PSAT2 RW 2.24

In 1942, the 1,500-mile Alaska Highway was constructed in under nine months, largely due to the skilled work of nearly 4,000 African American soldiers from US Army engineering regiments. The soldiers’ contribution was overlooked for decades. _____ in 2017, lawmakers declared October 25 a day of recognition—“Alaska Highway Day”—for the troops who helped build this critical roadway.

Which choice completes the text with the most logical transition?

- A. Lastly,
- B. Then,
- C. Similarly,
- D. For example,

ID: e0bd4f8a Answer

Correct Answer: B

Rationale

Choice B is the best answer. "Then" logically signals that the event described in this sentence—lawmakers’ 2017 declaration of Alaska Highway Day—is part of a chronological sequence of events, occurring after the decades-long period in which the soldiers’ contribution was overlooked.

Choice A is incorrect because "lastly" illogically signals that this sentence presents the last of a series of points or reasons. Instead, it describes a later event occurring in a chronological sequence of events. Choice C is incorrect because "similarly" illogically signals that the information in this sentence is similar to the previous information about the soldiers’ contribution being overlooked for decades. Instead, it describes an event occurring after that decades-long period. Choice D is incorrect because "for example" illogically signals that this sentence provides an example illustrating the previous information about the soldiers’ contribution being overlooked for decades. Instead, it describes an event occurring after that decades-long period.

Question Difficulty: Easy

Question ID 570dd854

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	

ID: 570dd854

PSAT2 RW 2.25

While researching a topic, a student has taken the following notes:

Here I Have Returned is a sculpture by Egyptian American artist Sherin Guirguis. It is a large, curved strip of wood inspired by the shape of a sistrum. A sistrum is a curved musical instrument played by ancient Egyptian priestesses in ceremonies. Guirguis says that the sculpture symbolizes “women who have lifted and supported Egyptian society and culture.” Overall, Guirguis wants her works to “engage audiences in a dialogue about power, agency, and social transformation.”

The student wants to use a quotation from Guirguis to explain what the sculpture represents. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- Guirguis, whose works include a sculpture that is a large, curved strip of wood, has explained that she wants her work to
- A. create a dialogue with audiences.
- Inspired by the sistrum played by Egyptian priestesses, *Here I Have Returned* symbolizes “women who have lifted and
- B. supported Egyptian society and culture,” according to Guirguis.
- According to Guirguis, the curved strip of wood used in *Here I Have Returned* was inspired by the sistrum, a musical
- C. instrument played by ancient Egyptian priestesses in ceremonies.
- Guirguis, the sculptor of *Here I Have Returned*, wants her works to “engage audiences in a dialogue about power, agency,
- D. and social transformation.”

ID: 570dd854 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The sentence uses a quotation from Guirguis to explain what *Here I Have Returned* represents, noting that Guirguis said the sculpture symbolizes "women who have lifted and supported Egyptian society and culture."

Choice A is incorrect. The sentence explains what Guirguis says she wants her work to achieve and provides an example of her work; it doesn't use a quotation to explain what the sculpture represents. Choice C is incorrect. The sentence mentions the instrument whose shape inspired the sculpture but doesn't use a quotation to explain what the sculpture represents. Choice D is incorrect. While the sentence does use a quotation from Guirguis, the quotation explains what she hopes her works in general achieve, not what the sculpture in particular represents.

Question Difficulty: Medium

Question ID 9336f63b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	■ ■ ■

ID: 9336f63b

PSAT2 RW 2.26

While researching a topic, a student has taken the following notes:

- A small number of US Navy sailors of Filipino descent served during the US Civil War (1861–1865).
- Stephen Amos was born in the Philippines around 1830. He enlisted in the US Navy in November 1863.
- Raphael Ignases was born in the Philippines around 1834. He enlisted in the US Navy in July 1861.

The student wants to emphasize the historical significance of Stephen Amos’s enlistment date. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Both Stephen Amos and Raphael Ignases were US Navy sailors of Filipino descent, but Amos enlisted in the Navy in
A. 1863, two years later than Ignases.

Stephen Amos was a US Navy sailor of Filipino descent, along with Raphael Ignases, who was born in the Philippines
B. around 1834.

Stephen Amos enlisted in the US Navy in 1863, making him one of the few sailors of Filipino descent to serve in the US
C. Civil War (1861–1865).

When Stephen Amos enlisted in the US Navy in November 1863, he joined sailors such as Raphael Ignases, who had
D. been born in the Philippines around 1834.

ID: 9336f63b Answer

Correct Answer: C

Rationale

Choice C is the best answer. Noting the time frame of the US Civil War and the fact that Amos enlisted during this time frame, the sentence emphasizes that Amos’s enlistment date places him among the historically significant group of US Navy sailors of Filipino descent who served during the Civil War.

Choice A is incorrect. The sentence identifies Amos and Ignases and notes that Amos’s enlistment date was later than Ignases’s; the sentence doesn’t explain the historical significance of the date. Choice B is incorrect. The sentence identifies Amos and Ignases as US Navy sailors of Filipino descent, noting Ignases’s birth year; the sentence doesn’t provide Amos’s enlistment date or its historical significance. Choice D is incorrect. The sentence provides Amos’s enlistment date and mentions that he served alongside a sailor of Filipino descent; it doesn’t explain the historical significance of the date.

Question Difficulty: Hard

Question ID de01cccf

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	■ ■ ■

ID: de01cccf

PSAT2 RW 2.27

While researching a topic, a student has taken the following notes:

- The background colors of US and UK road signs are used to denote each sign’s purpose.
- Directional signs are a type of sign containing information such as route names, distance to a destination, etc.
- Highways, major roadways, and minor roadways in the US generally use green for directional signs.
- Highways in the UK generally use blue for directional signs.
- Major roadways in the UK generally use green for directional signs.

The student wants to contrast how green backgrounds are used in US and UK road signs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- Blue backgrounds are generally used on directional highway signs in the UK, in contrast to the green highway signs used A. in the US.
- In the US, green signs indicate directional information on both highways and major roadways, but in the UK, directional B. signs of this color generally appear only on major roadways.
- Green road signs are used in both the US and UK to denote directional information on major roadways, like distance to a C. destination or route names.
- D. Both the UK and the US use directional signs, which include information on route names and distance to a destination.

ID: de01cccf Answer

Correct Answer: B

Rationale

Choice B is the best answer. The sentence contrasts how green backgrounds are used in US and UK road signs, noting that the US uses them for both highways and major roadways while the UK uses them only for major roadways.

Choice A is incorrect. While the sentence does contrast US and UK road signs, the contrast focuses on the use of blue versus green backgrounds rather than on the use of green backgrounds specifically. Choice C is incorrect. The sentence indicates a similarity between the uses of green road signs in the US and the UK rather than contrasting them. Choice D is incorrect. The sentence indicates a similarity between road signs in the US and those in the UK; it doesn’t contrast how green backgrounds are used in each country’s signs.

Question Difficulty: Hard

Question ID 4830ab87

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 4830ab87

PSAT2 RW 3.1

The way in which individual elements are balanced within a photographic image tends to affect how viewers perceive it: symmetry tends to give the elements equal importance, asymmetry emphasizes differences, and radial balance (organizing the elements around a central point) emphasizes the center over the periphery. What a photograph conveys is therefore largely _____ how it is balanced.

Which choice completes the text with the most logical and precise word or phrase?

- A. inhibited by
- B. contingent on
- C. obligated to
- D. reserved for

ID: 4830ab87 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of visual balance in photography. In this context, "contingent on" means conditional upon, or determined by. The text indicates that different ways of balancing elements in a photograph convey different ideas to viewers (for example, symmetrical composition tends to emphasize similarity among elements, whereas asymmetry tends to emphasize difference). Thus, there is a strong connection between what a photograph conveys and how its elements are balanced. Therefore, what a photo conveys is contingent on how it is balanced.

Choice A is incorrect. In this context, "inhibited by" most nearly means hindered or undermined by. The text indicates that different ways of balancing elements in a photograph convey different ideas to viewers, thereby indicating a relationship between image balance and what is conveyed. Although this explanation is consistent with a photographer choosing asymmetrical balance even while intending to convey similarity (associated with symmetrical balance), there is nothing to indicate that messages generally are inhibited by balance. Rather the implication is that balance is a tool photographers can use to craft their messages. Choice C is incorrect. In this context, "obligated to" means duty bound toward or answerable to. In context, it would not make sense to assert that a photograph’s message is obligated to how that photograph is balanced. Choice D is incorrect. In this context, "reserved for" most nearly means solely available to someone or held aside for some purpose. The text as a whole does not support the claim that a photograph’s message is reserved for how that photograph is balanced.

Question Difficulty: Medium

Question ID 710799ae

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 710799ae

PSAT2 RW 3.2

Cuttlefish appear to be surprisingly _____ at exercising self-control: in a 2021 study conducted by behavioral ecologist Alexandra Schnell, these cephalopods routinely demonstrated restraint by delaying gratification, waiting for a favorite treat instead of instantly devouring a readily available meal.

Which choice completes the text with the most logical and precise word or phrase?

- A. awkward
- B. imaginative
- C. manageable
- D. competent

ID: 710799ae Answer

Correct Answer: D

Rationale

Choice D is the best answer because it most logically completes the text’s discussion of self-control in cuttlefish. In this context, "competent" means capable. The text describes a study in which cuttlefish demonstrated an ability to exercise restraint by waiting for a favorite treat instead of eating a meal that was already available to them. That is, by delaying gratification, the cuttlefish in the study demonstrated that they were capable of, or competent at, exercising self-control.

Choice A is incorrect because nothing in the text suggests that in demonstrating self-control, cuttlefish are "awkward," or lacking skill or grace. If anything, the text indicates that cuttlefish appear to be surprisingly skilled at exercising self-control. Choice B is incorrect because in this context, "imaginative" would mean creative, and there’s nothing in the text to suggest that cuttlefish demonstrate creativity. Instead, the text focuses on the fact that cuttlefish seem to be capable of exercising restraint and self-control. Choice C is incorrect because in this context, "manageable" would mean capable of being controlled. Although the text indicates that the cuttlefish in the study demonstrated restraint by waiting for a favorite treat instead of devouring a readily available meal (which suggests their ability to control themselves), it doesn’t suggest that cuttlefish are thus also able to be managed or controlled by others.

Question Difficulty: Medium

Question ID 1107e7dc

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	

ID: 1107e7dc

PSAT2 RW 3.3

Economists often assert that countries looking to increase their reliance on solar energy should expand their capacity for storage; having an ample reserve of stored energy can mitigate the effects of _____ solar energy collection caused by unpredictable shifts in cloud cover and haze.

Which choice completes the text with the most logical and precise word or phrase?

- A. developments of
- B. fluctuations in
- C. calibrations with
- D. incentives for

ID: 1107e7dc Answer

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text’s discussion of solar energy collection and storage. In this context, "fluctuations in" means irregular changes in. The text notes that economists recommend that countries that want to rely more on solar energy (energy captured from sunlight) should attempt to expand their storage capacity. The text goes on to explain that having a large amount of stored solar energy can lessen the negative effect of "unpredictable shifts in cloud cover and haze"—that is, unpredictable changes in environmental conditions that at times limit sunlight. This context suggests that having ample solar energy stored and ready to use can mitigate the issue of fluctuations in solar energy collection caused by irregular patterns of sunlight availability.

Choice A is incorrect because in this context, "developments of solar energy collection" would most clearly refer to the creation of ways to collect solar energy (energy captured from sunlight). The text focuses on collection and the idea that having stored energy can mitigate, or make less bad, the effects of a problem caused by unpredictable changes that affect sunlight ("cloud cover and haze"); there’s no reason to think that the creation of new collection methods would be viewed as a problem that would need to be mitigated. Choice C is incorrect because in this context, "calibrations with" would most clearly refer to precise adjustments for a particular purpose. The text focuses on the collection of solar energy (energy captured from sunlight) and the idea that having stored energy can mitigate, or make less bad, the effects of a problem caused by unpredictable changes that affect sunlight ("cloud cover and haze"); there’s no reason to think that precision in something related to solar energy collection would be viewed as a problem that would need to be mitigated. Choice D is incorrect. The text focuses on the collection of solar energy (energy captured from sunlight) and the idea that having stored energy can mitigate, or make less bad, the effects of a problem caused by unpredictable changes that affect sunlight ("cloud cover and haze"); nothing in the text suggests that "incentives for," or rewards for, solar energy collection would be viewed as a problem that would need to be mitigated, especially in countries that want to rely more on solar energy than they already do.

Question Difficulty: Medium

Question ID a1ff8ce3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	■ ■ ■

ID: a1ff8ce3

PSAT2 RW 3.4

As discussed by scholar Anna Mladentseva, many artworks produced in the mid-1990s to the early 2000s exclusively for exhibition on the internet, such as Sinae Kim’s *Genesis* (2001), have become inaccessible because viewing them requires the use of _____ software (most notably Adobe Flash, discontinued in 2021).

Which choice completes the text with the most logical and precise word or phrase?

- A. defunct
- B. arcane
- C. ubiquitous
- D. extraneous

ID: a1ff8ce3 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most logically completes the text’s discussion of artworks that were produced for exhibition on the internet. In this context, "defunct" means no longer existing or functioning. According to the text, many artworks that were produced in the mid-1990s to the early 2000s for exhibition on the internet have become inaccessible due to their reliance on certain software. As an example, the text cites Sinae Kim’s *Genesis*, which relied on software that was discontinued in 2021. This context supports the idea that the software is defunct.

Choice B is incorrect because the text indicates that viewing artworks produced in the mid-1990s to the early 2000s for exhibition on the internet requires the use of software. Artists generally want their works to be seen by many people, so it wouldn’t make sense for creators of internet art to require the use of software that is "arcane," or known or knowable to only a few people. Moreover, the text states that a prominent example of software used to view these artworks is Adobe Flash, which was discontinued in 2021, meaning it’s now defunct. Choice C is incorrect because "ubiquitous" would mean found everywhere, which wouldn’t make sense in this context. The text indicates that the reason why many artworks that were produced in the mid-1990s to the early 2000s for exhibition on the internet have become inaccessible has to do with the software required for viewing them. According to the text, one example of such software is Adobe Flash, which was discontinued in 2021, meaning it’s now defunct. Choice D is incorrect because nothing in the text suggests that the software used to view artworks produced in the mid-1990s to the early 2000s for exhibition on the internet is "extraneous," or irrelevant. Instead, the text indicates that use of specific software is required to view certain artworks from this period and that the discontinuation of the software renders the works inaccessible.

Question Difficulty: Hard

Question ID c7265342

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	

ID: c7265342

PSAT2 RW 3.5

The following text is from the 1895 poem “Marshlands” by Emily Pauline Johnson, a Kanienkahagen (Mohawk) writer also known as Tekahionwake.

Among the wild rice in the still lagoon, In monotone the lizard shrills his tune.
The wild goose, homing, seeks a sheltering, Where rushes grow, and oozing lichens cling.
Late cranes with heavy wing, and lazy flight, Sail up the silence with the nearing night.
And like a spirit, swathed in some soft veil, Steals twilight and its shadows o’er the swale.
Hushed lie the sedges, and the vapours creep, Thick, grey and humid, while the marshes sleep.

Which choice best describes the overall structure of the text?

- A. It names animal species found in a place, then names plant species there.
- B. It sketches a setting by presenting a series of images of nature.
- C. It makes an extended comparison of nature to human emotions.
- D. It identifies a location, then refers to a person living there.

ID: c7265342 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it best describes the overall structure of the text. The text first establishes the setting, a "still lagoon," and then goes on to provide more details about the lagoon by presenting several descriptive images of nature, including a lizard that "shrills his tune," "oozing lichens," and "thick, grey and humid" vapors.

Choice A is incorrect. Instead of simply naming species, the text presents descriptive images of nature; further, instead of naming animals and then moving on to name plants, the text refers first to a plant ("the wild rice"), then to two animals ("the lizard," "the wild goose"), then to another plant ("rushes"), and so on. Choice C is incorrect because the text does not draw any comparison between nature and human emotions; it does not refer to human emotions at all. Choice D is incorrect. While the text does begin by identifying a specific location, a "still lagoon," it makes no mention of a person living there.

Question Difficulty: Medium

Question ID d168e7c5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	

ID: d168e7c5

PSAT2 RW 3.6

The following text is from Annie Dillard’s 1987 autobiographical novel *An American Childhood*. The narrator is a young girl living in Pittsburgh.

I walked. My mother had given me the freedom of the streets as soon as I could say our telephone number. I walked and memorized the neighborhood. I made a mental map and located myself upon it. At night in bed I rehearsed the small world’s scheme and set challenges: Find the store using backyards only. Imagine a route from the school to my friend’s house.

©1987 by Annie Dillard

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It describes the narrator trying to memorize her telephone number.
- B. It provides examples of what the narrator thinks about at night.
- C. It gives directions to the narrator’s favorite local store.
- D. It portrays the narrator’s relationship with her mother.

ID: d168e7c5 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it accurately describes how the underlined portion functions in the text as a whole. In the text, the narrator explains that she walks around her neighborhood, making a "mental map" of the area that she imagines herself navigating through. She then states that she rehearses "the small world’s scheme"—that is, imagines moving through her mental map—and challenges herself to use the map in her mind while lying in bed at night. The underlined portion presents two such challenges (getting to the store through backyards and traveling from school to a friend’s house) and thus provides examples of what the narrator thinks about at night.

Choice A is incorrect because the underlined portion makes no mention of memorizing a telephone number. Although the narrator mentions that she had to learn the home telephone number before her mother would give her permission to walk around the neighborhood, there is nothing in the underlined portion or the rest of the text about memorizing the telephone number. Choice C is incorrect. Although the underlined portion refers to navigation tasks like finding a route to a store only through backyards, the text contains no specific directions to any store, nor is any store identified as the narrator’s favorite. Choice D is incorrect because the underlined portion makes no mention of the narrator’s mother and doesn’t address the narrator’s relationship with her. Although the narrator mentions that her mother gave her permission as a child to walk around the neighborhood, there is nothing in the underlined portion or the rest of the text about the mother or her relationship to her child.

Question Difficulty: Medium

Question ID d2c2a7ef

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: d2c2a7ef

PSAT2 RW 3.7

The following text is from Anton Chekhov’s 1898 short story “Ionitch” (translated by Marian Fell in 1915). The text is set in a Russian city referred to as the city of S.

If newcomers to the little provincial city of S. complained that life there was monotonous and dull, its inhabitants would answer that, on the contrary, S. was a very amusing place, indeed, that it had a library and a club, that balls were given there, and finally, that very pleasant families lived there with whom one might become acquainted. And they always pointed to the Turkins as the most accomplished and most enlightened family of all.

What does the text suggest about the Turkins?

- A. They are relative newcomers to the city of S.
- B. They have a unique status in the city of S.
- C. They have long disliked living in the city of S.
- D. They are amused by the other residents of the city of S.

ID: d2c2a7ef Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents a statement about the Turkins that is suggested by the text. The text explains that while many pleasant families live in the city of S., inhabitants of the city consider the Turkins to be "the most accomplished and most enlightened family of all." Thus, by indicating that they are seen as more accomplished and more enlightened than all the other families in the city of S., the text suggests that the Turkins have a unique status there.

Choice A is incorrect because the text doesn’t indicate that the Turkins have recently moved to the city of S.; it provides no information about when the Turkins arrived. Choice C is incorrect because the text explains how the Turkins are viewed by others in the city of S. but gives no indication of how the Turkins feel about living there. Choice D is incorrect because the text explains how the Turkins are viewed by other residents of the city of S. but gives no indication of how the Turkins view the other residents.

Question Difficulty: Medium

Question ID f64ff4fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	

ID: f64ff4fb

PSAT2 RW 3.8

In her 1998 book *Blues Legacies and Black Feminism*, Angela Y. Davis bases her analysis in part on recordings of songs sung in the 1920s by Gertrude “Ma” Rainey and Bessie Smith. Davis focuses on how Rainey and Smith improvised the lyrics—replacing the original lines with mischievous jokes and wordplay. Davis’s work was particularly labor intensive because in order to transcribe, or write down, the lyrics as Rainey and Smith sang them, Davis had to listen repeatedly to the vinyl recordings, which weren’t very clear.

What does the text most strongly suggest about the songs sung by Rainey and Smith?

- A. The songs have grown in popularity since Rainey and Smith first sang them.
- B. There were more recordings made of Rainey’s songs than there were of Smith’s.
- C. There were few, if any, reliable transcriptions of Smith’s and Rainey’s improvised lyrics when Davis began her research.
- D. According to Davis, the songs sung by Rainey were more musically innovative than those sung by Smith typically were.

ID: f64ff4fb Answer

Correct Answer: C

Rationale

Choice C is the best answer because it describes the songs sung by Gertrude "Ma" Rainey and Bessie Smith in a way the text implies is accurate. The text describes Angela Y. Davis’s research on Rainey and Smith for her book *Blues Legacies and Black Feminism*, focusing on her efforts to transcribe the improvised lyrics in Rainey’s and Smith’s songs. The text calls Davis’s transcription process "labor intensive" since the lack of clarity in the recordings required her to listen to each repeatedly to verify the accuracy of her transcripts. The fact that Davis undertook a painstaking transcription process using only fairly low-fidelity recordings suggests that reliable transcriptions were otherwise unavailable to her.

Choice A is incorrect. The text doesn’t discuss the popularity of Rainey’s and Smith’s songs either in the 1920s or after. Although it is plausible that the music of Rainey and Smith is more widely enjoyed than it was in the 1920s, this isn’t supported by the text. Choice B is incorrect. The text doesn’t discuss which of Rainey and Smith was the more prolific recording artist and so provides no support for such a claim. Choice D is incorrect. The text discusses the creativity of both Rainey and Smith, but it does so only to note similarities between them in terms of improvisation and wit, not to emphasize differences between them or Davis’s relative views of the artists’ inventiveness.

Question Difficulty: Medium

Question ID 7f0be746

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	■ ■ ■

ID: 7f0be746

PSAT2 RW 3.9

The following text is from Milan Kundera’s 1984 novel *The Unbearable Lightness of Being* (translated by Michael Henry Heim in 1984). Karenin is a dog that belongs to Tomas and Tereza.

Karenin was not overjoyed by the move to Switzerland [from Prague]. Karenin hated change. Dog time cannot be plotted along a straight line; it does not move on and on, from one thing to the next. It moves in a circle like the hands of a clock, which—they, too, unwilling to dash madly ahead—turn round and round the face, day in and day out following the same path. In Prague, when Tomas and Tereza bought a new chair or moved a flower pot, Karenin would look on in displeasure. It disturbed his sense of time. It was as though they were trying to dupe the hands of the clock by changing the numbers on its face.

©1984 by Milan Kundera. Translation ©1984 by Harper & Row, Publishers, Inc.

Which choice best states the main idea of the text?

- As a dog, Karenin possesses a sense of time that involves a strong preference for predictability and an aversion to
- A. disruption.
- B. After he’s moved to a new home, Karenin’s negative response to changes has become more pronounced.
- C. Similar to Tomas and Tereza, Karenin comprehends time as circular rather than as a straightforward progression.

- As is the case for other dogs, Karenin’s sense of time seems to accelerate depending on the objects and places that
- D. surround him.

ID: 7f0be746 Answer

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately states the main idea of the text. The text begins by stating that Tomas and Tereza's dog Karenin felt disrupted by a recent move because of his dislike of change. The text then goes on to suggest that this is because the way a dog experiences time differs from the way humans experience time: time for a dog doesn't move linearly, going "on and on, from one thing to the next," but instead moves circularly, "like the hands of a clock." That is, time for a dog is experienced as a cyclical pattern characterized by routine and predictability, with each day "following the same path." The text then concludes by providing examples of seemingly insignificant changes in routine that profoundly "disturbed [Karenin's] sense of time," causing him to feel displeasure. Thus, the main idea of the text is that Karenin's sense of time as a dog involves a strong preference for predictability and an aversion to disruption.

Choice B is incorrect. Although the text emphasizes Karenin's displeasure with the recent move to a new home, it doesn't suggest that the move has made his negative responses more pronounced than they once were. Rather, in accounting for Karenin's displeasure with the move to Switzerland, the text explains that Karenin generally has a negative response to any kind of change. Choice C is incorrect because the text doesn't suggest that Karenin comprehends time similarly to how Tomas and Tereza comprehend it. On the contrary, the text strongly implies a contrast between dogs' circular experience of time with the way humans experience time as a straightforward progression that can be "plotted on a straight line." Choice D is incorrect because the text provides no indication that a change in the places and objects surrounding Karenin causes him to feel as though time is accelerating. Although the text does use the language of "dash[ing] madly ahead" in relation to time, the phrase appears in the context of a comparison illustrating how dogs experience time: time for a dog moves just as the hands of a clock do, in a circle and "unwilling to dash madly ahead"—that is, always in a regular and predictable way.

Question Difficulty: Hard

Question ID 4d38c2ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 4d38c2ea

PSAT2 RW 3.10

Gorget, or necklaces with large pendants, have been part of the ceremonial attire of tribes from the US Southeast for centuries. One of the oldest examples, the Fairfield Gorget, was found in Fairfield, Missouri, in 1958. Its overall design resembles that of other art from the region during the Mississippian period (900–1600 CE). Yet the image on the gorget is of a jaguar—a species whose range doesn’t extend to Missouri. Jaguar images are common in ancient Mexican art, and Mexico lies squarely in the species’ range. Therefore, some scholars argue that long-distance trade in Mexican art objects brought the imagery to Missouri, where a local artist could have adopted it and incorporated it into the Fairfield Gorget.

Which finding, if true, would most strongly support the underlined explanation?

- The Fairfield Gorget is dated not to the Mississippian period but instead to the earlier Woodland period, which ended
- A. around 900 CE.
- The range of the jaguar is shown to have expanded dramatically after the Mississippian period came to a close around
- B. 1600 CE.
- An ancient Mexican art object is found at a site that dates to the Mississippian period and is close to where the Fairfield
- C. Gorget was found.
- Certain works by present-day artists from Southeastern tribes reflect outside cultural influences, including contemporary
- D. Mexican art.

ID: 4d38c2ea Answer

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most strongly support the underlined explanation that the jaguar imagery on the Fairfield Gorget could have been inspired by art objects brought to Missouri from Mexico. The text describes the Fairfield Gorget as one of the oldest examples of these necklaces, noting that it's similar to other art from the Mississippian period except for its image of a jaguar. The text goes on to explain that this is unusual because the jaguar's range didn't include Missouri. The Fairfield Gorget's artist likely wouldn't have encountered a live jaguar, but since such jaguar images were common in ancient Mexican art (and the jaguar's range did include Mexico), perhaps some of that art inspired the gorget's artist. Therefore, evidence that an ancient Mexican art object had been found at a site that was close to the Fairfield Gorget in both location and time period would strongly support the explanation that long-distance trade from Mexico took place during the Mississippian period and that jaguar imagery could have reached Missouri by those means.

Choice A is incorrect. The underlined explanation indicates that Mexican artistic influence preceded the creation of the Fairfield Gorget. An artifact from Mexico that appeared in Missouri only after the gorget was created cannot support such a claim. Choice B is incorrect. The underlined explanation pertains to whether the jaguar image on the Fairfield Gorget was suggested to the Mississippian artist who made it by encounters with depictions of jaguars in Mexican art. The appearance of jaguars in Missouri after the Mississippian period cannot provide evidence for such a claim. Choice D is incorrect. The underlined explanation pertains to influence that Mexican art might have had on Mississippian art, restricted to the period from 900 CE to 1600 CE. The actions of contemporary artists cannot provide evidence for centuries-old networks of artistic influence and exchange.

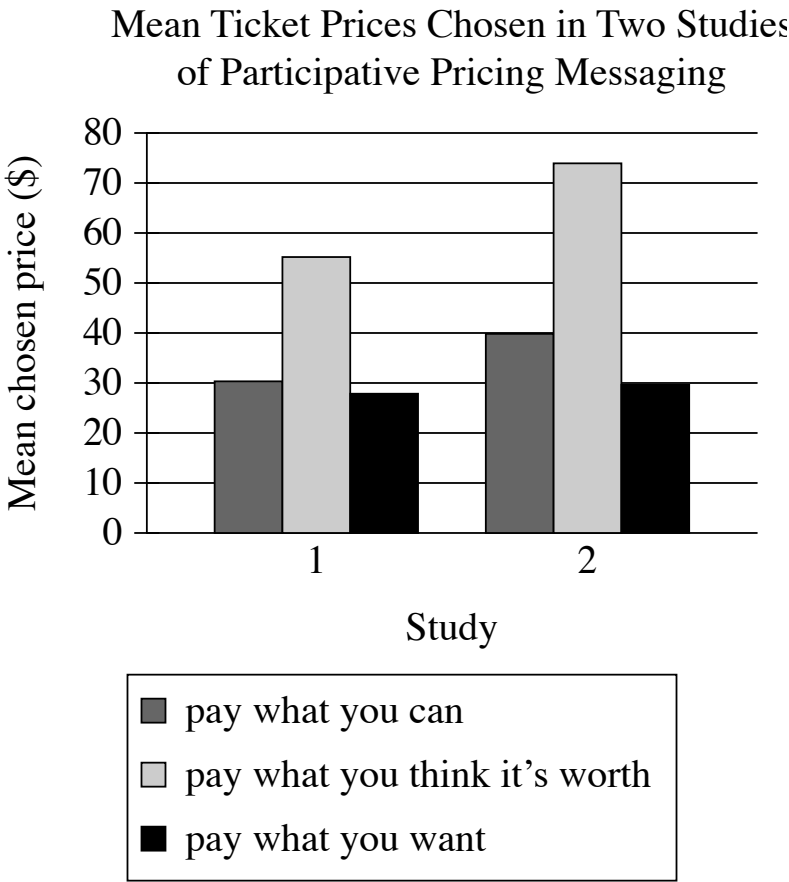
Question Difficulty: Hard

Question ID 3430be35

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 3430be35

PSAT2 RW 3.11



Participative pricing, in which purchasers choose the prices they pay for products, can enable sellers to capitalize on the heterogeneous values consumers assign to the same goods and services, but doing so requires careful messaging. Annie Peng Cui and Jennifer Wiggins recruited 171 participants (ages 18–60) online for an initial study and 83 students (ages 18–31) at a state university for a second study to test the effect of three different messages—“pay what you can,” “pay what you think it’s worth,” and “pay what you want”—on how much participants would pay for concert tickets. Their results illustrate both the heterogeneity of consumer valuations and how sellers can benefit by prompting consumers to consider their own valuations: _____

Which choice most effectively uses data from the graph to complete the text?

- the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but
- A. when considering what they could afford to pay, the students tended to choose a lower price than did the other group.
- in all three messaging conditions, the group of participants recruited online tended to choose lower prices than did the students, but both groups tended to choose prices closest to the actual cost of the tickets when prompted to consider
- B. the tickets’ value.

the students tended to value the concert tickets more highly than did the more age-diverse group recruited online, but both groups tended to choose a higher price when considering the value of the tickets than when considering what they C. could afford or wanted to pay.

within each group of participants, there was wide variation in the value that individuals assigned to the concert tickets, D. but the students tended to assign a higher value to the tickets than did the more age-diverse group recruited online.

ID: 3430be35 Answer

Correct Answer: C

Rationale

Choice C is the best answer because it most effectively uses data from the graph to complete the text about the effect of messaging on participative pricing. The graph shows mean ticket prices chosen by participants in response to three messages across two studies: Study 1, which the text indicates was conducted with an age-diverse group recruited online, and Study 2, which was conducted with student participants. The graph indicates that in the "pay what you think it's worth" condition, the mean price of the concert tickets in Study 2 was about \$74, which is greater than the mean price of about \$55 in Study 1. In other words, when participants were asked to consider their valuation of the tickets, the response was heterogeneous, or mixed. Moreover, according to the graph, both Study 1 and Study 2 show higher prices for the tickets under the "pay what you think it's worth" condition than they do under both the "pay what you can" and the "pay what you want" conditions. That is, the data suggest that both groups of participants named higher prices when considering the value of the tickets than when considering either what they could afford or wanted to pay, a finding that supports the idea that sellers can benefit when prompting consumers to consider their own valuations when they choose prices.

Choice A is incorrect because it contradicts information in the graph. Although the graph shows that students in Study 2 assigned a higher value to the tickets than did the age-diverse group in Study 1, which would support the idea that consumer valuations were heterogeneous, the graph shows that in the "pay what you can" (i.e., what you can afford) condition, the students in Study 2 assigned a higher price (about \$40), not a lower price, than the age-diverse group in Study 1 did (about \$30). Moreover, even if it were true that the students had assigned a lower price in this condition, it wouldn't support the result described in the text, only that the participants across the two studies had different ideas of what they can afford to pay. Choice B is incorrect. Although a finding that participants tended to choose prices that were closest to the actual ticket costs in the "pay what you think it's worth" condition would support the idea that sellers benefit by prompting consumers to think about their own valuations (since it's implied that sellers would lose money in the other conditions, where chosen prices were lower than the participants' valuations), neither the text nor the graph addresses how any of the prices chosen by the study participants relate to the tickets' actual market price. Choice D is incorrect. Although the wide variation in participant valuations would support the idea that consumer valuations tend to be heterogeneous, neither the text nor the graph provides any information from which to discern the relative levels of variance among the responses from participants in either study.

Question Difficulty: Hard

Question ID 6d69ab93

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	■ ■ ■

ID: 6d69ab93

PSAT2 RW 3.12

Initially observed in 2017, the interstellar object ‘Oumuamua is the first object of its kind to be seen in our solar system. Researchers have been puzzled because its acceleration cannot be entirely explained by the gravitational pull of nearby bodies: there must be a nongravitational influence on its velocity and trajectory. Some previously suggested explanations for this nongravitational acceleration involve mechanisms that are unlikely or unrealistic, such as geometric effects from ‘Oumuamua being potentially composed of several spatially separated bodies. Now, Jennifer Bergner and colleagues propose that the nongravitational acceleration is due to the gaseous expulsion of entrapped hydrogen from ‘Oumuamua’s water-rich icy body.

Which statement, if true, would most strongly support the claim made by Bergner and colleagues about the cause of ‘Oumuamua’s acceleration?

- Existing proposed models of outgassing from ‘Oumuamua include the direct conversion of nitrogen or carbon monoxide from a solid to a gaseous state without becoming liquid, but these models have theoretical or observational
- A. inconsistencies.
- ‘Oumuamua’s trajectory is inconsistent with a nongravitational acceleration that would be caused by the release of hydrogen gas resulting from the processing of water ice (H₂O), but the interstellar object’s observable properties can be
- B. explained if it has a significant component of molecular hydrogen ice (H₂).
- Since nongravitational accelerations of interstellar objects are several orders of magnitude weaker than gravitational accelerations, deviation from behavior that could be fully attributed to gravitational pull has been detected on a limited
- C. number of objects similar to ‘Oumuamua.
- Exposure to interstellar cosmic radiation can result in the formation of embedded pockets of hydrogen gas in water ice; moreover, when traveling through the solar system, ‘Oumuamua experienced warming sufficient to alter its icy structure
- D. and allow for outgassing.

ID: 6d69ab93 Answer

Correct Answer: D

Rationale

Choice D is the best answer because it presents findings that, if true, would support the claim made by Bergner and colleagues that the nongravitational acceleration of 'Oumuamua is due to the expulsion of entrapped hydrogen. The text first introduces the observation of a unique interstellar object named 'Oumuamua and goes on to explain that the object exhibited nongravitational acceleration that could not be fully attributed to the expected cause: gravitational pull of nearby celestial bodies. The text concludes by stating that Bergner and colleagues claim that the nongravitational acceleration is caused by expulsion of hydrogen gas from 'Oumuamua's water-rich icy body. To support this claim requires evidence that hydrogen gas could be present within 'Oumuamua at all, which this answer choice presents: cosmic radiation can result in embedded pockets of hydrogen gas in water ice. Additionally, evidence that this gas can be released from such a body is required to fully support the claim, which this answer choice goes on to provide: 'Oumuamua experienced sufficient warming as it traveled through the solar system to alter its icy structure and release the hydrogen gas. Thus, this answer choice provides the best evidence to support Bergner and colleagues' claim.

Choice A is incorrect because this answer choice concerns faults with previous models of outgassing from 'Oumuamua of carbon monoxide and nitrogen, which would not support a claim regarding hydrogen outgassing. Furthermore, inconsistencies in other models would not provide evidence in support of a different model or explanation. Choice B is incorrect. The evidence presented in this answer choice would weaken the claim proposed by Bergner and colleagues that the nongravitational acceleration of 'Oumuamua is caused by the expulsion of hydrogen gas because this answer choice suggests that there is evidence that refutes this claim: 'Oumuamua's trajectory is inconsistent with a nongravitational acceleration caused by the release of hydrogen gas. Furthermore, the remaining portion of this answer choice is unrelated to the claim. Choice C is incorrect because the claim being made by Bergner and colleagues concerns the expulsion of entrapped hydrogen gas, but this answer choice is concerned solely with the differences in magnitude of gravitational and nongravitational acceleration, which would not support Bergner's claim. Furthermore, this answer choice discusses interstellar objects similar to 'Oumuamua, but the text states that 'Oumuamua is the first observed object of its kind in our solar system, so evidence from other, similar bodies would not be available.

Question Difficulty: Hard

Question ID 93cad661

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	

ID: 93cad661

PSAT2 RW 3.13

A main goal of the Association for the Advancement of Creative Musicians (AACM), an arts organization founded in 1965, is to advance new works by Black musicians. The AACM achieves this goal in part by focusing on young artists. By having established musicians and composers serve as mentors, the AACM gives young artists the benefits of expert technical training and creative guidance. Numerous organizations offer similar kinds of support to new generations of painters, writers, and other artists, suggesting that _____

Which choice most logically completes the text?

- A. artists of all ages benefit more from technical training than from creative guidance.
- many arts organizations recognize the importance of providing opportunities for young artists to learn from experienced
- B. mentors.
- C. most established artists could become even better artists by serving as mentors.
- D. finding a mentor is more important for musicians than it is for painters, writers, and other types of artists.

ID: 93cad661 Answer

Correct Answer: B

Rationale

Choice B is the best answer because it presents the conclusion that most logically follows from the text’s discussion of the relationship between the AACM’s use of mentors to support young Black musicians and similar support that other organizations offer their young artists. According to the text, in service of AACM’s goal to support new works by Black musicians, AACM mentors provide beneficial training and guidance to young artists. The text goes on to say that many other art organizations support new artists in similar ways, suggesting that these organizations recognize the importance of providing opportunities for young artists to learn from mentors who are established in their field.

Choice A is incorrect because the text gives no indication that technical training is more beneficial than creative guidance. The text states that AACM mentors offer both technical training and creative guidance to young artists and that other arts organizations offer similar kinds of support; however, nothing in the text suggests that young artists, or artists of any age, benefit more from technical training. Choice C is incorrect because the text is mainly concerned with the fact that established artists are mentoring young artists through AACM and other arts organizations. The text states that young musicians benefit from AACM mentors but does not address what mentors gain from the relationship, so there is no reason to conclude that most established artists could improve as artists by serving as mentors. Choice D is incorrect. Although the text indicates that mentors are beneficial to young musicians and that young people pursuing other kinds of art (including painting and writing) can also find mentors through arts organizations, the text gives no indication that mentors are more important for musicians than for other types of artists.

Question Difficulty: Easy

Question ID 96d1c1fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	■ ■ ■

ID: 96d1c1fe

PSAT2 RW 3.14

Tatiana R. Feuerborn and colleagues analyzed the genomes of more than a hundred domesticated dogs from sites in Siberia dating from 11,000 years ago to the present. They found that the dogs constituted a genetically isolated population of Arctic breeds until approximately 2,000 years ago, at which point there was substantial interbreeding with Near Eastern dog breeds. Furthermore, beginning around 2,000 years ago, some sites contain artifacts consistent with a Near East origin, like glass beads, but the people show no evidence of having traveled extensively outside Siberia. From this, Feuerborn and colleagues concluded that around 2,000 years ago _____

Which choice most logically completes the text?

- A. dogs and artifacts like glass beads began to be transported from the Near East to Siberia.
- B. people from Siberia began to reach the Near East, where they acquired dogs and artifacts such as glass beads.
- C. glass beads and other artifacts from the Near East began to be exchanged for dogs from Siberia.
- D. dogs from the Near East began to be exchanged for glass beads and other artifacts from Siberia.

ID: 96d1c1fe Answer

Correct Answer: A

Question ID 6df020e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	

ID: 6df020e6

PSAT2 RW 3.15

A study showed that a solar park caused nearby land to cool, though the ecological impact of this temperature decrease isn't yet known. Before the park's construction, the surface temperature 30 meters outside of the park boundary was 0.1°C cooler than that of a control ____ construction, the temperature 30 meters outside of the boundary was 1.7°C cooler than that of the control area.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. area after
- B. area after,
- C. area. After
- D. area, after

ID: 6df020e6 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period is used to correctly mark the boundary between one sentence ("Before...area") and another ("After...area"). The phrase "After construction" modifies the next sentence to indicate that the temperature was cooler after construction.

Choice A is incorrect because it results in a run-on sentence that fuses the two sentences ("Before...area" and "after...area") without punctuation and/or a conjunction. Choice B is incorrect because it results in a run-on sentence that connects the two sentences ("Before...area" and "after...area") without punctuation and/or a conjunction. Additionally, no punctuation is needed between "after" and "construction." Choice D is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between the two sentences ("Before...area" and "after...area").

Question Difficulty: Medium

Question ID 8790d061

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	

ID: 8790d061

PSAT2 RW 3.16

While constructing the Limyra Bridge, ancient Roman builders relied on temporary wooden beams to support the stone and brick structure. These temporary supports, a construction system known as _____ were removed when the bridge was completed.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. falsework;
- B. falsework
- C. falsework,
- D. falsework:

ID: 8790d061 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The convention being tested is the use of punctuation within a sentence. The comma after "falsework" pairs with the comma after "supports" to separate the supplementary element "a construction system known as falsework" from the rest of the sentence. This supplementary element functions to clarify the term "these temporary supports," and the pair of commas indicates that this element could be removed without affecting the grammatical coherence of the sentence.

Choice A is incorrect because a semicolon can't be paired with a comma in this way to separate the supplementary element from the rest of the sentence. Choice B is incorrect because it fails to use appropriate punctuation to separate the supplemental element from the rest of the sentence. Choice D is incorrect because a colon can't be paired with a comma in this way to separate the supplementary element from the rest of the sentence.

Question Difficulty: Medium

Question ID 56315bd0

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Form, Structure, and Sense	

ID: 56315bd0

PSAT2 RW 3.17

Solarpunk is an art movement that imagines renewable energy–powered technology infused complementarily into nature. In Paolo Bacigalupi’s solarpunk short story “Efficiency,” an artificial intelligence that absorbs sustainable energies, redistributing them through intricate networks of weights and generators, _____ Chicago’s energy grid.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. have been powering
- B. power
- C. powers
- D. are powering

ID: 56315bd0 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The convention being tested is subject-verb agreement. The singular verb "powers" agrees in number with the singular subject "an artificial intelligence." Choice A is incorrect because the plural verb "have been powering" doesn’t agree in number with the singular subject "an artificial intelligence." Choice B is incorrect because the plural verb "power" doesn’t agree in number with the singular subject "an artificial intelligence." Choice D is incorrect because the plural verb "are powering" doesn’t agree in number with the singular subject "an artificial intelligence."

Question Difficulty: Medium

Question ID 5ee7fb04

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	■ ■ ■

ID: 5ee7fb04

PSAT2 RW 3.18

In modern plays, actors typically won't acknowledge the _____ do so breaks the fourth wall, a metaphorical barrier between actors and audiences that allows viewers to suspend the knowledge that they're watching a staged performance.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. audience. As to
- B. audience to
- C. audience. To
- D. audience, to

ID: 5ee7fb04 Answer

Correct Answer: C

Rationale

Choice C is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period is used correctly to mark the boundary between one sentence ("In modern...audience") and another ("To do...performance").

Choice A is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "as." Choice B is incorrect because it results in a run-on sentence. The sentences ("In modern...audience" and "To do...performance") are fused without punctuation and/or a conjunction. Choice D is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences.

Question Difficulty: Hard

Question ID 6fac7f45

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	■ ■ ■

ID: 6fac7f45

PSAT2 RW 3.19

Butterfly is a 1988 painting by the Japanese artist Ay-O. Like many of Ay-O’s paintings, *Butterfly*, which portrays a swimmer performing the butterfly stroke, attempts to make use of the entire visual light _____ sporting rainbow-striped goggles, the rainbow-hued swimmer splashes through a wavy rainbow of water.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. spectrum
- B. spectrum:
- C. spectrum while
- D. spectrum, while

ID: 6fac7f45 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The convention being tested is the use of a colon within a sentence. In this choice, the colon correctly introduces the following description of how the painting makes use of the entire visual light spectrum by depicting a rainbow-hued swimmer.

Choice A is incorrect because it results in a run-on sentence. The main clauses ("Like...spectrum" and "sporting...water") are fused without punctuation and/or a conjunction. Choice C is incorrect because it results in a run-on sentence. The main clauses ("Like...spectrum" and "while...water") are fused without punctuation. Furthermore, the conjunction "while" fails to indicate that what follows is a description of how the painting makes use of the entire visual light spectrum. Choice D is incorrect because it results in a logically confusing sentence. The conjunction "while," which suggests that what follows is occurring at the same time as or despite what came before, fails to indicate that what follows is a description of how the painting makes use of the entire visual light spectrum.

Question Difficulty: Hard

Question ID eb95235b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Standard English Conventions	Boundaries	■ ■ ■

ID: eb95235b

PSAT2 RW 3.20

The Limón technique, developed by Mexican-born dancer and choreographer Jose Limón, is known for its emphasis on breath control and its interplay of weight and _____ dancers may explore, for example, the moment of mid-air suspension at the top of a jump.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. weightlessness
- B. weightlessness which
- C. weightlessness,
- D. weightlessness;

ID: eb95235b Answer

Correct Answer: D

Rationale

Choice D is the best answer. The convention being tested is the use of punctuation within a sentence. This choice uses a semicolon in a conventional way to join the first main clause ("The Limón...weightlessness") and the second main clause ("dancers...jump").

Choice A is incorrect because it results in a run-on sentence. The two main clauses are fused without punctuation and/or a conjunction. Choice B is incorrect. The relative pronoun "which" can't be used in this way to join two main clauses. Choice C is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join two main clauses.

Question Difficulty: Hard

Question ID e0bd4f8a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: e0bd4f8a

PSAT2 RW 3.21

In 1942, the 1,500-mile Alaska Highway was constructed in under nine months, largely due to the skilled work of nearly 4,000 African American soldiers from US Army engineering regiments. The soldiers’ contribution was overlooked for decades. _____ in 2017, lawmakers declared October 25 a day of recognition—“Alaska Highway Day”—for the troops who helped build this critical roadway.

Which choice completes the text with the most logical transition?

- A. Lastly,
- B. Then,
- C. Similarly,
- D. For example,

ID: e0bd4f8a Answer

Correct Answer: B

Rationale

Choice B is the best answer. "Then" logically signals that the event described in this sentence—lawmakers’ 2017 declaration of Alaska Highway Day—is part of a chronological sequence of events, occurring after the decades-long period in which the soldiers’ contribution was overlooked.

Choice A is incorrect because "lastly" illogically signals that this sentence presents the last of a series of points or reasons. Instead, it describes a later event occurring in a chronological sequence of events. Choice C is incorrect because "similarly" illogically signals that the information in this sentence is similar to the previous information about the soldiers’ contribution being overlooked for decades. Instead, it describes an event occurring after that decades-long period. Choice D is incorrect because "for example" illogically signals that this sentence provides an example illustrating the previous information about the soldiers’ contribution being overlooked for decades. Instead, it describes an event occurring after that decades-long period.

Question Difficulty: Easy

Question ID 0d088ae0

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Transitions	

ID: 0d088ae0

PSAT2 RW 3.22

Observing that a fire in a closed container soon went out, leading eighteenth-century scientists did not conclude that fresh air (specifically, oxygen) is necessary for combustion; instead, many theorized that the container’s air had become saturated with a substance called phlogiston. _____ when Joseph Priestley first isolated oxygen gas in 1774, he termed it “dephlogisticated air.”

Which choice completes the text with the most logical transition?

- A. In other words,
- B. For this reason,
- C. Alternatively,
- D. Nevertheless,

ID: 0d088ae0 Answer

Correct Answer: B

Rationale

Choice B is the best answer. "For this reason" logically signals that the reason Joseph Priestley termed oxygen "dephlogisticated air" was that he accepted the theory mentioned in the previous sentence—that the presence of phlogiston, rather than the absence of oxygen, causes fire in a closed container to go out.

Choice A is incorrect because "in other words" illogically signals that the information about Priestley terming oxygen "dephlogisticated air" is a restatement of the previous theory concerning phlogiston. Instead, Priestley chose the term as a result of this theory. Choice C is incorrect because "alternatively" illogically signals that Priestley termed oxygen "dephlogisticated air" as an alternative to the previous theory concerning phlogiston. Instead, Priestley chose the term as a result of this theory. Choice D is incorrect because "nevertheless" illogically signals that Priestley termed oxygen "dephlogisticated air" despite the previous theory concerning phlogiston. Instead, Priestley chose the term as a result of this theory.

Question Difficulty: Medium

Question ID 23da9791

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	

ID: 23da9791

PSAT2 RW 3.23

While researching a topic, a student has taken the following notes:

- Scientists have long sought to determine the origin of glass in Chile’s Atacama Desert.
- A 2017 study concluded that ancient grass fires had melted the area’s sandy soil into glass.
- In 2021, a different study revealed that the mineral signatures of glass samples were consistent with the mineral signatures of comet samples collected by NASA.
- That study concluded that the glass had formed as a result of a cometary explosion close to the desert’s surface.

The student wants to describe how scientific understanding about the glass’s origin has evolved. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- Scientists have long sought to determine the origin of the glass, with one study concluding that it formed when ancient
- A. grass fires melted the area’s sandy soil.
 - B. Studies in 2017 and 2021 offered different explanations for the origin of the glass.
 - C. research.
 - D. glass formed as a result of a cometary explosion close to the desert’s surface.

ID: 23da9791 Answer

Correct Answer: D

Rationale

Choice D is the best answer. The sentence describes how scientific understanding of the glass’s origin has evolved, explaining that new research suggests the glass formed as a result of a cometary explosion instead of being caused by grass fires (as was previously believed).

Choice A is incorrect. The sentence explains the conclusion of the older study; it doesn’t describe how scientific understanding of the glass’s origin has evolved. Choice B is incorrect. While the sentence indicates that the two studies provided different explanations, it doesn’t describe how scientific understanding of the glass’s origin has evolved. Choice C is incorrect. The sentence references a recent finding but doesn’t effectively describe how scientific understanding of the glass’s origin has evolved.

Question Difficulty: Medium

Question ID 4c26f18a

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	

ID: 4c26f18a

PSAT2 RW 3.24

While researching a topic, a student has taken the following notes:

- By interlocking their bodies, ants can form bridges to help fellow ants cross gaps.
- In 2020, Yasemin Ozkan-Aydin was inspired by ant behavior to design collaborative quadruped robots.
- Over the course of a year, she designed, built, tested, and refined her robots.
- Each robot is programmed to send a signal to another robot upon encountering a gap in a path.
- The signaled robot connects to the back of the signaler robot via magnetic sensors and pushes it across the gap.

The student wants to begin a narrative about the creation of the robots. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- When one of Yasemin Ozkan-Aydin’s robots encounters a gap in its path, it sends a signal to another robot; the signaled
- A. robot connects to the back of the signaler and pushes it across the gap.
 - B. After a year, Yasemin Ozkan-Aydin had designed, built, tested, and refined her robots.
 - C. set out to design quadruped robots capable of similarly collaborative behavior.
 - D. Ants, which have inspired the design of robots, form bridges by interlocking their bodies.

ID: 4c26f18a Answer

Correct Answer: C

Rationale

Choice C is the best answer. The sentence effectively begins a narrative about the creation of the robots, explaining that ants’ gap-crossing technique inspired Yasemin Ozkan-Aydin to "set out to design" robots that could collaborate in a similar manner.

Choice A is incorrect. The sentence describes how the robots work together to cross gaps; it doesn’t effectively begin a narrative about the creation of the robots. Choice B is incorrect. The sentence provides an overview of the steps Ozkan-Aydin took in creating the robots; it doesn’t effectively begin a narrative about their creation. Choice D is incorrect. The sentence explains how ants form bridges; it doesn’t effectively begin a narrative about the creation of the robots.

Question Difficulty: Medium

Question ID 113f16da

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	

ID: 113f16da

PSAT2 RW 3.25

While researching a topic, a student has taken the following notes:

- Some animals have evolved to physically resemble another animal, plant, or object. This is known as mimicry.
- Crab spiders mimic the appearance of flowers. This helps crab spiders ambush their prey.
- Katydids mimic the appearance of leaves. This helps katydids hide from their predators.

The student wants to emphasize a difference in how katydids and crab spiders use mimicry. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Katydids mimic the appearance of flowers, and crab spiders mimic that of leaves.
- B. Katydids and crab spiders are two examples of animals that use mimicry.
- C. Unlike crab spiders, which use mimicry to ambush prey, katydids use mimicry to hide from predators.
- D. Animals that use mimicry have evolved to resemble another animal, plant, or object.

ID: 113f16da Answer

Correct Answer: C

Rationale

Choice C is the best answer. The sentence uses "unlike" to emphasize a difference in how katydids and crab spiders use mimicry, noting that crab spiders use mimicry to ambush prey while katydids use it to hide from predators.

Choice A is incorrect. While the sentence does contrast katydids and crab spiders, it misrepresents the information in the notes. Katydids mimic the appearance of leaves (not flowers), whereas crab spiders mimic the appearance of flowers (not leaves). Choice B is incorrect. While the sentence indicates that katydids and crab spiders use mimicry, it doesn't emphasize a difference in how they use it. Choice D is incorrect. The sentence describes what mimicry is; it doesn't emphasize a difference in how katydids and crab spiders specifically use mimicry.

Question Difficulty: Medium

Question ID cfade68d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	■ ■ ■

ID: cfade68d

PSAT2 RW 3.26

While researching a topic, a student has taken the following notes:

- Marcela Guerrero is a curator at the Whitney Museum of American Art in New York.
- She curated the Whitney’s 2018 exhibition *Pacha, Llaqta, Wasichay: Indigenous Space, Modern Architecture, New Art*. This exhibition featured works by seven emerging Latino artists.
- She curated the Whitney’s 2020 exhibition *Vida Americana: Mexican Muralists Remake American Art, 1925–1945*. This exhibition included nearly 200 works by twentieth-century Latino and Mexican artists.

The student wants to describe the exhibition that Guerrero curated in 2018. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- Held at the Whitney Museum of American Art, the exhibition *Vida Americana: Mexican Muralists Remake American Art, 1925–1945* included nearly 200 works by twentieth-century Mexican and Latino artists.
- Pacha, Llaqta, Wasichay: Indigenous Space, Modern Architecture, New Art*, an exhibition at the Whitney Museum of American Art, featured works by seven emerging Latino artists.
- In both 2018 and 2020, Marcela Guerrero curated exhibitions at the Whitney Museum of American Art in New York.
- While one exhibition that Marcela Guerrero curated featured works by emerging artists, another included works by twentieth-century artists.

ID: cfade68d Answer

Correct Answer: B

Rationale

Choice B is the best answer. The sentence describes the 2018 exhibition Guerrero curated, noting that the exhibition, which was titled *Pacha, Llaqta, Wasichay: Indigenous Space, Modern Architecture, New Art*, featured the works of seven emerging Latino artists.

Choice A is incorrect. The sentence describes the exhibition Guerrero curated in 2020; it doesn’t describe her 2018 exhibition. Choice C is incorrect. The sentence emphasizes a similarity between the two exhibitions Guerrero curated; it doesn’t describe her 2018 exhibition. Choice D is incorrect. The sentence emphasizes a difference between the two exhibitions Guerrero curated; it doesn’t describe her 2018 exhibition.

Question Difficulty: Hard

Question ID ec03f090

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	■ ■ ■

ID: ec03f090

PSAT2 RW 3.27

While researching a topic, a student has taken the following notes:

- A sestina is a thirty-nine-line poetic form.
- Each line of the poem ends with one of six end words, which alternate according to a set pattern.
- “Forage Sestina” is a sestina by Marilyn Hacker. Its end words are *words, structure, wire, beam, wall, and room*.
- “Towards Autumn” is a sestina by Marilyn Hacker. Its end words are *daughter, friend, bread, mother, lover, and myself*.

The student wants to use one of the poems to illustrate the sestina form. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- Hacker employs the sestina, a poetic form with thirty-nine lines and six end words, in both “Forage Sestina” and “Towards A. Autumn.”
- As a sestina, “Towards Autumn” contains thirty-nine lines and six end words—in this case, *daughter, friend, bread, mother, B. lover, and myself*—that alternate in a set pattern.
- The thirty-nine-line sestina form uses the words *daughter, friend, bread, mother, lover, and myself*, which are found in the C. poem “Forage Sestina.”
- Hacker has used the sestina form multiple times, as in “Towards Autumn,” which contains these six words: *words, D. structure, wire, beam, wall, and room*.

ID: ec03f090 Answer

Correct Answer: B

Rationale

Choice B is the best answer. The sentence uses the poem "Towards Autumn" to illustrate the form of a sestina, explaining that a sestina’s thirty-nine lines all end in one of six alternating words and showcasing that poem’s specific end words.

Choice A is incorrect. The sentence identifies both of Hacker’s poems as sestinas; it doesn’t use one of the poems to illustrate the form of a sestina. Choice C is incorrect. While the sentence appears to use the poem "Forage Sestina" to illustrate a feature of the sestina form, it misrepresents the information in the notes. According to the notes, "Forage Sestina" doesn’t use these six end words; it uses six other words instead. Choice D is incorrect. While the sentence uses the poem "Towards Autumn" as an example of one of Hacker’s sestinas, it misrepresents the information in the notes. According to the notes, "Towards Autumn" doesn’t use these six end words; it uses six other words instead.

Question Difficulty: Hard

Question ID b52e5b6f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: b52e5b6f

PSAT2 M 1.1

A mixture consisting of only vitamin D and calcium has a total mass of **150** grams. The mass of vitamin D in the mixture is **50** grams. What is the mass, in grams, of calcium in the mixture?

- A. 200
- B. 150
- C. 100
- D. 50

ID: b52e5b6f Answer

Correct Answer: C

Rationale

Choice C is correct. Let d represent the mass, in grams, of vitamin D in the mixture, and let c represent the mass, in grams, of calcium in the mixture. It's given that the mixture consists of only vitamin D and calcium and that the total mass of the mixture is 150 grams. Therefore, the equation $d + c = 150$ represents this situation. It's also given that the mass of vitamin D in the mixture is 50 grams. Substituting 50 for d in the equation $d + c = 150$ yields $50 + c = 150$. Subtracting 50 from both sides of this equation yields $c = 100$. Therefore, the mass of calcium in the mixture is 100 grams.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the total mass, in grams, of the mixture, not the mass, in grams, of calcium in the mixture.

Choice D is incorrect. This is the mass, in grams, of vitamin D in the mixture, not the mass, in grams, of calcium in the mixture.

Question Difficulty: Easy

Question ID fa80893a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: fa80893a

PSAT2 M 1.2

If $2x + 3 = 9$, what is the value of $6x - 1$?

ID: fa80893a Answer

Correct Answer: 17

Rationale

The correct answer is 17. It’s given that $2x + 3 = 9$. Multiplying each side of this equation by 3 yields $3(2x + 3) = 3(9)$, or $6x + 9 = 27$. Subtracting 10 from each side of this equation yields $6x + 9 - 10 = 27 - 10$, or $6x - 1 = 17$. Therefore, the value of $6x - 1$ is 17.

Question Difficulty: Easy

Question ID a478f9f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: a478f9f5

PSAT2 M 1.3

Each of **157** gemstones can be classified as one of three classifications, as shown in the frequency table.

Classification	Frequency
color X	119
color Y	3
color Z	35

If one of the gemstones is selected at random, what is the probability of selecting a gemstone of color Y?

- A. $\frac{3}{157}$
- B. $\frac{35}{157}$
- C. $\frac{119}{157}$
- D. $\frac{154}{157}$

ID: a478f9f5 Answer

Correct Answer: A

Rationale

Choice A is correct. If one of the gemstones is selected at random, the probability of selecting a gemstone of color Y is equal to the number of gemstones of color Y divided by the total number of gemstones. According to the table, there are 3 gemstones of color Y, and it's given that the total number of gemstones is 157. Therefore, if one of the gemstones is selected at random, the probability of selecting a gemstone of color Y is $\frac{3}{157}$.

Choice B is incorrect. This is the probability of selecting a gemstone of color Z.

Choice C is incorrect. This is the probability of selecting a gemstone of color X.

Choice D is incorrect. This is the probability of selecting a gemstone that's not of color Y.

Question Difficulty: Easy

Question ID ed856e9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: ed856e9c

PSAT2 M 1.4

What is the y-intercept of the graph of $y = 34x + 81$ in the xy-plane?

- A. $(0, 81)$
- B. $(0, 34)$
- C. $(0, -34)$
- D. $(0, -81)$

ID: ed856e9c Answer

Correct Answer: A

Rationale

Choice A is correct. In the xy-plane, the graph of an equation in the form $y = mx + b$, where m and b are constants, has a slope of m and a y-intercept of $(0, b)$. Therefore, the y-intercept of the graph of $y = 34x + 81$ is $(0, 81)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

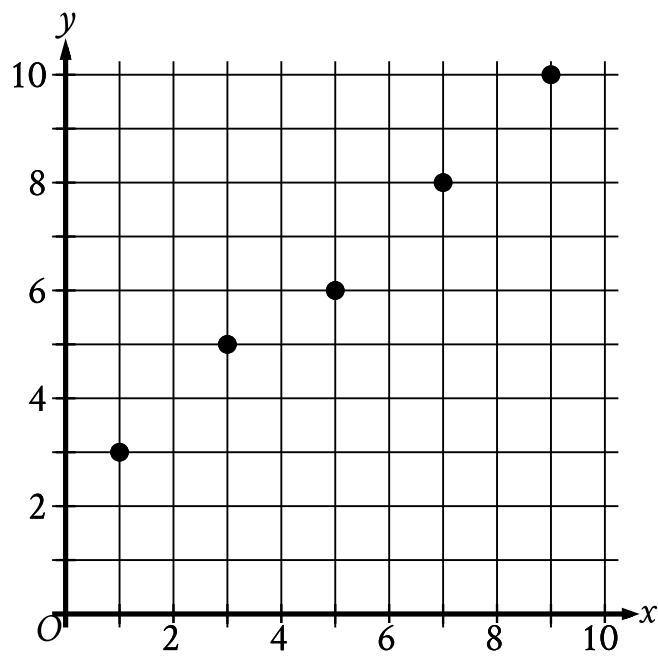
Question ID 16988f9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 16988f9c

PSAT2 M 1.5

The scatterplot shows the relationship between two variables, x and y .



Which equation is the most appropriate linear model for this relationship?

- A. $y = -0.9x - 2.2$
- B. $y = -0.9x + 2.2$
- C. $y = -0.9x$
- D. $y = 0.9x + 2.2$

ID: 16988f9c Answer

Correct Answer: D

Rationale

Choice D is correct. A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The graph of an appropriate linear model for this relationship passes near the points $(1, 3)$ and $(9, 10)$ in the xy -plane. Two points on a line, x_1, y_1 and x_2, y_2 , can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(1, 3)$ and $(9, 10)$ for x_1, y_1 and x_2, y_2 , respectively, in the slope formula yields $m = \frac{10 - 3}{9 - 1}$, or $m = 0.875$. Therefore, the value of m for an appropriate linear model is approximately 0.875. Substituting 0.875 for m in $y = mx + b$ yields $y = 0.875x + b$. Since an appropriate linear model passes near the point $(1, 3)$, the approximate value of b can be found by substituting 1 for x and 3 for y in the equation $y = 0.875x + b$, which yields $3 = (0.875)(1) + b$, or $3 = 0.875 + b$. Subtracting 0.875 from both sides of this equation yields $2.125 = b$. Therefore, the value of b for an appropriate linear model is approximately 2.125. Thus, of the given choices, $y = 0.9x + 2.2$ is the most appropriate linear model for this relationship.

Alternate approach: A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The scatterplot shows that as the x -values of the data points increase, the y -values of the data points increase, which means the graph of an appropriate linear model has a positive slope. Of the given choices, $y = 0.9x + 2.2$ is the only linear model whose graph has a positive slope.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice C is incorrect. The graph of this model has a negative slope, not a positive slope.

Question Difficulty: Easy

Question ID 0700a2d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 0700a2d5

PSAT2 M 1.6

How many yards are equivalent to 77 rods? (5.5 yards = 1 rod)

ID: 0700a2d5 Answer

Correct Answer: 423.5, 847/2

Rationale

The correct answer is 423.5. It's given that 5.5 yards = 1 rod. Therefore, 77 rods is equivalent to $77 \text{ rods} \times \frac{5.5 \text{ yards}}{1 \text{ rod}}$, or 423.5 yards. Note that 423.5 and 847/2 are examples of ways to enter a correct answer.

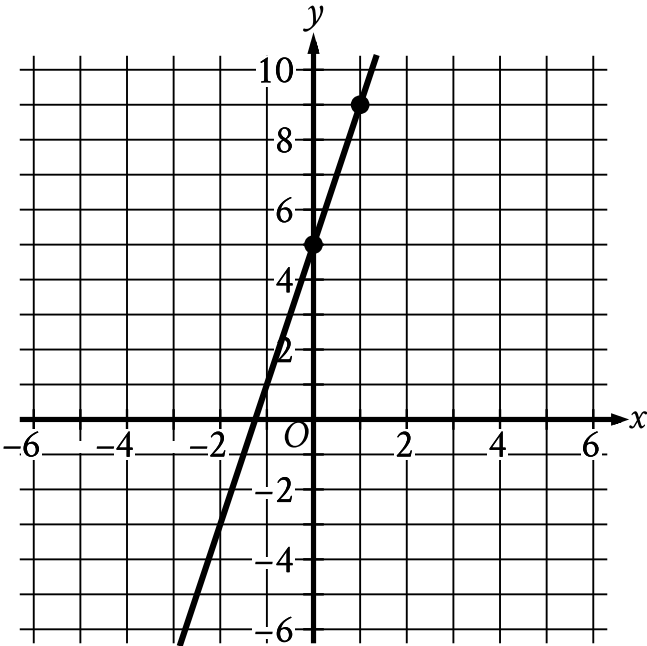
Question Difficulty: Easy

Question ID b8fa27db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: b8fa27db

PSAT2 M 1.7



Line j is shown in the xy -plane. Line k (not shown) is parallel to line j . What is the slope of line k ?

ID: b8fa27db Answer

Correct Answer: 4

Rationale

The correct answer is 4. It's given that line k is parallel to line j . It follows that the slope of line k is equal to the slope of line j . Given two points on a line in the xy -plane, x_1, y_1 and x_2, y_2 , the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. In the xy -plane shown, the points $0, 5$ and $1, 9$ are on line j . It follows that the slope of line j is $\frac{9 - 5}{1 - 0}$, or 4. Since the slope of line j is equal to the slope of line k , the slope of line k is also 4.

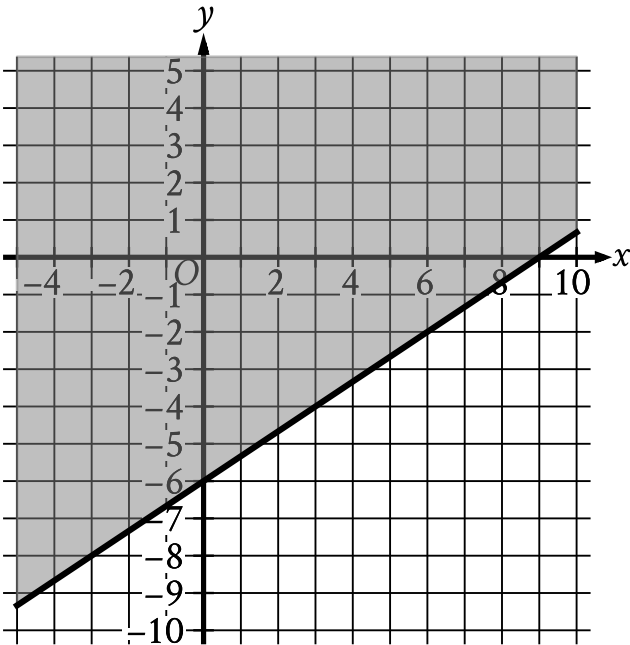
Question Difficulty: Easy

Question ID c9355dec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: c9355dec

PSAT2 M 1.8



The shaded region shown represents the solutions to which inequality?

- A. $y \geq \frac{2}{3}x - 6$
- B. $y \geq \frac{2}{3}x + 6$
- C. $y \geq \frac{2}{3}x - 9$
- D. $y \geq \frac{2}{3}x + 9$

ID: c9355dec Answer

Correct Answer: A

Rationale

Choice A is correct. The equation for the line representing the boundary of the shaded region can be written in slope-intercept form $y = mx + b$, where m is the slope and b is the y -intercept of the line. For the graph shown, the boundary line passes through the points $(0, -6)$ and $(9, 0)$. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, -6)$ and $(9, 0)$ for (x_1, y_1) and (x_2, y_2) , respectively, in this equation yields $m = \frac{0 - (-6)}{9 - 0}$, which is equivalent to $m = \frac{6}{9}$, or $m = \frac{2}{3}$. Since the point $(0, -6)$ represents the y -intercept, it follows that $b = -6$. Substituting $\frac{2}{3}$ for m and -6 for b in the equation $y = mx + b$ yields $y = \frac{2}{3}x - 6$ as the equation of the boundary line. Since the shaded region represents all the points on and above this boundary line, it follows that the shaded region shown represents the solutions to the inequality $y \geq \frac{2}{3}x - 6$.

Choice B is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, 6)$, not $(0, -6)$.

Choice C is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, -9)$, not $(0, -6)$.

Choice D is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, 9)$, not $(0, -6)$.

Question Difficulty: Easy

Question ID e86f0651

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e86f0651

PSAT2 M 1.9

A circle has a radius of **43** meters. What is the area, in square meters, of the circle?

- A. $\frac{43\pi}{2}$
- B. 43π
- C. 86π
- D. $1,849\pi$

ID: e86f0651 Answer

Correct Answer: D

Rationale

Choice D is correct. The area, A , of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the circle has a radius of 43 meters. Substituting 43 for r in the formula $A = \pi r^2$ yields $A = \pi 43^2$, or $A = 1,849\pi$. Therefore, the area, in square meters, of the circle is $1,849\pi$.

Choice A is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{\frac{43}{2}}$ meters.

Choice B is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{43}$ meters.

Choice C is incorrect. This is the circumference, in meters, of the circle.

Question Difficulty: Medium

Question ID 3a3b95df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3a3b95df

PSAT2 M 1.10

$d = 16 - \frac{x}{30}$

The equation shown gives the estimated amount of diesel d , in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \leq x \leq 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

- A. 0
- B. 6
- C. 14
- D. 16

ID: 3a3b95df Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $d = 16 - \frac{x}{30}$ gives the estimated amount of diesel d , in gallons, that remains in the gas tank of the truck after being driven x miles. Substituting 300 for x in the given equation yields $d = 16 - \frac{300}{30}$, which is equivalent to $d = 16 - 10$, or $d = 6$. Therefore, the estimated amount of diesel that remains in the gas tank of the truck when $x = 300$ is 6 gallons.

Choice A is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 480$, not when $x = 300$.

Choice C is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 60$, not when $x = 300$.

Choice D is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 0$, not when $x = 300$.

Question Difficulty: Easy

Question ID 739f1bbc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 739f1bbc

PSAT2 M 1.11

In triangle ABC , $AB = 4,680$ millimeters (mm) and $BC = 4,680$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

- A. $AC = 4,680$ mm
- B. $AC = 468$ mm
- C. $AC = 46.8$ mm
- D. $AC = 4.68$ mm

ID: 739f1bbc Answer

Correct Answer: A

Rationale

Choice A is correct. In an equilateral triangle, all three sides have the same length. It’s given that in triangle ABC , $AB = 4,680$ mm and $BC = 4,680$ mm. Therefore, if $AC = 4,680$ mm, then all three sides of triangle ABC have the same length, so triangle ABC is equilateral. Therefore, $AC = 4,680$ mm is sufficient to prove that triangle ABC is equilateral.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 07bcecac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 07bcecac

PSAT2 M 1.12

$$P(t) = 24.8(1.036)^t$$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement “ $P(8)$ is approximately equal to 32.91”?

- A. In 1984, the predicted population of this country was approximately 8 million.
- B. In 1984, the predicted population of this country was approximately 32.91 million.
- C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.
- D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.

ID: 07bcecac Answer

Correct Answer: C

Rationale

Choice C is correct. The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. Since the value of $P8$ is the value of Pt when $t = 8$, it follows that “ $P8$ is approximately equal to 32.91” means that the value of Pt is approximately equal to 32.91 when $t = 8$. Therefore, the best interpretation of the statement “ $P8$ is approximately equal to 32.91” is that 8 years after 1984, the predicted population of this country was approximately 32.91 million.

Choice A is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 8 million.

Choice B is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 32.91 million.

Choice D is incorrect. 32.91 years after 1984, the predicted population of this country was $24.8(1.036)^{32.91}$ million, or approximately 79.42 million, not approximately 8 million.

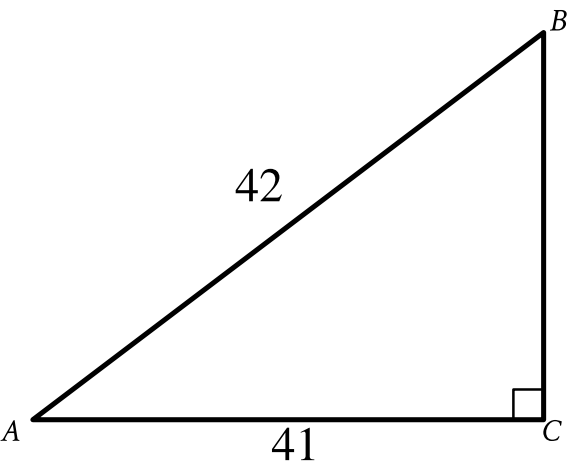
Question Difficulty: Easy

Question ID 2bddbc1b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 2bddbc1b

PSAT2 M 1.13



Note: Figure not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

- A. $\frac{42}{41}$
- B. $\frac{41}{42}$
- C. $\frac{1}{42}$
- D. $\frac{1}{41}$

ID: 2bddbc1b Answer

Correct Answer: B

Rationale

Choice B is correct. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to angle A is 41, and the length of the hypotenuse is 42. Therefore, $\cos A = \frac{41}{42}$.

Choice A is incorrect. This is the value of $\frac{1}{\cos A}$. Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 79cf8505

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 79cf8505

PSAT2 M 1.14

A line segment that has a length of **115 centimeters (cm)** is divided into three parts. One part is **47 cm** long. The other two parts have lengths that are equal to each other. What is the length, in **cm**, of one of the other two parts of equal length?

ID: 79cf8505 Answer

Correct Answer: 34

Rationale

The correct answer is 34. It’s given that a line segment has a length of 115 cm and is divided into three parts, where one part is 47 cm long and the other two parts have lengths that are equal. If x represents the length, in cm, of each of the two parts of equal length, then the equation $47 + x + x = 115$, or $47 + 2x = 115$, represents this situation. Subtracting 47 from each side of this equation yields $2x = 68$. Dividing each side of this equation by 2 yields $x = 34$. Therefore, the length, in cm, of one of the two parts of equal length is 34.

Question Difficulty: Medium

Question ID a29e89fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: a29e89fc

PSAT2 M 1.15

The list gives the mass, in grams, of 5 alpine marmots. 4,010; 4,010; 3,030; 4,050; 3,050
What is the mean mass, in grams, of these 5 alpine marmots?

ID: a29e89fc Answer

Correct Answer: 3630

Rationale

The correct answer is 3,630. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The sum of the masses, in grams, of these alpine marmots is $4,010 + 4,010 + 3,030 + 4,050 + 3,050$, or 18,150 grams. The number of alpine marmots in the data set is 5. Therefore, the mean mass, in grams, of these 5 alpine marmots is $\frac{18,150}{5}$, or 3,630.

Question Difficulty: Medium

Question ID b4acba95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b4acba95

PSAT2 M 1.16

$x^2 - 12x + 27 = 0$ How many distinct real solutions does the given equation have?

- A. Exactly two
- B. Exactly one
- C. Zero
- D. Infinitely many

ID: b4acba95 Answer

Correct Answer: A

Rationale

Choice A is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $x^2 - 12x + 27 = 0$, $a = 1$, $b = -12$, and $c = 27$. Substituting these values for a , b , and c in $b^2 - 4ac$ yields $-12^2 - 4(1)(27)$, or $-144 - 108$, or -252 . Since the value of its discriminant is negative, the given equation has zero distinct real solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 946ab892

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 946ab892

PSAT2 M 1.17

For the linear function g , the graph of $y = g(x)$ in the xy -plane has a slope of 2 and passes through the point $(1, 14)$. Which equation defines g ?

- A. $g(x) = 2x$
- B. $g(x) = 2x + 2$
- C. $g(x) = 2x + 12$
- D. $g(x) = 2x + 14$

ID: 946ab892 Answer

Correct Answer: C

Rationale

Choice C is correct. An equation defining a linear function can be written in the form $gx = mx + b$, where m is the slope and $0, b$ is the y -intercept of the graph of $y = gx$ in the xy -plane. It's given that the graph of $y = gx$ has a slope of 2 . Therefore, $m = 2$. It's also given that the graph of $y = gx$ passes through the point $1, 14$. It follows that when $x = 1$, $gx = 14$. Substituting 1 for x , 14 for gx , and 2 for m in the equation $gx = mx + b$ yields $14 = 2(1) + b$, or $14 = 2 + b$. Subtracting 2 from each side of this equation yields $12 = b$. Therefore, $b = 12$. Substituting 2 for m and 12 for b in the equation $gx = mx + b$ yields $gx = 2x + 12$. Therefore, the equation that defines g is $gx = 2x + 12$.

Choice A is incorrect. For this function, the graph of $y = gx$ in the xy -plane passes through the point $1, 2$, not $1, 14$.

Choice B is incorrect. For this function, the graph of $y = gx$ in the xy -plane passes through the point $1, 4$, not $1, 14$.

Choice D is incorrect. For this function, the graph of $y = gx$ in the xy -plane passes through the point $1, 16$, not $1, 14$.

Question Difficulty: Medium

Question ID 1b403590

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 1b403590

PSAT2 M 1.18

An object has a mass of **168** grams and a volume of **24** cubic centimeters. What is the density, in grams per cubic centimeter, of the object?

- A. **7**
- B. **144**
- C. **192**
- D. **4,032**

ID: 1b403590 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the object has a mass of 168 grams and a volume of 24 cubic centimeters. Dividing the mass, in grams, of the object by the volume, in cubic centimeters, of the object gives the density, in grams per cubic centimeter, of the object. It follows that the density of the object is $\frac{168 \text{ grams}}{24 \text{ cubic centimeters}}$, which is equivalent to $\frac{168}{24}$ grams per cubic centimeter, or 7 grams per cubic centimeter.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 02add2d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 02add2d2

PSAT2 M 1.19

A company has a newsletter. In January 2018, there were 1,300 customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c , subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

- A. $c = 1,300^{msup}$
- B. $c = 1,300^{msup}$
- C. $c = 1,300^{msup}$
- D. $c = 1,300^{msup}$

ID: 02add2d2 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that in January 2018, there were 1,300 customers subscribed to a company's newsletter and for the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. It follows that this situation can be represented by the equation $c = a1 + \frac{r}{100}^m$, where c is the total number of customers subscribed to the company's newsletter m months after January 2018, a is the number of customers subscribed to the newsletter in January 2018, and the total number of customers subscribed to the newsletter each month was $r\%$ greater than the total number subscribed the previous month. Substituting 1,300 for a and 7 for r in this equation yields $c = 1,3001 + \frac{7}{100}^m$, or $c = 1,3001.07^m$.

Choice A is incorrect. This equation represents a situation where the total number of customers subscribed each month was 93% less, not 7% greater, than the total number subscribed the previous month.

Choice C is incorrect. This equation represents a situation where the total number of customers subscribed each month was 70%, not 7%, greater than the total number subscribed the previous month.

Choice D is incorrect. This equation represents a situation where the total number of customers subscribed each month was 600%, not 7%, greater than the total number subscribed the previous month.

Question Difficulty: Medium

Question ID 39714777

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 39714777

PSAT2 M 1.20

$p(x) + 57 = x^2$

The given equation relates the value of x and its corresponding value of $p(x)$ for the function p . What is the minimum value of the function p ?

- A. $-3,249$
- B. -57
- C. 57
- D. $3,249$

ID: 39714777 Answer

Correct Answer: B

Rationale

Choice B is correct. For a quadratic function defined by an equation of the form $px = ax - h^2 + k$, where a , h , and k are constants and $a > 0$, the minimum value of the function is k . Subtracting 57 from both sides of the given equation yields $px = x^2 - 57$. This function is in the form $px = ax - h^2 + k$, where $a = 1$, $h = 0$, and $k = -57$. Therefore, the minimum value of the function p is -57.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 1baffbcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 1baffbcf

PSAT2 M 1.21

The expression $0.35x$ represents the result of decreasing a positive quantity x by what percent?

- A. 3.5%
- B. 35%
- C. 6.5%
- D. 65%

ID: 1baffbcf Answer

Correct Answer: D

Rationale

Choice D is correct. Let $n\%$ represent the percent by which the positive quantity x is decreased to result in $0.35x$. The value of n can be found by solving the equation $x - \frac{n}{100}x = 0.35x$. Since x is a common factor of each of the terms on the left-hand side of this equation, the equation can be rewritten as $x(1 - \frac{n}{100}) = 0.35x$. Dividing each side of this equation by x yields $1 - \frac{n}{100} = 0.35$. Multiplying each side of this equation by 100 yields $100 - n = 35$. Subtracting 100 from each side of this equation yields $-n = -65$. Dividing each side of this equation by -1 yields $n = 65$. Therefore, the expression $0.35x$ represents the result of decreasing the positive quantity x by 65%.

Choice A is incorrect. Decreasing the quantity x by 3.5% yields $x - 0.035x$, or $0.965x$, not $0.35x$.

Choice B is incorrect. Decreasing the quantity x by 35% yields $x - 0.35x$, or $0.65x$, not $0.35x$.

Choice C is incorrect. Decreasing the quantity x by 6.5% yields $x - 0.065x$, or $0.935x$, not $0.35x$.

Question Difficulty: Hard

Question ID 7866a908

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■ ■ ■

ID: 7866a908

PSAT2 M 1.22

A sample of a certain alloy has a total mass of **50.0** grams and is **50.0%** silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was **30.0%** silicon by mass and the second piece was **80.0%** silicon by mass. What was the mass, in grams, of the silicon in the second piece?

- A. **9.0**
- B. **16.0**
- C. **20.0**
- D. **30.0**

ID: 7866a908 Answer

Correct Answer: B

Rationale

Choice B is correct. Let x represent the total mass, in grams, of the first piece, and let y represent the total mass, in grams, of the second piece. It's given that the sample has a total mass of 50.0 grams. Therefore, the equation $x + y = 50.0$ represents this situation. It's also given that the sample is 50.0% silicon by mass. Therefore, the total mass of the silicon in the sample is 0.500×50.0 , or 25.0, grams. It's also given that the first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. Therefore, the masses, in grams, of the silicon in the first and second pieces can be represented by the expressions $0.300x$ and $0.800y$, respectively. Since the sample was created by combining the first and second pieces, and the total mass of the silicon in the sample is 25.0 grams, the equation $0.300x + 0.800y = 25.0$ represents this situation. Subtracting y from both sides of the equation $x + y = 50.0$ yields $x = 50.0 - y$. Substituting $50.0 - y$ for x in the equation $0.300x + 0.800y = 25.0$ yields $0.300(50.0 - y) + 0.800y = 25.0$. Distributing 0.300 on the left-hand side of this equation yields $15.0 - 0.300y + 0.800y = 25.0$. Combining like terms on the left-hand side of this equation yields $15.0 + 0.500y = 25.0$. Subtracting 15.0 from both sides of this equation yields $0.500y = 10.0$. Dividing both sides of this equation by 0.500 yields $y = 20.0$. Substituting 20.0 for y in the expression representing the mass, in grams, of the silicon in the second piece, $0.800y$, yields $0.800(20.0)$, or 16.0. Therefore, the mass, in grams, of the silicon in the second piece is 16.0.

Choice A is incorrect. This is the mass, in grams, of the silicon in the first piece, not the second piece.

Choice C is incorrect. This is the total mass, in grams, of the second piece, not the mass, in grams, of the silicon in the second piece.

Choice D is incorrect. This is the total mass, in grams, of the first piece, not the mass, in grams, of the silicon in the second piece.

Question Difficulty: Hard

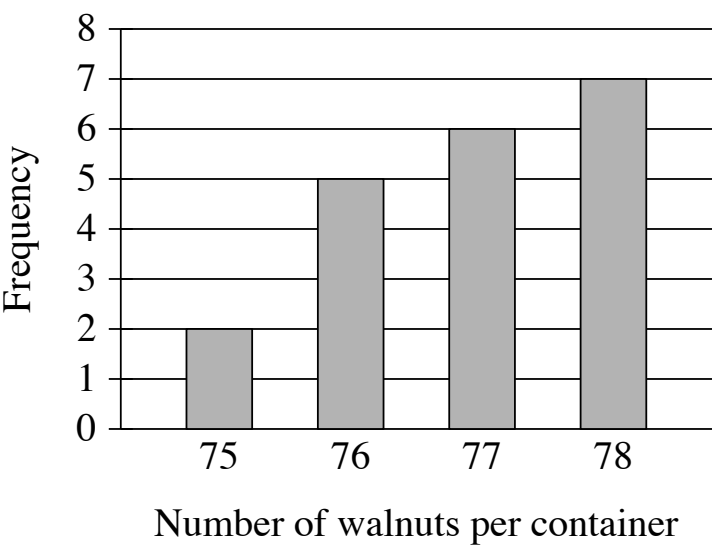
Question ID fe6a49d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: fe6a49d6

PSAT2 M 2.1

The bar graph shows the distribution of the number of walnuts per container for **20** containers at a grocery store.



How many of these containers of walnuts contain exactly **78** walnuts?

- A. **2**
- B. **7**
- C. **20**
- D. **78**

ID: fe6a49d6 Answer

Correct Answer: B

Rationale

Choice B is correct. The height of each bar in the graph shown represents the number of containers that contain the number of walnuts specified at the bottom of the bar. The bar for 78 walnuts has a height of 7. Therefore, 7 of these containers of walnuts contain exactly 78 walnuts.

Choice A is incorrect. This is the number of containers that contain exactly 75 walnuts, not 78 walnuts.

Choice C is incorrect. This is the total number of containers of walnuts represented in the bar graph, not the number that contain exactly 78 walnuts.

Choice D is incorrect. This is the number of walnuts in a container that contains exactly 78 walnuts, not the number of containers that contain exactly 78 walnuts.

Question Difficulty: Easy

Question ID 771ee744

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 771ee744

PSAT2 M 2.2

There are **320** marbles in a container. Of these marbles, **10%** are red. How many marbles in the container are red?

- A. **32**
- B. **288**
- C. **320**
- D. **352**

ID: 771ee744 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that 10% of the 320 marbles in a container are red. Therefore, the number of red marbles can be calculated by multiplying the number of marbles in the container by $\frac{10}{100}$, which gives $320\frac{10}{100}$, or 32.

Choice B is incorrect. This is the number of marbles in the container that aren't red.

Choice C is incorrect. This is the total number of marbles in the container.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID fe4c1c9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: fe4c1c9e

PSAT2 M 2.3

A mechanical device in a workshop produces items at a constant rate of **60** items per hour. At this rate, how many items will the mechanical device produce in **3** hours?

ID: fe4c1c9e Answer

Correct Answer: 180

Rationale

The correct answer is 180. It's given that a mechanical device produces items at a constant rate of 60 items per hour. This rate can be written as $\frac{60 \text{ items}}{1 \text{ hour}}$. Let x represent the number of items the mechanical device will produce in 3 hours at the given rate. It follows that $\frac{60 \text{ items}}{1 \text{ hour}} = \frac{x \text{ items}}{3 \text{ hours}}$, which can be written as $\frac{60}{1} = \frac{x}{3}$, or $60 = \frac{x}{3}$. Multiplying each side of this equation by 3 yields $180 = x$. Therefore, at the given rate, the mechanical device will produce 180 items in 3 hours.

Alternate approach: It's given that a mechanical device produces items at a constant rate of 60 items per hour. At this rate, the mechanical device will produce $\frac{60 \text{ items}}{1 \text{ hour}} \cdot 3 \text{ hours}$, or 180 items in 3 hours.

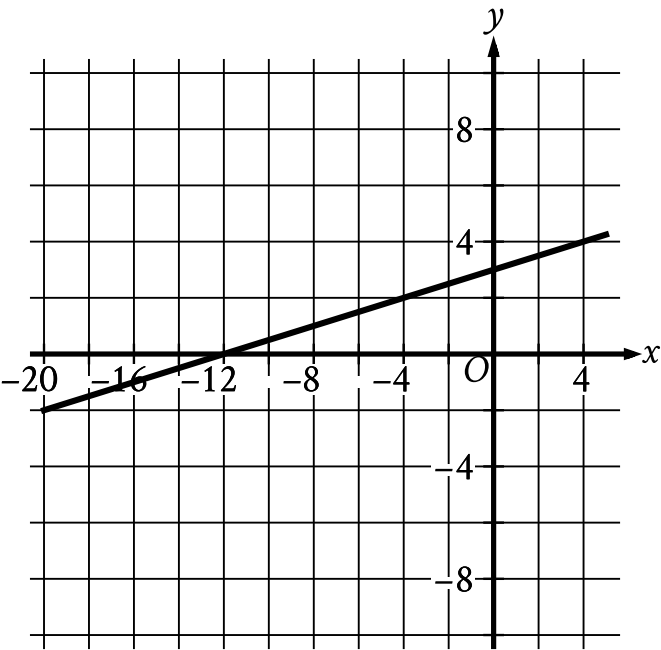
Question Difficulty: Easy

Question ID c10ad793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c10ad793

PSAT2 M 2.4



The graph of the linear function f is shown, where $y = f(x)$. What is the x-intercept of the graph of f ?

- A. $(-12, 0)$
- B. $(0, 0)$
- C. $(\frac{1}{4}, 0)$
- D. $(12, 0)$

ID: c10ad793 Answer

Correct Answer: A

Rationale

Choice A is correct. The x-intercept of a graph is the point where the graph intersects the x-axis. The graph of function f , where $y = f(x)$, intersects the x-axis at -12, 0. Therefore, the x-intercept of the graph of f is -12, 0.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c6a26e14

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: c6a26e14

PSAT2 M 2.5

$|x + 45| = 48$ What is the positive solution to the given equation?

- A. 3
- B. 48
- C. 93
- D. 96

ID: c6a26e14 Answer

Correct Answer: A

Rationale

Choice A is correct. The given absolute value equation can be rewritten as two linear equations: $x + 45 = 48$ and $-x + 45 = 48$, or $x + 45 = -48$. Subtracting 45 from both sides of the equation $x + 45 = 48$ yields $x = 3$. Subtracting 45 from both sides of the equation $x + 45 = -48$ yields $x = -93$. Thus, the given equation has two possible solutions, 3 and -93. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

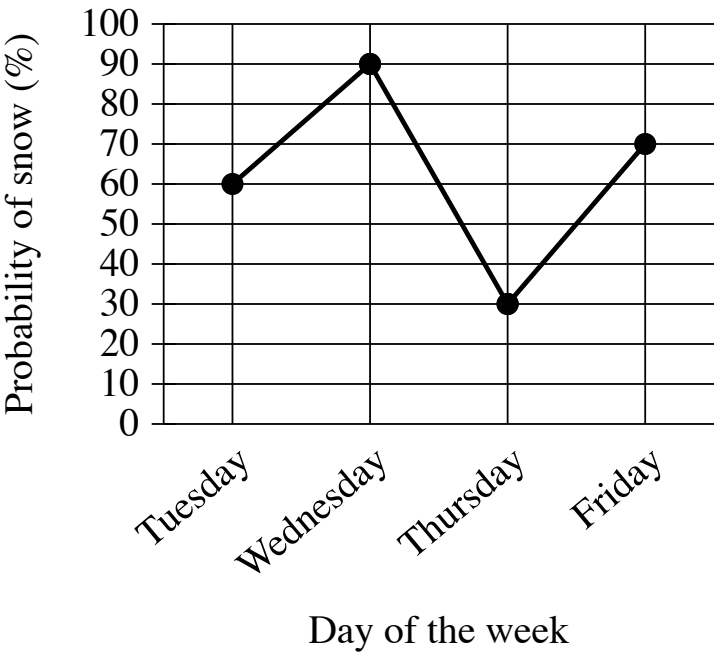
Question Difficulty: Easy

Question ID 13f67ddc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 13f67ddc

PSAT2 M 2.6



The line graph shows the probability of snow, as a percent, at a certain location for each day during a four-day period. According to the line graph, for which day during this four-day period is the probability of snow **30%**?

- A. Tuesday
- B. Wednesday
- C. Thursday
- D. Friday

ID: 13f67ddc Answer

Correct Answer: C

Rationale

Choice C is correct. For the line graph shown, the probability of snow, as a percent, is represented on the vertical axis. According to the line graph, during this four-day period, the probability of snow is 30% for Thursday.

Choice A is incorrect. The probability of snow on Tuesday is 60%.

Choice B is incorrect. The probability of snow on Wednesday is 90%.

Choice D is incorrect. The probability of snow on Friday is 70%.

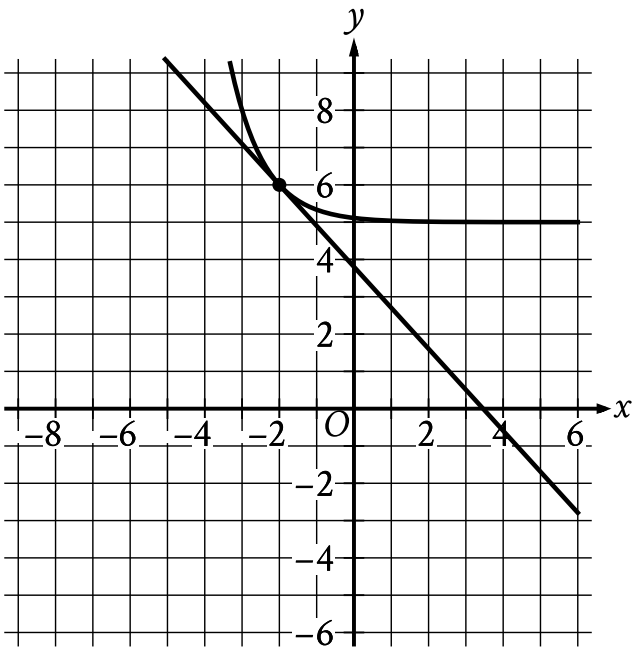
Question Difficulty: Easy

Question ID 5c7d5744

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5c7d5744

PSAT2 M 2.7



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

- A. $(6, 0)$
- B. $(-2, 6)$
- C. $(0, -2)$
- D. $(0, 0)$

ID: 5c7d5744 Answer

Correct Answer: B

Rationale

Choice B is correct. The solution x, y to the system of two equations corresponds to the point where the graphs of the equations intersect in the xy -plane. The graphs of the linear equation and the nonlinear equation shown intersect at the point $-2, 6$. Thus, the solution x, y to this system is $-2, 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e0177f5f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: e0177f5f

PSAT2 M 2.8

$x = 4$ $y = 5 - x$ The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 1
- B. 4
- C. 5
- D. 9

ID: e0177f5f Answer

Correct Answer: A

Rationale

Choice A is correct. The first equation in the given system of equations is $x = 4$. Substituting 4 for x in the second equation in the given system of equations yields $y = 5 - 4$, or $y = 1$.

Choice B is incorrect. This is the value of x in the solution to the given system of equations, not the value of y .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e006209c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e006209c

PSAT2 M 2.9

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

- A. **130**
- B. **63**
- C. **4**
- D. **0**

ID: e006209c Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the geologist has already collected 63 samples from the volcano. Let x represent the number of additional samples the geologist needs to collect. After collecting x additional samples, the geologist will have collected a total of $63 + x$ samples. It's given that the geologist needs to collect at least 67 samples. Therefore, $63 + x \geq 67$. Subtracting 63 from each side of this inequality yields the inequality $x \geq 4$. Thus, the geologist needs to collect a minimum of 4 additional samples.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of samples the geologist has already collected, rather than the minimum number of additional samples the geologist needs to collect.

Choice D is incorrect. If the geologist collects 0 additional samples, the geologist will have collected a total of 63 samples, which is less than 67 samples.

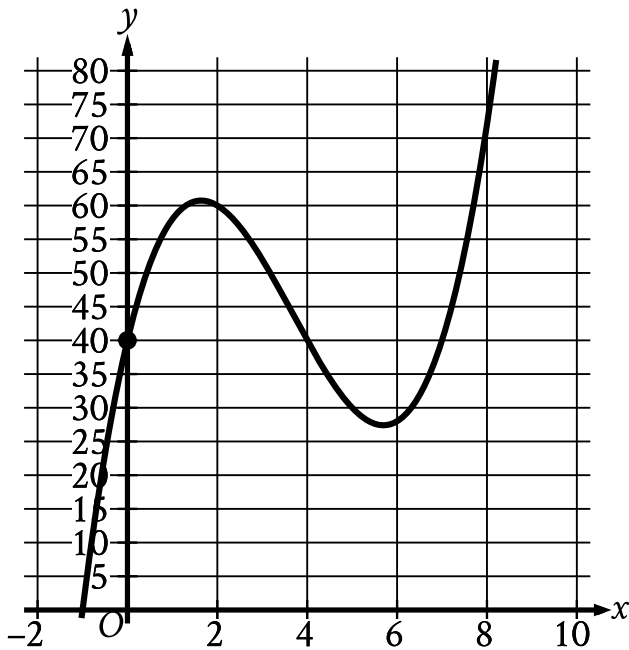
Question Difficulty: Easy

Question ID 26f5269a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 26f5269a

PSAT2 M 2.10



The y -intercept of the graph shown is (x, y) . What is the value of y ?

ID: 26f5269a Answer

Correct Answer: 40

Rationale

The correct answer is 40. The y -intercept of a graph in the xy -plane is the point x, y on the graph where $x = 0$. The y -intercept of the graph shown is 0, 40. Therefore, the value of y is 40.

Question Difficulty: Easy

Question ID f2b63f49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: f2b63f49

PSAT2 M 2.11

$8x - 7x + 130 = 260$ What value of x is the solution to the given equation?

ID: f2b63f49 Answer

Correct Answer: 130

Rationale

The correct answer is 130. It’s given that $8x - 7x + 130 = 260$. Combining like terms on the left-hand side of this equation yields $x + 130 = 260$. Subtracting 130 from each side of this equation yields $x = 130$. Therefore, the value of x that’s the solution to the given equation is 130.

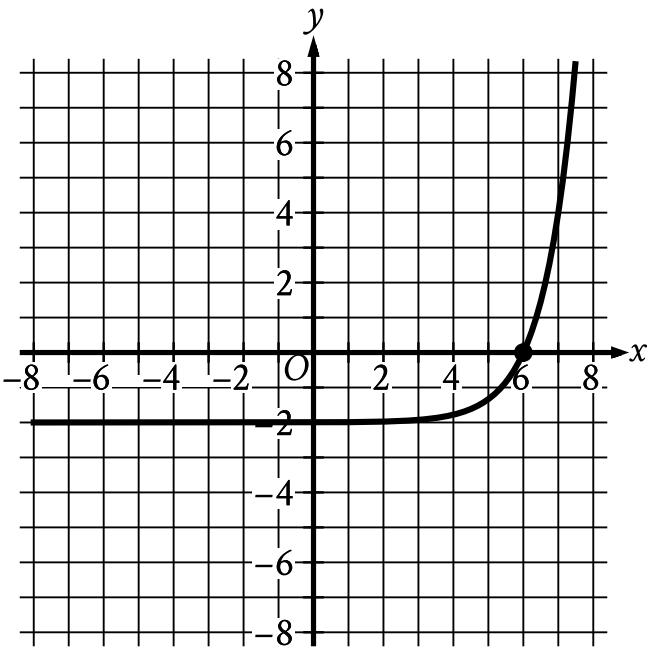
Question Difficulty: Easy

Question ID 2b6c12eb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2b6c12eb

PSAT2 M 2.12



What is the x-coordinate of the x-intercept of the graph shown?

ID: 2b6c12eb Answer

Correct Answer: 6

Rationale

The correct answer is 6. An x-intercept of a graph is a point on the graph where it intersects the x-axis, or where the value of y is 0. The graph shown intersects the x-axis at the point $(6, 0)$. Therefore, the x-coordinate of the x-intercept of the graph shown is 6.

Question Difficulty: Easy

Question ID 1d18794b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1d18794b

PSAT2 M 2.13

A contract for a certain service requires a onetime activation cost of **\$35** and a monthly cost of **\$23**. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

- A. $c = \frac{t}{23} + 35$
- B. $c = \frac{t}{35} + 23$
- C. $c = 23t + 35$
- D. $c = 35t + 23$

ID: 1d18794b Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that this service contract requires a monthly cost of \$ 23. A monthly cost of \$ 23 for t months results in a cost of \$ $23t$. It's also given that this service contract requires a onetime activation cost of \$ 35. Adding the onetime activation cost to the monthly cost of the service contract for t months yields the total cost c , in dollars, of this service contract for t months. Therefore, this situation can be represented by the equation $c = 23t + 35$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 4e97f862

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4e97f862

PSAT2 M 2.14

The function f is defined by $f(x) = 3x - 8$. What is the value of $f(7)$?

- A. 29
- B. 13
- C. -5
- D. -29

ID: 4e97f862 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 3x - 8$. The value of $f(7)$ is the value of $f(x)$ when $x = 7$. Substituting 7 for x in the given equation yields $f(7) = 3(7) - 8$, which is equivalent to $f(7) = 21 - 8$, or $f(7) = 13$.

Choice A is incorrect. This is the value of $f(7)$ when $f(x) = 3x + 8$, rather than $f(x) = 3x - 8$.

Choice C is incorrect. This is the value of $f(1)$, rather than $f(7)$. Choice D is incorrect. This is the value of $f(-7)$, rather than $f(7)$.

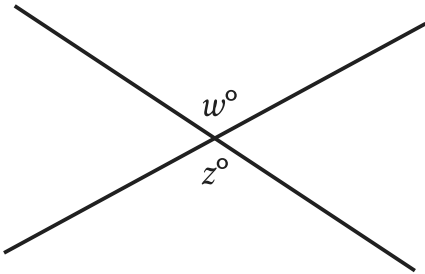
Question Difficulty: Easy

Question ID 64d1f49f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 64d1f49f

PSAT2 M 2.15



Note: Figure not drawn to scale.
In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

ID: 64d1f49f Answer

Correct Answer: D

Rationale

Choice D is correct. In the figure shown, the angles with measures w° and z° are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is 136.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure w° .

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 4efea6a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 4efea6a3

PSAT2 M 2.16

The area of a rectangle is **57** square inches. The length of the longest side of the rectangle is **19** inches. What is the length, in inches, of the shortest side of this rectangle?

ID: 4efea6a3 Answer

Correct Answer: 3

Rationale

The correct answer is 3. The area of a rectangle can be calculated by multiplying the length of its longest side by the length of its shortest side. It’s given that the area of the rectangle is 57 square inches and the length of the longest side of the rectangle is 19 inches. Let x represent the length, in inches, of the shortest side of this rectangle. It follows that $57 = 19x$. Dividing both sides of this equation by 19 yields $3 = x$. Therefore, the length, in inches, of the shortest side of the rectangle is 3.

Question Difficulty: Easy

Question ID 02489d55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 02489d55

PSAT2 M 2.17

Which expression is equivalent to $19(x^2 - 7)$?

- A. $19x^2 - 133$
- B. $19x^2 - 26$
- C. $19x^2 - 7$
- D. $19x^2 + 12$

ID: 02489d55 Answer

Correct Answer: A

Rationale

Choice A is correct. The expression $19x^2 - 7$ can be rewritten as $19x^2 - 197$, which is equivalent to $19x^2 - 133$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID bd4d0e0c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: bd4d0e0c

PSAT2 M 2.18

The function f is defined by $f(x) = 10x^2 - 32x - 152$. What is the value of $f(0)$?

- A. -152
- B. -32
- C. 0
- D. 10

ID: bd4d0e0c Answer

Correct Answer: A

Rationale

Choice A is correct. The value of $f0$ is the value of fx when $x = 0$. The function f is defined by $fx = 10x^2 - 32x - 152$. Substituting 0 for x in this equation yields $f0 = 100^2 - 320 - 152$. This equation can be rewritten as $f0 = 100 - 0 - 152$, or $f0 = -152$. Therefore, the value of $f0$ is -152.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 0bebc08c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 0bebc08c

PSAT2 M 2.19

$x = 3$ $y = (15 - x)^2$ A solution to the given system of equations is (x, y) . What is the value of xy ?

- A. 432
- B. 54
- C. 45
- D. 18

ID: 0bebc08c Answer

Correct Answer: A

Rationale

Choice A is correct. The first equation in the given system of equations is $x = 3$. Substituting 3 for x in the second equation in the given system of equations yields $y = 15 - 3^2$, or $y = 144$. Substituting 3 for x and 144 for y in the expression xy yields $(3)(144)$, or 432. Therefore, the value of xy is 432.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 301faf80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 301faf80

PSAT2 M 2.20

The product of two positive integers is **462**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

ID: 301faf80 Answer

Correct Answer: 14

Rationale

The correct answer is 14. Let x represent the first integer and y represent the second integer. If the first integer is 5 greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is 462; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting 462 from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924, and whose sum is 5. The two values whose product is -924 and whose sum is 5 are 33 and -28. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting 33 from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by 2 yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by 14 yields $x = 33$. Therefore, the two integers are 14 and 33, so the smaller of the two integers is 14.

Question Difficulty: Hard

Question ID df747160

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: df747160

PSAT2 M 2.21

Which expression is equivalent to $17(x^2 - 100y^2)$?

- A. $17(x - 2y)(x - 50y)$
- B. $17(x - 2y)(x + 50y)$
- C. $17(x - 10y)(x - 10y)$
- D. $17(x - 10y)(x + 10y)$

ID: df747160 Answer

Correct Answer: D

Rationale

Choice D is correct. Expressions in the form $a^2 - b^2$ follow the difference of two squares pattern and can be factored as $(a - b)(a + b)$. In the given expression, $17(x^2 - 100y^2)$, the expression $x^2 - 100y^2$ follows the difference of two squares pattern. It follows that the expression $x^2 - 100y^2$ can be rewritten as $(x - 10y)(x + 10y)$. Therefore, the expression $17(x - 10y)(x + 10y)$ is equivalent to $17(x^2 - 100y^2)$.

Choice A is incorrect. This expression is equivalent to $17(x^2 - 52xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Choice B is incorrect. This expression is equivalent to $17(x^2 + 48xy - 100y^2)$, not $17(x^2 - 100y^2)$.

Choice C is incorrect. This expression is equivalent to $17(x^2 - 20xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Question Difficulty: Easy

Question ID 797a81fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■ ■ ■

ID: 797a81fb

PSAT2 M 2.22

$-12x + 14y = 36$ $-6x + 7y = -18$ How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 797a81fb Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has zero solutions if the lines representing the equations in the xy -plane are distinct and parallel. Two lines are distinct and parallel if they have the same slope but different y -intercepts. Each equation in the given system can be written in slope-intercept form $y = mx + b$, where m is the slope of the line representing the equation in the xy -plane and b is the y -intercept. Adding $12x$ to both sides of the first equation in the given system of equations, $-12x + 14y = 36$, yields $14y = 12x + 36$. Dividing both sides of this equation by 14 yields $y = \frac{6}{7}x + \frac{18}{7}$. It follows that the first equation in the given system of equations has a slope of $\frac{6}{7}$ and a y -intercept of $\frac{18}{7}$. Adding $6x$ to both sides of the second equation in the given system of equations, $-6x + 7y = -18$, yields $7y = 6x - 18$. Dividing both sides of this equation by 7 yields $y = \frac{6}{7}x - \frac{18}{7}$. It follows that the second equation in the given system of equations has a slope of $\frac{6}{7}$ and a y -intercept of $-\frac{18}{7}$. Since the slopes of these lines are the same and the y -intercepts are different, it follows that the given system of equations has zero solutions.

Alternate approach: To solve the system by elimination, multiplying the second equation in the given system of equations, $-6x + 7y = -18$, by -2 yields $12x - 14y = 36$. Adding this equation to the first equation in the given system of equations, $-12x + 14y = 36$, yields $(-12x + 12x) + (-14y + 14y) = 36 + 36$, or $0 = 72$. Since this equation isn't true, the given system of equations has zero solutions.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

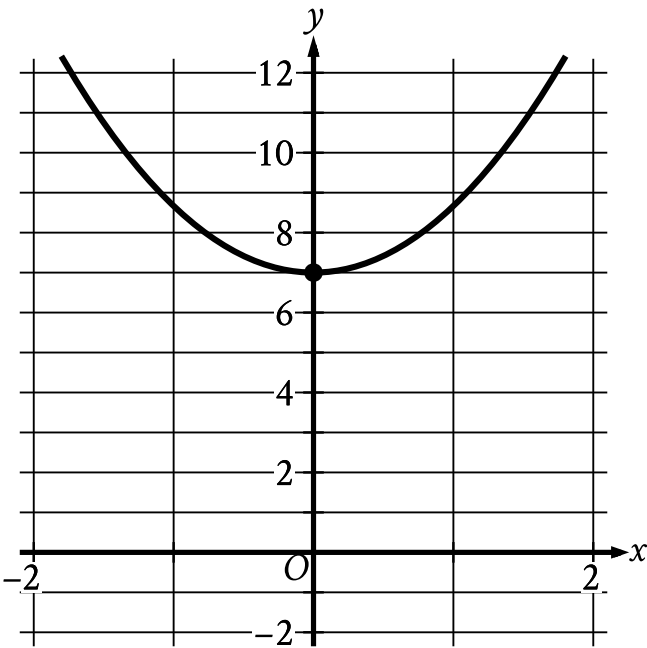
Question Difficulty: Hard

Question ID 2f4eafcc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2f4eafcc

PSAT2 M 3.1



The parabola shown intersects the y -axis at the point (x, y) . What is the value of y ?

ID: 2f4eafcc Answer

Correct Answer: 7

Rationale

The correct answer is 7. It's given that the parabola intersects the y -axis at the point x, y . The graph shows that the parabola intersects the y -axis at the point $0, 7$. Therefore, the value of y is 7.

Question Difficulty: Easy

Question ID 2e379126

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2e379126

PSAT2 M 3.2

The function g is defined by $g(x) = 4x - 6$. What is the value of $g(-7)$?

- A. -34
- B. -22
- C. $-\frac{13}{4}$
- D. $-\frac{1}{4}$

ID: 2e379126 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function g is defined by $gx = 4x - 6$. Substituting -7 for x into the given equation yields $g-7 = 4-7 - 6$, or $g-7 = - 34$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $gx = - 7$, not the value of $g-7$.

Question Difficulty: Easy

Question ID e006209c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e006209c

PSAT2 M 3.3

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

- A. **130**
- B. **63**
- C. **4**
- D. **0**

ID: e006209c Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the geologist has already collected 63 samples from the volcano. Let x represent the number of additional samples the geologist needs to collect. After collecting x additional samples, the geologist will have collected a total of $63 + x$ samples. It's given that the geologist needs to collect at least 67 samples. Therefore, $63 + x \geq 67$. Subtracting 63 from each side of this inequality yields the inequality $x \geq 4$. Thus, the geologist needs to collect a minimum of 4 additional samples.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of samples the geologist has already collected, rather than the minimum number of additional samples the geologist needs to collect.

Choice D is incorrect. If the geologist collects 0 additional samples, the geologist will have collected a total of 63 samples, which is less than 67 samples.

Question Difficulty: Easy

Question ID 0d43db90

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 0d43db90

PSAT2 M 3.4

The perimeter of triangle ABC is **17** inches, the length of side AB is **4** inches, and the length of side AC is **7** inches. What is the length, in inches, of side BC ?

- A. **4**
- B. **6**
- C. **7**
- D. **11**

ID: 0d43db90 Answer

Correct Answer: B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three sides of the triangle. It's given that the lengths of side AB and side AC are 4 inches and 7 inches, respectively. Let x represent the length, in inches, of side BC . The sum of the lengths, in inches, of all three sides of triangle ABC can be represented by the expression $4 + 7 + x$. Since it's given that the perimeter of triangle ABC is 17 inches, it follows that $17 = 4 + 7 + x$, or $17 = 11 + x$. Subtracting 11 from both sides of this equation yields $6 = x$. Therefore, the length, in inches, of side BC is 6.

Choice A is incorrect. This is the length, in inches, of side AB . Choice C is incorrect. This is the length, in inches, of side AC .

Choice D is incorrect. This is the sum of the lengths, in inches, of sides AB and AC .

Question Difficulty: Easy

Question ID a130fcdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a130fcdc

PSAT2 M 3.5

$g(x) = 11x + 4$

For the given linear function g , which table shows three values of x and their corresponding values of $g(x)$?

A.

x	$g(x)$
-1	7
0	11
1	15

B.

x	$g(x)$
-1	-4
0	0
1	4

C.

x	$g(x)$
-1	-7
0	4
1	15

D.

x	$g(x)$
-1	-11
0	0
1	11

ID: a130fcdc Answer

Correct Answer: C

Question ID c4259674

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c4259674

PSAT2 M 3.6

The function f is defined by $f(x) = 4x^{-1}$. What is the value of $f(21)$?

- A. -84
- B. $\frac{1}{84}$
- C. $\frac{4}{21}$
- D. $\frac{21}{4}$

ID: c4259674 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that function f is defined by the equation $f x = 4x^{-1}$. The value of $f21$ is the value of $f x$ when $x = 21$. Substituting 21 for x in the given equation yields $f21 = 421^{-1}$, which is equivalent to $f21 = 4\frac{1}{21}$, or $f21 = \frac{4}{21}$.

Choice A is incorrect. This is the value of $f21$ when $f x = - 4x$, rather than $f x = 4x^{-1}$.

Choice B is incorrect. This is the value of $f21$ when $f x = 4x^{-1}$, rather than $f x = 4x^{-1}$.

Choice D is incorrect. This is the value of $f21$ when $f x = 4^{-1}x$, rather than $f x = 4x^{-1}$.

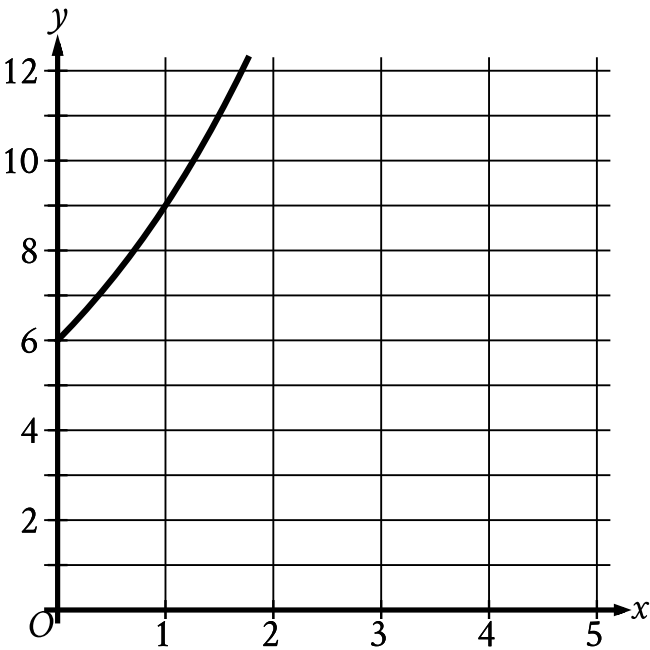
Question Difficulty: Medium

Question ID f1fa0821

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f1fa0821

PSAT2 M 3.7



The graph gives the estimated population y , in thousands, of a town x years since 2003, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

- A. The estimated population at $x = 1$ is 0.5 times the estimated population at $x = 0$.
- B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.
- C. The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.
- D. The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.

ID: f1fa0821 Answer

Correct Answer: B

Rationale

Choice B is correct. On the graph shown, the y -axis represents estimated population, in thousands. The graph shows that when $x = 0$, the y -coordinate is 6. Therefore, the estimated population at $x = 0$ is 6 thousand. The graph also shows that when $x = 1$, the y -coordinate is 9. Therefore, the estimated population at $x = 1$ is 9 thousand. Dividing 9 thousand by 6 thousand yields 1.5; therefore, 9 thousand is 1.5 times 6 thousand. It follows that the estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

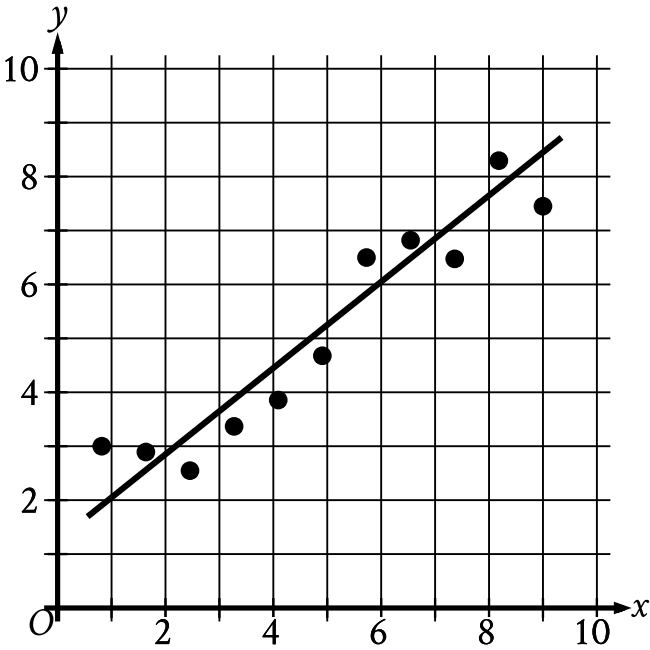
Question Difficulty: Medium

Question ID ad7dbb22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: ad7dbb22

PSAT2 M 3.8



The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown. For how many of the 11 data points does the line of best fit predict a greater y -value than the actual y -value?

ID: ad7dbb22 Answer

Correct Answer: 6

Rationale

The correct answer is 6. The line of best fit predicts a greater y -value than the actual y -value for any data point that's located below the line of best fit. For the scatterplot shown, 6 of the data points are below the line of best fit. Therefore, the line of best fit predicts a greater y -value than the actual y -value for 6 of the data points.

Question Difficulty: Medium

Question ID 55447be2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 55447be2

PSAT2 M 3.9

$y = \frac{1}{3}x - 14$ $y = -x + 18$ The solution to the given system of equations is (x, y) . What is the value of x ?

ID: 55447be2 Answer

Correct Answer: 24

Rationale

The correct answer is 24. The given system of equations can be solved by the substitution method. The first equation in the given system of equations is $y = \frac{1}{3}x - 14$. Substituting $\frac{1}{3}x - 14$ for y in the second equation in the given system yields $\frac{1}{3}x - 14 = -x + 18$. Adding 14 to both sides of this equation yields $\frac{1}{3}x = -x + 32$. Adding x to both sides of this equation yields $\frac{4}{3}x = 32$. Multiplying both sides of this equation by $\frac{3}{4}$ yields $x = 24$.

Question Difficulty: Medium

Question ID 8ad64841

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 8ad64841

PSAT2 M 3.10

A linear function f gives a company’s profit, in dollars, for selling x items. The company’s profit is \$220 when it sells 8 items, and its profit is \$320 when it sells 10 items. Which equation defines f ?

- A. $f(x) = 150x - 320$
- B. $f(x) = 32x$
- C. $f(x) = 50x - 10x$
- D. $f(x) = 50x - 180$

ID: 8ad64841 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the relationship between x and fx is linear. A linear function can be written in the form $fx = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane. Given two points on a line, x_1, y_1 and x_2, y_2 , the slope of the line can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It’s given that the company’s profit is \$ 220 when it sells 8 items and the profit is \$ 320 when it sells 10 items. Since fx represents the company's profit, in dollars, for selling x items, the graph of $y = fx$ in the xy -plane passes through the points 8, 220 and 10, 320. Substituting 8, 220 and 10, 320 for x_1, y_1 and x_2, y_2 , respectively, in the slope formula yields $m = \frac{320 - 220}{10 - 8}$, which gives $m = \frac{100}{2}$, or $m = 50$. Substituting 50 for m , 8 for x , and 220 for fx in $fx = mx + b$ yields $220 = 508 + b$, or $220 = 400 + b$. Subtracting 400 from each side of this equation yields $-180 = b$. Substituting 50 for m and -180 for b in $fx = mx + b$ yields $fx = 50x - 180$. Therefore, the equation that defines f is $fx = 50x - 180$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 40049d49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 40049d49

PSAT2 M 3.11

$4x + 12 = \frac{a(x+3)}{2}$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a ?

- A. 0
- B. 3
- C. 8
- D. 12

ID: 40049d49 Answer

Correct Answer: C

Rationale

Choice C is correct. If an equation has infinitely many solutions, then the two sides of the equation must be equivalent. Multiplying each side of the given equation by 2 yields $8x + 24 = ax + 3$. Since 8 is a common factor of both terms on the left-hand side of this equation, the equation can be rewritten as $8x + 3 = ax + 3$. The two sides of this equation are equivalent when $a = 8$. Therefore, if the given equation has infinitely many solutions, the value of a is 8.

Alternate approach: If the given equation, $4x + 12 = \frac{ax + 3}{2}$, has infinitely many solutions, then both sides of this equation are equal for any value of x . If $x = 0$, then substituting 0 for x in the given equation yields $40 + 12 = \frac{a0 + 3}{2}$, or $12 = \frac{3}{2}a$. Dividing both sides of this equation by $\frac{3}{2}$ yields $8 = a$.

Choice A is incorrect. If the value of a is 0, the given equation is equivalent to $4x + 12 = 0$, which has one solution, not infinitely many solutions.

Choice B is incorrect. If the value of a is 3, the given equation is equivalent to $4x + 12 = \frac{3x + 3}{2}$, or $4x + 12 = \frac{3}{2}x + \frac{9}{2}$, which has one solution, not infinitely many solutions.

Choice D is incorrect. If the value of a is 12, the given equation is equivalent to $4x + 12 = \frac{12x + 3}{2}$, or $4x + 12 = 6x + 18$, which has one solution, not infinitely many solutions.

Question Difficulty: Medium

Question ID d72a2b4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: d72a2b4d

PSAT2 M 3.12

In March, the price of a collectible card was **\$15.50**. In April, the price of the collectible card was **\$17.36**. The price of the collectible card in April was *p*% of the price of the collectible card in March. What is the value of *p*?

- A. 12
- B. 88
- C. 112
- D. 188

ID: d72a2b4d Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the price of the collectible card was \$ 15.50 in March and \$ 17.36 in April. It's also given that the price of the collectible card in April was *p*% of the price in March. It follows that \$ 17.36 is *p*% of \$ 15.50. Therefore, the value of *p* can be calculated by solving the equation $17.36 = \frac{p}{100}15.50$, or $17.36 = \frac{15.50p}{100}$. Multiplying each side of this equation by 100 yields $1,736 = 15.50p$. Dividing each side of this equation by 15.50 yields $112 = p$. Therefore, the value of *p* is 112.

Choice A is incorrect. 12% is the percent increase in the price of the collectible card from March to April.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3148fe3e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 3148fe3e

PSAT2 M 3.13

$x^2 + y + 10 = 10$ $8x + 16 - y = 0$ The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -16
- B. -4
- C. 2
- D. 8

ID: 3148fe3e Answer

Correct Answer: B

Rationale

Choice B is correct. Adding y to each side of the second equation in the given system of equations yields $8x + 16 = y$. Substituting $8x + 16$ for y in the first equation yields $x^2 + 8x + 16 + 10 = 10$. Subtracting 10 from each side of this equation yields $x^2 + 8x + 16 = 0$. This equation can be rewritten as $x + 4^2 = 0$. Taking the square root of each side of this equation yields $x + 4 = 0$. Subtracting 4 from each side of this equation yields $x = -4$. Therefore, the value of x is -4.

Choice A is incorrect. This is the value of y , not x . Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID be1b8c74

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: be1b8c74

PSAT2 M 3.14

$x = 8a(b + 9)$

The given equation relates the positive numbers a , b , and x . Which equation correctly expresses a in terms of b and x ?

- A. $a = \frac{x}{8} - (b + 9)$
- B. $a = \frac{x}{8(b+9)}$
- C. $a = \frac{8(b+9)}{x}$
- D. $a = 8x(b + 9)$

ID: be1b8c74 Answer

Correct Answer: B

Rationale

Choice B is correct. To express a in terms of b and x , the given equation can be rewritten such that a is isolated on one side of the equation. Since it's given that b is a positive number, $b + 9$ is not equal to zero. Therefore, dividing both sides of the given equation by $8b + 9$ yields the equivalent equation $\frac{x}{8b + 9} = a$, or $a = \frac{x}{8b + 9}$.

Choice A is incorrect. This equation is equivalent to $x = 8a + b + 9$.

Choice C is incorrect. This equation is equivalent to $x = \frac{8b + 9}{a}$. Choice D is incorrect. This equation is equivalent to $x = \frac{a}{8b + 9}$.

Question Difficulty: Medium

Question ID fde6f3bb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: fde6f3bb

PSAT2 M 3.15

$g(x) = \frac{3}{5}x + \frac{7}{6}$ $h(x) = 6x - 5$

The functions g and h are defined by the equations shown. Which expression is equivalent to $g(x) \cdot h(x)$?

- A. $\frac{18x^2}{5} - \frac{35}{6}$
- B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$
- C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$
- D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

ID: fde6f3bb Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that $gx = \frac{3}{5}x + \frac{7}{6}$ and $hx = 6x - 5$. Substituting $\frac{3}{5}x + \frac{7}{6}$ for gx and $6x - 5$ for hx in the expression $gx \cdot hx$ yields $\frac{3}{5}x + \frac{7}{6}6x - 5$. This expression can be rewritten as $\frac{3}{5}x6x - 5 + \frac{7}{6}6x - 5$, or $\frac{18x^2}{5} - 3x + 7x - \frac{35}{6}$, which is equivalent to $\frac{18x^2}{5} + 4x - \frac{35}{6}$.

Choice A is incorrect. This expression is equivalent to $\frac{3}{5}x6x + \frac{7}{6}-5$, not $\frac{3}{5}x + \frac{7}{6}6x - 5$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $\frac{3}{5}x - \frac{7}{6}6x + 5$, not $\frac{3}{5}x + \frac{7}{6}6x - 5$.

Question Difficulty: Medium

Question ID 301faf80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 301faf80

PSAT2 M 3.16

The product of two positive integers is **462**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

ID: 301faf80 Answer

Correct Answer: 14

Rationale

The correct answer is 14. Let x represent the first integer and y represent the second integer. If the first integer is 5 greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is 462; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting 462 from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924, and whose sum is 5. The two values whose product is -924 and whose sum is 5 are 33 and -28. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting 33 from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by 2 yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by 14 yields $x = 33$. Therefore, the two integers are 14 and 33, so the smaller of the two integers is 14.

Question Difficulty: Hard

Question ID 40491607

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 40491607

PSAT2 M 3.17

$f(x) = (x - 1)(x + 3)(x - 2)$

In the xy -plane, when the graph of the function f , where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g . If the graph of $y = g(x)$ crosses through the point $(4, b)$, where b is a constant, what is the value of b ?

ID: 40491607 Answer

Correct Answer: 48

Rationale

The correct answer is 48. It's given that in the xy -plane, when the graph of the function f , where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g . Therefore, function g can be defined by the equation $g(x) = f(x) + 6$. It's given that $f(x) = (x - 1)(x + 3)(x - 2)$. Substituting $(x - 1)(x + 3)(x - 2)$ for $f(x)$ in the equation $g(x) = f(x) + 6$ yields $g(x) = (x - 1)(x + 3)(x - 2) + 6$. For the point $(4, b)$, the value of x is 4. Substituting 4 for x in the equation $g(x) = (x - 1)(x + 3)(x - 2) + 6$ yields $g(4) = (4 - 1)(4 + 3)(4 - 2) + 6$, or $g(4) = 48$. It follows that the graph of $y = g(x)$ crosses through the point $(4, 48)$. Therefore, the value of b is 48.

Question Difficulty: Hard

Question ID 50b99b2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	■ ■ ■

ID: 50b99b2d

PSAT2 M 3.18

Objects R and S each travel at a constant speed. The speed of object R is half the speed of object S. Object R travels a distance of $4x$ inches in y seconds. Which expression represents the time, in seconds, it takes object S to travel a distance of $24x$ inches?

- A. $12y$
- B. $3y$
- C. $16y$
- D. $6y$

ID: 50b99b2d Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that object R travels a distance of $4x$ inches in y seconds. This speed can be written as $\frac{4x \text{ inches}}{y \text{ seconds}}$. It's given that the speed of object R is half the speed of object S. It follows that the speed of object S is twice the speed of object R, which is $2\frac{4x \text{ inches}}{y \text{ seconds}}$, or $\frac{8x \text{ inches}}{y \text{ seconds}}$. Let n represent the time, in seconds, it takes object S to travel a distance of $24x$ inches. The value of n can be found by solving the equation $\frac{8x \text{ inches}}{y \text{ seconds}} = \frac{24x \text{ inches}}{n \text{ seconds}}$, which can be written as $\frac{8x}{y} = \frac{24x}{n}$. Multiplying each side of this equation by ny yields $8xn = 24xy$. Dividing each side of this equation by $8x$ yields $n = 3y$. Therefore, the expression $3y$ represents the time, in seconds, it takes object S to travel a distance of $24x$ inches.

Choice A is incorrect. This expression represents the time, in seconds, it would take object S to travel a distance of $24x$ inches if the speed of object R were twice, not half, the speed of object S.

Choice C is incorrect. This expression represents the time, in seconds, it takes object S to travel a distance of $128x$ inches, not $24x$ inches.

Choice D is incorrect. This expression represents the time, in seconds, it takes object R, not object S, to travel a distance of $24x$ inches.

Question Difficulty: Hard

Question ID 7355b9d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■ ■ ■

ID: 7355b9d9

PSAT2 M 3.19

If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, where n and k are constants and $k > 0$, what is the value of n ?

- A. -29
- B. $-\frac{1}{29}$
- C. $\frac{1}{29}$
- D. 29

ID: 7355b9d9 Answer

Correct Answer: D

Rationale

Choice D is correct. If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, then the expression can be written as $k - xax + b$, where a and b are constants. This expression can be rewritten as $akx + bk - ax^2 - bx$, or $-ax^2 + ak - bx + bk$. Since this expression is equivalent to $-x^2 + \frac{1}{29}nk^2$, it follows that $-a = -1$, $ak - b = 0$, and $bk = \frac{1}{29}nk^2$. Dividing each side of the equation $-a = -1$ by -1 yields $a = 1$. Substituting 1 for a in the equation $ak - b = 0$ yields $k - b = 0$. Adding b to each side of this equation yields $k = b$. Substituting k for b in the equation $bk = \frac{1}{29}nk^2$ yields $k^2 = \frac{1}{29}nk^2$. Since k is positive, dividing each side of this equation by k^2 yields $1 = \frac{1}{29}n$. Multiplying each side of this equation by 29 yields $29 = n$.

Alternate approach: The expression $x^2 - y^2$ can be written as $x - yx + y$, which is a difference of two squares. It follows that $\frac{1}{29}nk^2 - x^2$ is equivalent to $\sqrt{\frac{1}{29}nk} - x\sqrt{\frac{1}{29}nk} + x$. It's given that $k - x$ is a factor of $-x^2 + \frac{1}{29}nk^2$, so the factor $\sqrt{\frac{1}{29}nk} - x$ is equal to $k - x$. Adding x to both sides of the equation $\sqrt{\frac{1}{29}nk} - x = k - x$ yields $\sqrt{\frac{1}{29}nk} = k$. Since k is positive, dividing both sides of this equation by k yields $\sqrt{\frac{1}{29}n} = 1$. Squaring both sides of this equation yields $\frac{1}{29}n = 1$. Multiplying both sides of this equation by 29 yields $n = 29$.

Choice A is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}29k^2$, or $-x^2 - k^2$. This expression doesn't have $k - x$ as a factor.

Choice B is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}(-\frac{1}{29})k^2$, or $-x^2 + \frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

Choice C is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}(\frac{1}{29})k^2$, or $-x^2 + \frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

Question Difficulty: Hard

Question ID dcf63c94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: dcf63c94

PSAT2 M 3.20

$f(x) = 272(2)^x$

The function f is defined by the given equation. If $h(x) = f(x - 4)$, which of the following equations defines function h ?

- A. $h(x) = 17(2)^x$
- B. $h(x) = 68(2)^x$
- C. $h(x) = 272(16)^x$
- D. $h(x) = 272(8)^x$

ID: dcf63c94 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $f(x) = 272(2)^x$ and $h(x) = f(x - 4)$. Substituting $x - 4$ for x in $f(x) = 272(2)^x$ yields $f(x - 4) = 272(2)^{x - 4}$. Substituting hx for $f(x - 4)$ in this equation yields $hx = 272(2)^{x - 4}$. Using the properties of exponents, the function $h(x) = 272(2)^{x - 4}$ can be rewritten as $hx = \frac{2722^x}{2^4}$, which is equivalent to $hx = \frac{2722^x}{16}$, or $h(x) = 17(2)^x$. Therefore, of the given choices, an equation that defines function h is $h(x) = 17(2)^x$.

Choice B is incorrect. This equation defines function h if $h(x) = f(x - 2)$, not $h(x) = f(x - 4)$.

Choice C is incorrect. This equation defines function h if $h(x) = f(4x)$, not $h(x) = f(x - 4)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f731d88b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	■ ■ ■

ID: f731d88b

PSAT2 M 3.21

In convex pentagon $ABCDE$, segment AB is parallel to segment DE . The measure of angle B is 139 degrees, and the measure of angle D is 174 degrees. What is the measure, in degrees, of angle C ?

ID: f731d88b Answer

Correct Answer: 47

Rationale

The correct answer is 47. It's given that the measure of angle B is 139 degrees. Therefore, the exterior angle formed by extending segment AB at point B has measure $180 - 139$, or 41, degrees. It's given that segment AB is parallel to segment DE . Extending segment BC at point C and extending segment DE at point D until the two segments intersect results in a transversal that intersects two parallel line segments. One of these intersection points is point B , and let the other intersection point be point X . Since segment AB is parallel to segment DE , alternate interior angles are congruent. Angle CXD and the exterior angle formed by extending segment AB at point B are alternate interior angles. Therefore, the measure of angle CXD is 41 degrees. It's given that the measure of angle D in pentagon $ABCDE$ is 174 degrees. Therefore, angle CDX has measure $180 - 174$, or 6, degrees. Since angle C in pentagon $ABCDE$ is an exterior angle of triangle CDX , it follows that the measure of angle C is the sum of the measures of angles CDX and CXD . Therefore, the measure, in degrees, of angle C is $6 + 41$, or 47.

Alternate approach: A line can be created that's perpendicular to segments AB and DE and passes through point C . Extending segments AB and DE at points B and D , respectively, until they intersect this line yields two right triangles. Let these intersection points be point X and point Y , and the two right triangles be triangle BXC and triangle DYC . It's given that the measure of angle B is 139 degrees. Therefore, angle CBX has measure $180 - 139$, or 41, degrees. Since the measure of angle CBX is 41 degrees and the measure of angle BXC is 90 degrees, it follows that the measure of angle XCB is $180 - 90 - 41$, or 49, degrees. It's given that the measure of angle D is 174 degrees. Therefore, angle YDC has measure $180 - 174$, or 6, degrees. Since the measure of angle YDC is 6 degrees and the measure of angle CYD is 90 degrees, it follows that the measure of angle DCY is $180 - 90 - 6$, or 84, degrees. Since angles XCB , DCY , and angle C in pentagon $ABCDE$ form segment XY , it follows that the sum of the measures of those angles is 180 degrees. Therefore, the measure, in degrees, of angle C is $180 - 49 - 84$, or 47.

Question Difficulty: Hard

Question ID 2c288148

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 2c288148

PSAT2 M 3.22

$\sqrt{k - x} = 58 - x$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

ID: 2c288148 Answer

Correct Answer: 231

Rationale

The correct answer is 231. It's given that $\sqrt{k - x} = 58 - x$. Squaring both sides of this equation yields $k - x = (58 - x)^2$, which is equivalent to the given equation if $58 - x > 0$. It follows that if a solution to the equation $k - x = (58 - x)^2$ satisfies $58 - x > 0$, then it's also a solution to the given equation; if not, it's extraneous. The equation $k - x = (58 - x)^2$ can be rewritten as $k - x = 3,364 - 116x + x^2$. Adding x to both sides of this equation yields $k = x^2 - 115x + 3,364$. Subtracting k from both sides of this equation yields $0 = x^2 - 115x + (3,364 - k)$. The number of solutions to a quadratic equation in the form $0 = ax^2 + bx + c$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. Substituting -115 for b , 1 for a , and $3,364 - k$ for c in $b^2 - 4ac$ yields $(-115)^2 - 4(1)(3,364 - k)$, or $4k - 231$. The equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution if the discriminant is equal to zero, or $4k - 231 = 0$. Subtracting 231 from both sides of this equation yields $4k = 231$. Dividing both sides of this equation by 4 yields $k = 57.75$. Therefore, if $k = 57.75$, then the equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution. Substituting 57.75 for k in this equation yields $0 = x^2 - 115x + (3,364 - 57.75)$, or $0 = x^2 - 115x + 3,306.25$, which is equivalent to $0 = (x - 57.5)^2$. Taking the square root of both sides of this equation yields $0 = x - 57.5$. Adding 57.5 to both sides of this equation yields $57.5 = x$. To check whether this solution satisfies $58 - x > 0$, the solution, 57.5 , can be substituted for x in $58 - x > 0$, which yields $58 - 57.5 > 0$, or $0.5 > 0$. Since 0.5 is greater than 0 , it follows that if $k = 57.75$, or $4k = 231$, then the given equation has exactly one real solution. If $4k < 231$, then the discriminant, $4k - 231$, is negative and the given equation has no solutions. Therefore, the minimum possible value of $4k$ is 231 .

Question Difficulty: Hard